

COURSE NOTES

Abstract Algebra III

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Chapter 1

Lecture 1: Modules

Message: QQ, BB. HW 30%, Mid 20%, Final 50%

1.1 Basic Definitions and Examples

Definition 1.1.1. Let R be a ring with 1, and let M be an abelian group. We say that M is a left R -module if R “left acts” on M :

$$R \times M \rightarrow M, \quad (r, m) \mapsto rm$$

such that

$$\begin{aligned} r(m + n) &= rm + rn, \\ (r + s)m &= rm + sm, \\ (rs)m &= r(sm), \\ 1m &= m. \end{aligned} \quad \text{for any } r, s \in R, m, n \in M$$

A **right R -module** is defined by $(r, m) \mapsto mr$ with the axiom $m(rs) = (mr)s$. If R is commutative, left-modules and right-modules coincide. Otherwise, one can transform using the opposite ring R^{op} .

Theorem 1.1.2. Let M be an abelian group. Then M is an R -module if and only if there is a ring homomorphism $R \rightarrow \text{End}(M)$.

Proof. ($\Rightarrow$) Suppose M is an R -module. Define $\rho : R \rightarrow \text{End}(M)$ by

$$\rho(r)(m) = rm, \quad \forall r \in R, m \in M.$$

For each $r \in R$, the map $\rho(r) : M \rightarrow M$ is a group homomorphism since $r(m_1 + m_2) = rm_1 + rm_2$. Moreover, ρ is a ring homomorphism:

- $\rho(r + s)(m) = (r + s)m = rm + sm = \rho(r)(m) + \rho(s)(m)$, so $\rho(r + s) = \rho(r) + \rho(s)$.
- $\rho(rs)(m) = (rs)m = r(sm) = \rho(r)(\rho(s)(m))$, so $\rho(rs) = \rho(r) \circ \rho(s)$.

- $\rho(1)(m) = 1m = m$, so $\rho(1) = \text{id}_M$.

($\Leftarrow$) Suppose $\rho : R \rightarrow \text{End}(M)$ is a ring homomorphism. Define the scalar multiplication $R \times M \rightarrow M$ by

$$(r, m) \mapsto \rho(r)(m).$$

We verify the module axioms:

- $r(m_1 + m_2) = \rho(r)(m_1 + m_2) = \rho(r)(m_1) + \rho(r)(m_2) = rm_1 + rm_2$.
- $(r + s)m = \rho(r + s)(m) = (\rho(r) + \rho(s))(m) = \rho(r)(m) + \rho(s)(m) = rm + sm$.
- $(rs)m = \rho(rs)(m) = (\rho(r) \circ \rho(s))(m) = \rho(r)(\rho(s)(m)) = r(sm)$.
- $1m = \rho(1)(m) = \text{id}_M(m) = m$.

Hence M is an R -module. □

Example 1.1.3. (1) Let M be an abelian group. Then M is an $\text{End}(M)$ -module via

$$\text{End}(M) \times M \rightarrow M, \quad (\varphi, m) \mapsto \varphi(m).$$

(Use the theorem: $\text{End}(M) \rightarrow \text{End}(M)$.)

(2) Let V be a vector space over a field K , and let $A : V \rightarrow V$ be a linear transformation. Then V is a $K[x]$ -module via

$$K[x] \times V \rightarrow V, \quad (f(x), v) \mapsto f(A)v.$$

(3) If $|M| = 1$, then M is called the zero module.

(4) Let R be a ring with 1 and $A \in M_n(R)$. The solution of $AX = 0$ where $X \in R^n$ is a right R -module.

Definition 1.1.4 (R -submodule). Let R be a ring (with 1) and let M be a left R -module. A subset $N \subseteq M$ is called an R -submodule of M if:

1. $0 \in N$;
2. $n_1, n_2 \in N \implies n_1 + n_2 \in N$;
3. $n \in N \implies -n \in N$;
4. $r \in R, n \in N \implies rn \in N$.

Equivalently, N is a subgroup of the abelian group $(M, +)$ that is closed under the R -action.

Question: Are submodules of a finitely generated R -module finitely generated?

Answer: No, not in general. This property holds if and only if the ring R is left Noetherian.

Definition 1.1.5. Let M be an R -module. A subgroup N of M is called a submodule of M if N is closed under scalar multiplication by elements of R . In other words, $rn \in N$ for all $r \in R$ and $n \in N$.

Example 1.1.6 (Submodule of a f.g. module that is not f.g.). Let k be a field and consider the polynomial ring $R = k[x_1, x_2, x_3, \dots]$ in infinitely many variables.

- The ring R itself can be viewed as a left R -module, which we denote by $M = R$. M is finitely generated, for instance by the single element $1 \in R$.
- Consider the ideal $I = \langle x_1, x_2, x_3, \dots \rangle$ generated by all the variables. This ideal I is a submodule of $M = R$.
- This submodule I is not finitely generated. To see this, suppose for contradiction that I is finitely generated by a set of polynomials $\{f_1, \dots, f_n\}$. Let N be the largest index of any variable appearing in any of the polynomials $f_1, \dots, f_n$. Then any element in the ideal generated by $\{f_1, \dots, f_n\}$ must be a polynomial in variables $x_1, \dots, x_N$. However, the polynomial x_{N+1} is in I but cannot be expressed as a combination of $f_1, \dots, f_n$. This is a contradiction.

Therefore, the ideal I is a submodule of the finitely generated R -module R which is not itself finitely generated. This shows that the ring R is not Noetherian.

Definition 1.1.7. Let M be an R -module, $M_i, i \in I$ are R -submodules of M . Then

$$\bigcap_{i \in I} M_i$$

is an R -submodule of M .

$$\sum_{i \in I} = \left\{ \sum_{i \in I} f_i m_i \mid m_i \in M_i \right\}$$

is an R -submodule of M .

Definition 1.1.8. Let M be an R -module, $S \subseteq M$.

$$\langle S \rangle = \left\{ \sum_{finite} a_i s_i \mid a_i \in R, s_i \in S \right\}$$

or

$$\langle S \rangle = \bigcap_{N \text{ is an } R\text{-submodule of } M, S \subseteq N} N$$

is an R -submodule of M , called the R -submodule of M generated by S .

If $M = \langle S \rangle$, then we say M is generated by S . In particular, if $|S| = 1$ then we

call M a cyclic module. If $|S| < \infty$ then we call M a finite generated module.

1.2 Quotient Modules and Homomorphism Theorems

Proposition 1.2.1. *Let M be an R -module, N a submodule of M , M' an R -module, and $\varphi : M \rightarrow M'$ a morphism such that $N \subseteq \ker(\varphi)$. Then there exists a unique morphism*

$$\varphi_N : M/N \rightarrow M'$$

such that the following diagram commutes. Moreover, $\ker(\varphi_N) = \ker \varphi / N$.

$$\begin{array}{ccc} M & \xrightarrow{\varphi} & M' \\ \pi_N \downarrow & \nearrow \varphi_N & \\ M/N & & \end{array}$$

Remark 1.2.2. *The pair $(M/N, \pi_N)$ is unique in the following sense: if (S, θ) is a pair such that*

- S is an R -module,
- $\theta : M \rightarrow S$ is a surjective morphism satisfying the property of $(M/N, \pi_N)$ described in the above proposition,

then there exists a unique isomorphism $\psi : M/N \rightarrow S$ such that the following diagram commutes.

$$\begin{array}{ccc} M & \xrightarrow{\theta} & S \\ \pi_N \downarrow & \nearrow \exists! \psi & \\ M/N & & \end{array}$$

Proposition 1.2.3. *There is a unique isomorphism $\bar{\varphi} : \text{coim } \varphi \rightarrow \text{im } \varphi$ such that the following diagram commutes:*

$$\begin{array}{ccc} M & \xrightarrow{\varphi} & M' \\ \pi_{\ker(\varphi)} \downarrow & & \uparrow \\ \text{coim}(\varphi) & \xrightarrow{\bar{\varphi}} & \text{im}(\varphi) \end{array}$$

Recall:

$$\begin{aligned} \varphi : M &\rightarrow M', \quad R\text{-morphism} \\ \text{coker}(\varphi) &= M' / \text{im}(\varphi) \\ \text{coim}(\varphi) &= M / \ker \varphi \end{aligned}$$

Definition 1.2.4. Let M be an R -module and I an ideal of R . Denote by IM the submodule of M generated by all elements am where $a \in I$, $m \in M$.

$$IM = \left\{ \sum_{\text{finite}} a_i m_i \mid a_i \in I, m_i \in M \right\}$$

Proposition 1.2.5. Let M be an R -module. Then $R \rightarrow \text{End}(M)$ is a ring homomorphism.

1. $R \rightarrow \text{End}(M)$ factors through the natural surjection $R \rightarrow R/I$ if and only if $IM = 0$:

$$\begin{array}{ccc} R & \xrightarrow{\rho} & \text{End}(M) \\ & \searrow \pi_I & \nearrow \exists \rho_I \\ & R/I & \end{array}$$

$\exists \rho_I : R/I \rightarrow \text{End}(M)$ such that the diagram commutes.

2. The module M/IM is naturally endowed with a structure of R/I -module.
3. Let $\varphi : M \rightarrow M'$ be an R -homomorphism. Then $\varphi(IM) \subseteq IM'$, so φ induces a morphism of R/I -modules

$$M/IM \rightarrow M'/IM'.$$

1.3 Direct Sums and Free Modules

Definition 1.3.1. Direct sum & direct product ($\prod_{i \in I} M_i$ & $\bigoplus_{i \in I} M_i$)

$$\begin{cases} \text{direct product: } \prod_{i \in I} M_i \\ \text{direct sum: } \bigoplus_{i \in I} M_i \text{ (finitely many } \neq 0) \end{cases}$$

Equal when $|I| < \infty$. In a module category, direct product = product, direct sum = coproduct.

Definition 1.3.2 (free module). Let M be an R -module.

1. $x_1, \dots, x_n \in M$ are linearly independent if

$$a_1 x_1 + a_2 x_2 + \dots + a_n x_n = 0 \Rightarrow a_1 = a_2 = \dots = a_n = 0.$$

2. M is called free if it is generated by a linearly independent subset. This subset is called a basis of M .

3. If M is an R -module with a basis $x_1, \dots, x_n$, then $M = \langle x_1 \rangle \oplus \dots \oplus \langle x_n \rangle$.

Example 1.3.3. $R \oplus \dots \oplus R = R^n$.

Theorem 1.3.4. Let R be a commutative ring with 1. Let M be a finitely generated free R -module. Then any basis of M has the same cardinality. The number of elements in a basis is called the rank of the free module.

Proof. **Step 1 (Any basis is finite).** Let $B = \{x_i\}_{i \in I}$ be a basis of M . Since M is finitely generated, choose generators $y_1, \dots, y_n$ of M . Each y_k is a finite R -linear combination of elements of B , so there exists a finite subset $I_0 \subset I$ such that $y_1, \dots, y_n \in \sum_{i \in I_0} Rx_i$. Hence

$$M = \sum_{k=1}^n Ry_k \subseteq \sum_{i \in I_0} Rx_i \subseteq M,$$

so $M = \sum_{i \in I_0} Rx_i$. If $I \setminus I_0 \neq \emptyset$, pick $j \in I \setminus I_0$. Then $x_j \in M = \sum_{i \in I_0} Rx_i$, contradicting that B is linearly independent.

Thus $I = I_0$ is finite.

Step 2 (Reduce modulo a maximal ideal). Let $J \subset R$ be a maximal ideal and set $F := R/J$, a field. Then M/JM is naturally an F -vector space.

Step 3 (A basis descends to a basis mod J). Let $x_1, \dots, x_r$ be an R -basis of M , and write $\overline{m}$ for the class of $m \in M$ in M/JM , and $\overline{a}$ for the class of $a \in R$ in F .

Spanning. For any $m \in M$, write $m = \sum_{i=1}^r a_i x_i$. Then $\overline{m} = \sum_{i=1}^r \overline{a_i} \overline{x_i}$, so $\overline{x_1}, \dots, \overline{x_r}$ span M/JM .

Linear independence. Suppose $\sum_{i=1}^r \overline{a_i} \overline{x_i} = 0$ in M/JM . Then $\sum_{i=1}^r a_i x_i \in JM$, so there exist $a'_1, \dots, a'_r \in J$ such that $\sum_{i=1}^r a_i x_i = \sum_{i=1}^r a'_i x_i$. Hence $\sum_{i=1}^r (a_i - a'_i) x_i = 0$. By linear independence of $x_1, \dots, x_r$, we get $a_i - a'_i = 0$ for all i , so $a_i \in J$, thus $\overline{a_i} = 0$ for all i . Therefore $\overline{x_1}, \dots, \overline{x_r}$ are F -linearly independent, and form an F -basis of M/JM .

Step 4 (Compare cardinalities). Let $x_1, \dots, x_r$ and $y_1, \dots, y_s$ be two R -bases of M . By Step 3, their images $\overline{x_1}, \dots, \overline{x_r}$ and $\overline{y_1}, \dots, \overline{y_s}$ are both F -bases of the same F -vector space M/JM . Hence $r = \dim_F(M/JM) = s$. Therefore any two bases of M have the same cardinality. $\square$

Chapter 2

Lecture 2: Modules and Categories

Proposition 2.0.1 (Properties of free modules). *Let L be a free R -module.*

1. *Let $\pi : X \rightarrow Y$ be an epimorphism of R -modules. For any R -morphism $\varphi : L \rightarrow Y$, there exists a morphism $\tilde{\varphi} : L \rightarrow X$ such that the following diagram commutes (hence a free module is projective):*

$$\begin{array}{ccc} & L & \\ \tilde{\varphi} \swarrow & \downarrow \varphi & \\ X & \xrightarrow[\pi]{} & Y \end{array}$$

2. *In particular, for any R -module M , any epimorphism $\pi : M \rightarrow L$ has a section, that is, there is a morphism $s : L \rightarrow M$ such that $\pi \circ s = \text{id}_L$. In that case, s is injective, $\text{im}(s) \cong L$ and*

$$M = \ker \pi \oplus \text{im}(s).$$

Proof. Step 1: Prove the first property

Let $B = \{e_i\}_{i \in I}$ be a basis of the free module L .

Since π is surjective, for each $i \in I$ choose an element $x_i \in X$ such that

$$\pi(x_i) = \varphi(e_i).$$

(Here we use the axiom of choice if I is infinite.)

By the defining universal property of a free module: given any R -module X and any map $f : B \rightarrow X$, there exists a unique R -linear morphism $F : L \rightarrow X$ such that $F(e_i) = f(e_i)$ for all i . Apply this with $f(e_i) := x_i$. Then there exists a unique R -linear map

$$\tilde{\varphi} : L \rightarrow X \quad \text{such that} \quad \tilde{\varphi}(e_i) = x_i \quad \text{for all } i \in I.$$

Now check commutativity. Define two R -linear maps $L \rightarrow Y$:

$$\psi_1 := \pi \circ \tilde{\varphi}, \quad \psi_2 := \varphi.$$

For each basis element e_i we have

$$\psi_1(e_i) = (\pi \circ \tilde{\varphi})(e_i) = \pi(x_i) = \varphi(e_i) = \psi_2(e_i).$$

Since B generates L and both ψ_1, ψ_2 are R -linear, it follows that $\psi_1 = \psi_2$, i.e. $\pi \circ \tilde{\varphi} = \varphi$.

Step2: Use the first property prove the second one

Let $\pi : M \rightarrow L$ be an epimorphism. Apply (1) with $X = M$, $Y = L$, and take $\varphi = \text{id}_L : L \rightarrow L$. Since L is free, by (1) there exists an R -morphism

$$s : L \rightarrow M \quad \text{such that} \quad \pi \circ s = \text{id}_L.$$

So π has a section.

(i) s is injective. If $s(\ell) = 0$, then applying π gives

$$\ell = (\pi \circ s)(\ell) = \pi(0) = 0,$$

hence $\ker(s) = 0$ and s is injective. In particular, s induces an isomorphism

$$L \xrightarrow{\sim} \text{im}(s), \quad \ell \mapsto s(\ell),$$

so $\text{im}(s) \cong L$.

(ii) $M = \ker \pi \oplus \text{im}(s)$. First we show $\ker \pi \cap \text{im}(s) = 0$. If $m \in \ker \pi \cap \text{im}(s)$, then $m = s(\ell)$ for some $\ell \in L$ and also $\pi(m) = 0$. Thus

$$0 = \pi(m) = \pi(s(\ell)) = (\pi \circ s)(\ell) = \ell,$$

so $m = s(0) = 0$.

Next, we show $\ker \pi + \text{im}(s) = M$. For any $m \in M$, set

$$m_0 := m - s(\pi(m)).$$

Then

$$\pi(m_0) = \pi(m) - \pi(s(\pi(m))) = \pi(m) - (\pi \circ s)(\pi(m)) = \pi(m) - \pi(m) = 0,$$

so $m_0 \in \ker \pi$, and clearly $s(\pi(m)) \in \text{im}(s)$. Hence

$$m = m_0 + s(\pi(m)) \in \ker \pi + \text{im}(s).$$

Combining with $\ker \pi \cap \text{im}(s) = 0$, we conclude

$$M = \ker \pi \oplus \text{im}(s).$$

□

Definition 2.0.2 (Direct summand.). *Let M be a module. If $M = M_1 \oplus M_2$, then M_1 is called a direct summand of M .*

Proposition 2.0.3. *A direct summand of a free module is a projective module.*

Proof. Let F be a free R -module and suppose $F \cong P \oplus Q$ for some R -modules P, Q . Let

$$i : P \rightarrow F, \quad p : F \rightarrow P$$

be the canonical inclusion and projection, so that $p \circ i = \text{id}_P$.

To show that P is projective, let $\pi : X \twoheadrightarrow Y$ be an epimorphism of R -modules and let $\varphi : P \rightarrow Y$ be any R -morphism. Consider the composite

$$\varphi \circ p : F \rightarrow Y.$$

Since F is free, it is projective; hence there exists an R -morphism

$$\tilde{\Phi} : F \rightarrow X \quad \text{such that} \quad \pi \circ \tilde{\Phi} = \varphi \circ p.$$

Now define

$$\tilde{\varphi} := \tilde{\Phi} \circ i : P \rightarrow X.$$

Then

$$\pi \circ \tilde{\varphi} = \pi \circ \tilde{\Phi} \circ i = (\varphi \circ p) \circ i = \varphi \circ (p \circ i) = \varphi \circ \text{id}_P = \varphi.$$

Thus every map $\varphi : P \rightarrow Y$ lifts along every epimorphism $\pi : X \twoheadrightarrow Y$, so P is projective. $\square$

Definition 2.0.4 (Tensor product of modules: linearize the bilinear map.). *Let M, N be R -modules. The tensor product of M and N is a pair $(M \otimes N, t_{M,N})$ where $M \otimes N$ is an R -module and $t_{M,N} : M \times N \rightarrow M \otimes_R N$ is a bilinear map, usually denoted $(m, n) \mapsto m \otimes_R n$, which satisfies the following universal property: whenever X is an R -module and $f : M \times N \rightarrow X$ is a bilinear map, there exists a unique R -morphism $\mathcal{F} : M \otimes_R N \rightarrow X$ such that the following diagram commutes.*

Up to isomorphism, tensor product is unique: Suppose that $(T_1, t_1), (T_2, t_2)$ are two tensor products. We have $t_2 = \varphi_1 t_1$

$$\begin{aligned} t_1 &= \varphi_2 t_2 \\ \Rightarrow t_1 &= \varphi_2 t_2 = \varphi_2 \varphi_1 t_1 \end{aligned}$$

Now $M \times N \xrightarrow{t_1} T_1$. Since $t_1 = \varphi_2 \varphi_1 t_1$, by uniqueness $\varphi_2 \varphi_1 = \text{Id}_1 \Rightarrow T_1 \cong T_2$. Similarly $\varphi_1 \varphi_2 = \text{Id}$.

Proof of existence: Let F be a free R -module with basis $M \times N$, say $\{e(m, n)\}_{m \in M, n \in N}$.

Let B be the submodule of F generated by

$$\begin{aligned} &e(m_1 + m_2, n) - e(m_1, n) - e(m_2, n), \\ &e(m, n_1 + n_2) - e(m, n_1) - e(m, n_2), \\ &e(\lambda m, n) - \lambda e(m, n), \\ &e(m, \lambda n) - \lambda e(m, n). \end{aligned}$$

Then $M \otimes_R N = F/B$. Write $t(m, n) := m \otimes n$. The elements $\{m \otimes_R n\}_{m \in M, n \in N}$, called elementary tensors, generate $M \otimes_R N$.

Fix $m \in M$, the map $N \rightarrow M \otimes_R N, n \mapsto m \otimes_R n$ is a morphism of R -modules. Its image $m \otimes_R N = \{m \otimes_R n \mid n \in N\}$ is a submodule of $M \otimes_R N$ and is isomorphic to a quotient of N . The map might not be injective, i.e., we can have $m \neq 0$ but $m \otimes n = 0$.

For example, take $M = \mathbb{F}_2, N = \mathbb{F}_3$, then $M \otimes_{\mathbb{Z}} N = 0$ even though M and N are non-zero. This can be shown by constructing $M \times N \xrightarrow{\varphi} P$ bilinear such that $\varphi(m, n) \neq 0$.

Proposition 2.0.5. *Properties of tensor product. We have unique isomorphisms*

$$\begin{aligned} (M_1 \otimes_R M_2) \otimes_R M_3 &\longrightarrow M_1 \otimes_R (M_2 \otimes_R M_3) \text{ associative} \\ M_1 \otimes_R M_2 &\longrightarrow M_2 \otimes_R M_1 \text{ commutative} \\ M_1 \otimes_R (M_2 \oplus M_3) &\xrightarrow{\sim} (M_1 \otimes_R M_2) \oplus (M_1 \otimes_R M_3) \text{ tensor product of} \\ \text{Moreover,} \\ R \otimes_R M &\xrightarrow{\sim} M, \quad (\lambda \otimes_R m \mapsto \lambda m) \text{ sum of tensor product.} \end{aligned}$$

Proof:

$$(1) (M_1 \otimes_R M_2) \otimes_R M_3 \xrightarrow{\sim} M_1 \otimes_R (M_2 \otimes_R M_3)$$

Define $M_1 \times M_2 \xrightarrow{f_{m_3}} M_1 \otimes_R (M_2 \otimes_R M_3)$ by

$$(m_1, m_2) \mapsto m_1 \otimes (m_2 \otimes m_3).$$

f_{m_3} is bilinear, hence $\exists!$ R -module homomorphism

$$\begin{aligned} \psi_{m_3} : M_1 \otimes_R M_2 &\rightarrow M_1 \otimes_R (M_2 \otimes_R M_3) \\ m_1 \otimes m_2 &\mapsto m_1 \otimes (m_2 \otimes m_3). \end{aligned}$$

Now we can define $M_1 \otimes_R M_2 \times M_3 \xrightarrow{\varphi} M_1 \otimes_R (M_2 \otimes_R M_3)$ by

$$(m_1 \otimes m_2, m_3) \mapsto m_1 \otimes (m_2 \otimes m_3) = \psi_{m_3}(m_1 \otimes m_2).$$

φ is well-defined as f_m is, $\forall m \in M_3$.

Then $\exists!$ R -morphism $(M_1 \otimes_R M_2) \otimes_R M_3 \xrightarrow{\theta} M_1 \otimes_R (M_2 \otimes_R M_3)$ given by

$$(m_1 \otimes m_2) \otimes m_3 \mapsto m_1 \otimes (m_2 \otimes m_3).$$

Similarly, $\exists!$ R -morphism $M_1 \otimes_R (M_2 \otimes_R M_3) \xrightarrow{\rho} (M_1 \otimes_R M_2) \otimes_R M_3$ given by

$$m_1 \otimes (m_2 \otimes m_3) \mapsto (m_1 \otimes m_2) \otimes m_3.$$

$\theta\rho = \text{Id}$, $\rho\theta = \text{Id}$. Hence $M_1 \otimes_R (M_2 \otimes_R M_3) \simeq (M_1 \otimes_R M_2) \otimes_R M_3$.

(2) $M_1 \otimes_R M_2 \xrightarrow{\sim} M_2 \otimes_R M_1$

Define $M_1 \times M_2 \xrightarrow{\varphi} M_2 \otimes_R M_1$ bilinear by

$$(m_1, m_2) \mapsto m_2 \otimes m_1.$$

Hence $\exists!$ R -module morphism $M_1 \otimes_R M_2 \xrightarrow{\rho} M_2 \otimes_R M_1$ given by

$$m_1 \otimes m_2 \mapsto m_2 \otimes m_1.$$

If $\varphi\rho$ and $\rho\varphi$ are Id on $m_1 \otimes m_2$ and $m_2 \otimes m_1$ respectively. Since $m_1 \otimes m_2$ generates $M_1 \otimes M_2$, $m_2 \otimes m_1$ generates $M_2 \otimes M_1$, we have $\varphi\rho = \text{Id}$, $\rho\varphi = \text{Id}$. Hence $M_1 \otimes M_2 \xrightarrow{\sim} M_2 \otimes M_1$.

Theorem 2.0.6. *Canonical isomorphism.*

Let M, N, P be R -modules. We have a canonical isomorphism

$$\begin{aligned} \phi = \phi_{M,N,P} : \text{Hom}_R(M \otimes_R N, P) &\longrightarrow \text{Hom}_R(M, \text{Hom}_R(N, P)) \\ f &\longmapsto \phi(f)(m)(n) = f(m \otimes n) \end{aligned}$$

The inverse ϕ^{-1} is given by

$$\phi^{-1}(g)(m \otimes_R n) = g(m)(n) \quad \longleftarrow \quad g$$

Remark 2.0.7. *Canonical.*

If $f : A \rightarrow A'$, there exists a commutative diagram

$$\text{Hom}_R(A' \otimes_R B, C) \xrightarrow{\phi_{A',B,C}} \text{Hom}_R(A', \text{Hom}_R(B, C))$$

If $f : A \rightarrow A'$, there exists a commutative diagram

$$\begin{array}{ccc} \text{Hom}_R(A' \otimes_R B, C) & \xrightarrow{\phi_{A',B,C}} & \text{Hom}_R(A', \text{Hom}_R(B, C)) \\ \downarrow \text{Hom}_R(f \otimes \text{id}, C) & & \downarrow \text{Hom}_R(f, \text{Hom}_R(B, C)) \\ \text{Hom}_R(A \otimes_R B, C) & \xrightarrow{\phi_{A,B,C}} & \text{Hom}_R(A, \text{Hom}_R(B, C)) \end{array}$$

Notation. Let N, M_i ($1 \leq i \leq n$) be R -modules.

$$L^n(M_1, \dots, M_n; N) = \{f \mid f : M_1 \times \dots \times M_n \rightarrow N \text{ is an } n\text{-linear map}\}$$

In particular,

$$\begin{aligned} L^2(M_1, M_2; N) &\simeq L^1(M_1 \otimes M_2, N) \simeq \text{Hom}_R(M_1 \otimes M_2, N) \\ &\simeq L^1(M_1, L^1(M_2, N)) \end{aligned}$$

Proposition 2.0.8. $M^v \otimes_R N \simeq \text{Hom}_R(M, N)$.

Let M, N be finitely generated free R -modules. Define $M^v = \text{Hom}_R(M, R)$ (the dual of M). Then $M^v \otimes N \simeq \text{Hom}_R(M, N) = L(M, N)$.

Proof. Define $\varphi : M^v \times N \rightarrow \text{Hom}_R(M, N)$ by

$$\begin{aligned} (f, n) &\mapsto \varphi(f, n) : M \rightarrow N \\ m &\mapsto f(m)n \end{aligned}$$

On the other hand, let $V_1, \dots, V_n$ be a basis of M . Define $V_j^v(V_i) = \delta_{ij}$. $V_1^v, \dots, V_n^v$ is a dual basis of M^v .

Define $\psi : \text{Hom}_R(M, N) \rightarrow M^v \otimes_R N$ by

$$g \mapsto \sum_{i=1}^n V_i^v \otimes g(V_i).$$

Show $\varphi\psi = \text{id}$, $\psi\varphi = \text{id}$. □

Proposition 2.0.9. Lift R -module to R' -module.

(1) Assume that R is a subring of a ring R' . Let M be an R -module. Then the bilinear map

$$\begin{aligned} R' \times (R' \otimes_R M) &\longrightarrow R' \otimes_R M \\ (\lambda'_1, \lambda'_2 \otimes m) &\mapsto \lambda'_1 \lambda'_2 \otimes m \end{aligned}$$

defines a structure of R' -module on the R -module $R' \otimes_R M$.

(2) More generally, let $f : R \rightarrow T$ be a ring homomorphism. View T as an R -module defined by $R \times T \rightarrow T$, $(\lambda, \mu) \mapsto f(\lambda)\mu$. The bilinear map $T \times (T \otimes_R M) \rightarrow T \otimes_R M$,

$$(\mu_1, \mu_2 \otimes m) \mapsto \mu_1 \mu_2 \otimes m$$

defines a T -module structure on the R -module $T \otimes_R M$.

Example: Let V be an n -dimensional $\mathbb{Q}$ -vector space, $V \simeq \mathbb{Q}^n$. Then $\mathbb{C} \otimes_{\mathbb{Q}} V \simeq \mathbb{C} \otimes_{\mathbb{Q}} \mathbb{Q}^n \simeq \mathbb{C}^n$.

Lemma 2.0.10. $M/IM \simeq (R/I) \otimes_R M$.

Let I be an ideal of R and let M be an R -module. The morphism of R -modules $M \rightarrow R/I \otimes_R M$ induces an isomorphism $M/IM \simeq (R/I) \otimes_R M$, given by $m \mapsto 1 \otimes m$.

Example 2.0.11. *Counterexamples:*

1. *There is a module having no linearly independent elements.*
 2. *There is a module, some of whose submodules have no complement.*
 3. *There is a finitely generated module, one of its submodules is not finitely generated.*
2. There is a module, some of whose submodules have no complement.
3. There is a finitely generated module, one of its submodules is not finitely generated.
4. There is a module with a linearly dependent subset S such that no element is a linear combination of other elements of S .
5. There is a module with a minimal generating set that is not a basis and a maximal linearly independent set that is not a basis.
6. There is a free module with non-free submodules and non-free quotient modules.
7. There is an R -module M : $\exists r \in R, m \in M$ with $r \neq 0, m \neq 0$, but $rm = 0$.
8. There is a free module such that:
- a linearly independent subset is not contained in any basis;
 - a generating subset is not contained in any basis.

Definition 2.0.12. *Category*

A category $\mathcal{C}$ consists of:

- A class of objects.
- For each pair (A, B) , a set $\text{Hom}(A, B)$ whose elements are called morphisms from A to B .
- For each triple of objects (A, B, C) , a map

$$\text{Hom}(A, B) \times \text{Hom}(B, C) \rightarrow \text{Hom}(A, C)$$

denoted $(f, g) \mapsto g \circ f$.

It is assumed that the objects and morphisms satisfy the following conditions:

- If $(A, B) \neq (C, D)$, then $\text{Hom}(A, B)$ and $\text{Hom}(C, D)$ are disjoint.
- If $f \in \text{Hom}(A, B)$, $g \in \text{Hom}(B, C)$, $h \in \text{Hom}(C, D)$, then

$$h \circ (g \circ f) = (h \circ g) \circ f \quad (\text{associative}).$$

- For every object A we have an element $\text{id}_A \in \text{Hom}(A, A)$ such that

$$f \circ \text{id}_A = f \text{ for any } f \in \text{Hom}(A, B)$$

and $\text{id}_A \circ g = g$ for any $g \in \text{Hom}(B, A)$ (unital).

Remark 2.0.13. *Objects are in one-to-one correspondence with identity morphisms, which are uniquely determined by the property that they serve as two-sided identities for composition.*

Example 2.0.14. 1. **Set:** objects are sets; morphisms are maps.

2. **Group:** objects are groups; morphisms are group homomorphisms.

3. **Ab:** objects are abelian groups; morphisms are group homomorphisms.

4. **$R\text{-Mod}$:** objects are R -modules; morphisms are R -homomorphisms.

5. **Top:** objects are topological spaces; morphisms are continuous functions.

Lesson 3: Categories Highlight: Functor (covariant, contravariant), left exact/right exact, projective/injective/flat module

Notation:

- $R\text{-Mod}$...left R -modules
- $R\text{-mod}$...finitely generated left R -modules
- $\text{Mod-}R$...right R -modules
- $\text{mod-}R$...finitely generated right R -modules

Definition 2.0.15 (Subcategories). *A category $\mathcal{D}$ is called a **subcategory** of a category $\mathcal{C}$ if*

- (i) *the objects of $\mathcal{D}$ form a subclass of the objects of $\mathcal{C}$;*
- (ii) *for any objects A, B of $\mathcal{D}$, one has that $\text{Hom}_{\mathcal{D}}(A, B) \subseteq \text{Hom}_{\mathcal{C}}(A, B)$, and the identity morphisms and the composition of morphisms for $\mathcal{D}$ are the same as for $\mathcal{C}$.*

*The subcategory $\mathcal{D}$ is called **full** if $\text{Hom}_{\mathcal{D}}(A, B) = \text{Hom}_{\mathcal{C}}(A, B)$.*

Example 2.0.16. **Ab** is a full subcategory of **Grp**.

Definition 2.0.17 ($\mathcal{C}^{\text{op}}$). *Let $\mathcal{C}$ be a category. Define $\mathcal{C}^{\text{op}}$ as follows:*

- *The objects of $\mathcal{C}^{\text{op}}$ are the objects of $\mathcal{C}$.*
- $\text{Hom}_{\mathcal{C}^{\text{op}}}(A, B) = \text{Hom}_{\mathcal{C}}(B, A)$.

The composition satisfies: if $f \in \text{Hom}_{\mathcal{C}^{\text{op}}}(A, B)$ and $g \in \text{Hom}_{\mathcal{C}^{\text{op}}}(B, C)$, then $g \circ f$ in $\mathcal{C}^{\text{op}}$ equals $f \circ g$ in $\mathcal{C}$. The identity morphism id_A in $\mathcal{C}^{\text{op}}$ is the same as id_A in $\mathcal{C}$.

*We call $\mathcal{C}^{\text{op}}$ the **dual category** of $\mathcal{C}$.*

Definition 2.0.18 ((Covariant) Functor). Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A **(covariant) functor** $F: \mathcal{C} \rightarrow \mathcal{D}$ consists of:

- For each object A of $\mathcal{C}$, there is a unique object $F(A)$ of $\mathcal{D}$.
- For each pair (A, B) of objects of $\mathcal{C}$, F gives a map

$$\begin{aligned} \text{Hom}_{\mathcal{C}}(A, B) &\rightarrow \text{Hom}_{\mathcal{D}}(F(A), F(B)) \\ f &\mapsto F(f). \end{aligned}$$

It is assumed that the following conditions are satisfied:

- If $g \circ f$ is defined in $\mathcal{C}$, then $F(g) \circ F(f) = F(g \circ f)$.
- $F(\text{id}_A) = \text{id}_{F(A)}$.

Definition 2.0.19 (Contravariant Functor). A **contravariant functor** from $\mathcal{C}$ to $\mathcal{D}$ is a covariant functor from $\mathcal{C}^{\text{op}}$ to $\mathcal{D}$.

Example 2.0.20 ($\text{Hom}_R(M, -)$ and $\text{Hom}_R(-, N)$). Let M, N be R -modules. $\text{Hom}_R(M, N) = \{R\text{-homomorphisms from } M \text{ to } N\}$ is an abelian group.

Remark 2.0.21. If R is commutative, then $\text{Hom}_R(M, N)$ is an R -module.

In general, if R is noncommutative, $\text{Hom}_R(M, N)$ is not an R -module in a natural way. For $f \in \text{Hom}_R(M, N)$, define $\text{Hom}_R(M, -): R\text{-Mod} \rightarrow \mathbf{Ab}$.

$$\Rightarrow f \in \text{Hom}_R(M, N).$$

$$= f(srm)$$

$$N \longrightarrow \text{Hom}_R(N, M)$$

$$+ \downarrow \uparrow \text{Hom}_R(f, M) \quad \text{contravariant functor}$$

$$N' \longrightarrow \text{Hom}_R(N', M)$$

Definition 2.0.22. Tensor product (R non-commutative).

M : right R -module, N : left R -module. $M \otimes_R N$: abelian group satisfying $\rightarrow$ no R -module structure.

$$\begin{aligned} (rm) \otimes n &= m \otimes (rn) & (rs)(m \otimes n) &\rightarrow m(rs) \otimes n = r(s(m \otimes n)) \\ & & &\rightarrow m \otimes (rs)n = s(r(m \otimes n)) \end{aligned}$$

Definition 2.0.23. *Bimodule (双模).*

Let R and S be rings (with 1). An abelian group M is a left R -right S -bimodule if M is both a left R -module and a right S -module for which the two scalar multiplications jointly satisfy

$$r(xs) = (rx)s \quad \text{for } r \in R, x \in M, s \in S.$$

Notation:

$${}_R M := \text{left } R\text{-module}$$

$$M_R := \text{right } R\text{-module}$$

$${}_R M_S := \text{left } R\text{-right } S\text{-bimodule}.$$

Example 2.0.24. Let $U := {}_R U_S$ be a bimodule. Then for any R -module ${}_R M$, there are two S -modules:

- $\text{Hom}_R(U, M) \ni f, s \in S: (sf)(u) = f(us), \forall u \in U \rightarrow \text{left } S\text{-module}.$
- $\text{Hom}_R(M, U) \ni g, s \in S: (gs)(m) = g(m)s, \forall m \in M \rightarrow \text{right } S\text{-module}.$

$$\text{Hom}_R(U, -) : R\text{-}\mathbf{Mod} \rightarrow S\text{-}\mathbf{Mod}$$

$$M \mapsto \text{Hom}(U, M)$$

covariant functor.

$$\begin{pmatrix} M \\ \downarrow f \\ M' \end{pmatrix} \longmapsto \begin{matrix} \text{Hom}(U, M) \\ \downarrow \text{Hom}(U, f) \\ \text{Hom}(U, M') \end{matrix}$$

$\text{Hom}_R(-, U) : R\text{-}\mathbf{Mod} \rightarrow \mathbf{Mod}\text{-}S$ contravariant functor.

$$M \mapsto \text{Hom}(M, U)$$

$$\begin{pmatrix} M \\ \downarrow f \\ M' \end{pmatrix} \mapsto \begin{matrix} \text{Hom}(M, U) \\ \uparrow \text{Hom}(f, U) \\ \text{Hom}(M', U) \end{matrix}$$

Proposition 2.0.25. *Tensor product of bimodules: Let ${}_S M_R$ and ${}_R N$ be bimodules. Then $M \otimes_R N$ has a left S -module structure. If ${}_R N_T$, then $M \otimes_R N \in {}_S \mathbf{Mod}$. If ${}_R N_R$, then $M \otimes_R N \in \mathbf{Mod}\text{-}T$.*

Proof. We define the scalar multiplication as

$$s \left(\sum m_i \otimes n_i \right) = \left(\sum s m_i \right) \otimes n_i.$$

We need to show well-definedness:

$$\sum m_i \otimes n_i = \sum m'_j \otimes n'_j \Rightarrow s \left(\sum m_i \otimes n_i \right) = s \left(\sum m'_j \otimes n'_j \right).$$

Define $\varphi : M \times N \rightarrow M \otimes_R N$ by $(m, n) \mapsto sm \otimes n$. It is easy to check that φ is bilinear. Then there exists a unique $\psi : M \otimes_R N \rightarrow M \otimes_R N$ with $m \otimes n \mapsto sm \otimes n$, i.e., well-definedness is guaranteed. $\square$

Definition 2.0.26. Let $M_R, M'_R, {}_R N, {}_R N'$ be modules, and let $f : M \rightarrow M'$ and $g : N \rightarrow N'$ be R -homomorphisms. Consider the bilinear map $M \times N \rightarrow M' \otimes_R N'$ given by

$$(m, n) \mapsto f(m) \otimes g(n).$$

Then there exists a unique group homomorphism $f \otimes g : M \otimes_R N \rightarrow M' \otimes_R N'$ defined by

$$m \otimes n \mapsto f(m) \otimes g(n).$$

If ${}_S M_R$ and ${}_S M'_R$, then $f \otimes g$ is a left S -module homomorphism.

Note that $f \otimes g$ satisfies the following properties. For all $M_R, M'_R, {}_R N, {}_R N'$, and $f_1, f_2, f \in \text{Hom}_R(M, M')$, $g_1, g_2, g \in \text{Hom}_R(N, N')$:

1. $(f_1 + f_2) \otimes g = (f_1 \otimes g) + (f_2 \otimes g)$
2. $f \otimes (g_1 + g_2) = f \otimes g_1 + f \otimes g_2$
3. $f \otimes 0 = 0, 0 \otimes g = 0$
4. $1_M \otimes 1_N = 1_{M \otimes N}$

For all $M_R, M'_R, M''_R, {}_R N, {}_R N', {}_R N''$, and $f : M \rightarrow M', f' : M' \rightarrow M'', g : N \rightarrow N', g' : N' \rightarrow N''$:

$$(f' \otimes g') \circ (f \otimes g) = (f' \circ f) \otimes (g' \circ g).$$

Example 2.0.27. Given ${}_S U_R$ a bimodule. Then $U \otimes_R - : R\text{-Mod} \rightarrow S\text{-Mod}$ is a covariant functor.

$$\begin{aligned} M &\mapsto U \otimes M \\ \left(\begin{array}{c} M \\ f \downarrow \\ M' \end{array} \right) &\mapsto \left(\begin{array}{c} U \otimes M \\ \downarrow \text{id} \otimes f \\ U \otimes M' \end{array} \right) \end{aligned}$$

$- \otimes_S U : \text{Mod-}S \rightarrow \text{Mod-}R$ is a covariant functor

$$M \mapsto M \otimes_S U$$

$$\begin{pmatrix} M \\ f \downarrow \\ M' \end{pmatrix} \mapsto \begin{pmatrix} M \otimes U \\ \downarrow f \otimes \text{id} \\ M' \otimes U \end{pmatrix}$$

Definition 2.0.28 (left exact & right exact). Let $F : R\text{-Mod} \rightarrow S\text{-Mod}/\text{Ab}$ be a covariant functor. We say F is left exact if for any exact sequence

$$0 \rightarrow N \xrightarrow{\varphi} M \xrightarrow{\psi} P,$$

the sequence

$$0 \rightarrow F(N) \xrightarrow{F(\varphi)} F(M) \xrightarrow{F(\psi)} F(P)$$

is exact.

Let $F : R\text{-Mod} \rightarrow S\text{-Mod}/\text{Ab}$ be a covariant functor. We say F is left exact if for any exact sequence

$$0 \rightarrow N \xrightarrow{\varphi} M \xrightarrow{\psi} P,$$

the sequence

$$0 \rightarrow F(N) \xrightarrow{F(\varphi)} F(M) \xrightarrow{F(\psi)} F(P)$$

is exact. We say F is right exact if for any exact sequence

$$N \xrightarrow{\varphi} M \xrightarrow{\psi} P \rightarrow 0,$$

the sequence

$$F(N) \xrightarrow{F(\varphi)} F(M) \xrightarrow{F(\psi)} F(P) \rightarrow 0$$

is exact.

Let $G : R\text{-Mod} \rightarrow \text{Ab}$ be a contravariant functor. It is called left (or right) exact if it is left (or right) exact as a covariant functor $(R\text{-Mod})^{\text{op}} \rightarrow \text{Ab}$. (i.e., left and right are interchanged in the corresponding definitions for contravariant functors.) A functor is called exact if it is both left exact and right exact.

Theorem 2.0.29 (classical examples of left (right) exact functors). 1. ${}_R U_S$:

The hom functor $\text{Hom}_R(U, -) : R\text{-Mod} \rightarrow S\text{-Mod}$ is left exact.

2. ${}_S U_R$: The hom functor $\text{Hom}_R(-, U) : \text{Mod-}R \rightarrow \text{Mod-}S$ is left exact (contravariant functor).

3. ${}_S U_R$: The functor $U \otimes_R - : R\text{-Mod} \rightarrow S\text{-Mod}$ is right exact.

4. ${}_R U_S$: The functor $- \otimes_R U : \text{Mod-}R \rightarrow \text{Mod-}S$ is right exact.

Proof. (1) Aim: Prove if $0 \rightarrow N' \xrightarrow{f} N \xrightarrow{g} N''$ is exact, then $0 \rightarrow \text{Hom}_R(U, N') \xrightarrow{\text{Hom}_R(U, f)} \text{Hom}_R(U, N) \xrightarrow{\text{Hom}_R(U, g)} \text{Hom}_R(U, N'')$

$\text{Hom}_R(U, N) \xrightarrow{\text{Hom}_R(U, g)} \text{Hom}_R(U, N'')$ is exact.

Step 1: $\text{Hom}_R(U, f)$ is injective.

If $\text{Hom}_R(U, f)(\varphi) = 0$ for $\varphi \in \text{Hom}_R(U, N')$, then $f \circ \varphi = 0$. Since f is injective, $\varphi = 0$.

Step 2: Prove $\text{Im}(\text{Hom}_R(U, f)) \subseteq \text{Ker}(\text{Hom}_R(U, g))$.

Let $\varphi \in \text{Hom}_R(U, N')$. Then $\text{Hom}_R(U, g)(f\varphi) = gf\varphi = 0$ since $gf = 0$.

Step 3: Prove $\text{Im}(\text{Hom}_R(U, f)) \supseteq \text{Ker}(\text{Hom}_R(U, g))$.

Let $\psi \in \text{Ker}(\text{Hom}_R(U, g))$, so $g\psi = 0$. Then $\forall x \in U, g(\psi(x)) = 0 \Rightarrow \psi(x) \in \text{Ker}(g) = \text{Im}(f)$. Since f is injective, $\exists! y \in N'$ s.t. $f(y) = \psi(x)$. Define $\varphi \in \text{Hom}(U, N')$ by $\varphi(x) = y$. Then $\psi = f\varphi$.

(3) Aim: Prove if $N' \xrightarrow{f} N \xrightarrow{g} N'' \rightarrow 0$ is exact, then $U \otimes_R N' \xrightarrow{1 \otimes f} U \otimes_R N \xrightarrow{1 \otimes g} U \otimes_R N'' \rightarrow 0$ is exact.

Step 1: $1 \otimes g$ is surjective.

Every element of $U \otimes_R N''$ is of the form $\sum x_i \otimes g(y_i) = (1 \otimes g)(\sum x_i \otimes y_i) \in \text{Im}(1 \otimes g)$ (since g is surjective).

Step 2: $\text{Im}(1 \otimes f) = \text{Ker}(1 \otimes g)$.

$gf = 0 \Rightarrow (1 \otimes g)(1 \otimes f) = 1 \otimes (gf) = 0 \Rightarrow \text{Im}(1 \otimes f) \subseteq \text{Ker}(1 \otimes g)$.

$1 \otimes g$ induces $\theta : U \otimes N / \text{Im}(1 \otimes f) \rightarrow U \otimes N''$. Then $\text{Im}(1 \otimes f) = \text{Ker}(1 \otimes g) \Leftrightarrow \theta$ is an isomorphism.

Let $x \in U, y'' \in N''$. Choose $y \in N$ such that $g(y) = y''$.

Claim: $\overline{x \otimes y}$ is independent of the choice of y .

Let $y_1, y_2 \in N$ with $g(y_1) = g(y_2) = y''$. Then $y_1 - y_2 \in \text{Ker } g = \text{Im } f$. Thus $x \otimes y_1 - x \otimes y_2 = x \otimes (y_1 - y_2) \in \text{Im}(1 \otimes f)$. Hence the claim holds.

Therefore we can define

$$\begin{aligned} \tilde{\theta} : U \times N'' &\rightarrow U \otimes_R N / \text{Im}(1 \otimes f) \\ (x, y'') &\mapsto \overline{x \otimes y} \end{aligned}$$

□

$$(x, y'') \mapsto \pi \otimes y^k \quad \text{where } g(y) = y''$$

Check $\theta\theta' = \text{id}_{N''}$, $\theta'\theta = \text{id}_N / \text{Im}(1 \otimes f)$, hence θ is an isomorphism.

Definition 2.0.30 (Projective/Injective/Flat Module). *Let M be a left R -module.*

- M is called a **projective** (投射) R -module if $\text{Hom}_R(M, -)$ is exact.
- M is called an **injective** (内射) R -module if $\text{Hom}_R(-, M)$ is exact.
- M is called a **flat** (平坦) R -module if $M \otimes_R -$ is exact.

Equivalent Definitions of Projective Module (1) Whenever there is an R -epimorphism $M \xrightarrow{g} N$ and an R -homomorphism $P \xrightarrow{\gamma} N$, there exists an R -module homomorphism

$\bar{\gamma}: P \rightarrow M$ such that $\gamma = g\bar{\gamma}$.

$$\begin{array}{ccc} \bar{\gamma} \downarrow & & \gamma \\ M & \longrightarrow & N \longrightarrow 0 \end{array}$$

(2) For each epimorphism $f: {}_R M \rightarrow {}_R N$, the map $\text{Hom}_R(P, f): \text{Hom}_R(P, M) \rightarrow \text{Hom}_R(P, N)$ is an epimorphism.

(3) For every exact sequence $M' \xrightarrow{f} M \xrightarrow{g} M''$ in $R\text{-Mod}$, the sequence

$$\text{Hom}_R(P, M') \xrightarrow{\text{Hom}_R(P, f)} \text{Hom}_R(P, M) \xrightarrow{\text{Hom}_R(P, g)} \text{Hom}_R(P, M'')$$

is exact.

(4) P is projective.

(5) Every epimorphism $M \xrightarrow{f} P$ splits, i.e., there exists an R -homomorphism $g: P \rightarrow M$ such that $fg = \text{id}_P$.

(6) P is isomorphic to a direct summand of a free left R -module.

Proof. **Split Exact Sequences**

Let $0 \rightarrow A_1 \xrightarrow{f} B \xrightarrow{g} A_2 \rightarrow 0$ be exact. TFAE:

1. $\exists h: A_2 \rightarrow B$ with $g \circ h = \text{id}_{A_2}$
2. $\exists K: B \rightarrow A_1$ with $K \circ f = \text{id}_{A_1}$
3. The given sequence is isomorphic to $0 \rightarrow A_1 \xrightarrow{\iota_1} A_1 \oplus A_2 \xrightarrow{\pi_2} A_2 \rightarrow 0$

It is easy to show (1) $\Leftrightarrow$ (2) are equivalent.

(1) $\Leftrightarrow$ (4)

(4) $\Rightarrow$ (1) obvious.

(1) $\Rightarrow$ (4): If $M \rightarrow N \rightarrow 0$ is exact, then $\text{Hom}(P, M) \rightarrow \text{Hom}(P, N) \rightarrow 0$ is exact.

With identity maps on A_1 and A_2 , in particular, $B \simeq A_1 \oplus A_2$.

Let $0 \rightarrow A_1 \xrightarrow{f} B \xrightarrow{g} A_2 \rightarrow 0$ be exact with $\varphi_1, \varphi_2, \varphi_3$ isomorphisms.

(1) $\Leftrightarrow$ (4)

(4) $\Rightarrow$ (1) obvious.

(1) $\Rightarrow$ (4): We have $M \rightarrow N \rightarrow 0$ exact implies $\text{Hom}(P, M) \rightarrow \text{Hom}(P, N) \rightarrow 0$ exact, i.e., if $M' \xrightarrow{f} M \xrightarrow{g} M'' \rightarrow 0$ is exact, then consider

$$\text{Hom}(P, M') \xrightarrow{f^*} \text{Hom}(P, M) \xrightarrow{g^*} \text{Hom}(P, M'') \rightarrow 0.$$

We already have g^* is surjective. $g^* f^* = 0$ follows from $g \circ f = 0$.

Then it suffices to show $\ker g^* \subseteq \text{im } f^*$.

Let $\varphi \in \ker g^*$, i.e., $g \circ \varphi = 0 \Rightarrow \varphi(p) \in \ker g = \text{im } f$ for any $p \in P$.

Then for any $p \in P$, $\exists y \in M'$ s.t. $\varphi(p) = f(y)$.

We then can define $\theta: P \rightarrow M' / \ker f$; $p \mapsto y + \ker f$.

Then apply (2) again, θ lifts to $\theta': P \rightarrow M'$ satisfying $\varphi = f \circ \theta'$.

(3) $\Leftrightarrow$ (4) is similar to (1) $\Leftrightarrow$ (4).

Now we prove $(1) \Rightarrow (5) \Rightarrow (6) \Rightarrow (1)$.

$(1) \Rightarrow (5)$

$(5) \Rightarrow (6)$: Every module is a quotient of free modules.

Take $F \xrightarrow{\varphi} P \rightarrow 0$ where F is a free module.

Then by (6), $0 \rightarrow \ker \varphi \hookrightarrow F \rightarrow P \rightarrow 0$ splits, i.e., $F = \ker \varphi \oplus P$.

$(6) \Rightarrow (1)$: It follows since every free module is a projective module.

Equivalent Definition of Injective Module

1. U is injective.
2. For each monomorphism $f : K \rightarrow M$ and each homomorphism $\gamma : K \rightarrow U$ there is an R -homomorphism $\bar{\gamma} : M \rightarrow U$ such that the diagram

$$\begin{array}{ccc} & & U \\ & & \gamma \uparrow \\ 0 \rightarrow K & \rightarrow & \bar{\gamma} \\ & & 0 \rightarrow M \end{array}$$

commutes.

(3) For every monomorphism $f : K \rightarrow M$, the map $\text{Hom}(f, U) : \text{Hom}(M, U) \rightarrow \text{Hom}(K, U)$ is an epimorphism.

(4) For every exact sequence $M' \rightarrow M \xrightarrow{g} M''$ the sequence

$$\text{Hom}(M'', U) \xrightarrow{g^*} \text{Hom}(M, U) \rightarrow \text{Hom}(M', U)$$

is exact.

(5) Every monomorphism $U \xrightarrow{f} M$ splits, i.e., there exists $g : M \rightarrow U$ such that $g \circ f = \text{Id}_U$.

□

Theorem 2.0.31. (*Schanuel's lemma*) Let

$$0 \rightarrow K \rightarrow P \xrightarrow{\lambda} V \rightarrow 0$$

$$0 \rightarrow K' \rightarrow P' \xrightarrow{\lambda'} V \rightarrow 0$$

be short exact sequences of R -modules with P, P' projective. Then

$$K' \oplus P \simeq K \oplus P'.$$

Proof. Let

$$Y = \{(p, p') \in P \oplus P' \mid \lambda(p) = \lambda'(p')\},$$

called the pullback of (λ, λ') . Then Y is an R -module.

Define $\mu : Y \rightarrow P$ by $(p, p') \mapsto p$ and $\mu' : Y \rightarrow P'$ by $(p, p') \mapsto p'$.

Then

$$\ker \mu = \{(0, p') \in P \oplus P' \mid \lambda'(p') = 0\} \simeq \ker \lambda' = K'$$

and

$$\ker \mu' = \{(p, 0) \in P \oplus P' \mid \lambda(p) = 0\} \simeq \ker \lambda = K.$$

Then we have the short exact sequences

$$\begin{aligned} 0 \rightarrow K' \rightarrow Y \rightarrow P \rightarrow 0, \\ 0 \rightarrow K \rightarrow Y \rightarrow P' \rightarrow 0. \end{aligned}$$

Since P, P' are projective, we obtain $K' \oplus P \simeq K \oplus P'$.

□

Proposition 2.0.32. *Equivalent definitions of Noetherian module and Artinian module.*

For an R -module M , the following are equivalent:

1. *A.C.C. condition holds.*
2. *Every submodule of M is finitely generated.*
3. *Every nonempty set of submodules of M has a maximal element with respect to inclusion.*

A module satisfying the above conditions is called a Noetherian module.

For an R -module M , TFAE:

- (1) d.c.c. condition
- (2) every nonempty set of submodules of M has a minimal element with respect to inclusion.

A module satisfying the above conditions is called an *Artinian module*. If M is a Noetherian (or Artinian) module, we also say that M satisfies the maximal (minimal) / increasing (decreasing) chain condition on submodules.

Proposition 2.0.33. *For an R -module M , TFAE:*

- (1) *d.c.c. condition*
- (2) *every nonempty set of submodules of M has a minimal element with respect to inclusion.*

A module satisfying the above conditions is called an Artinian module. If M is a Noetherian (or Artinian) module, we also say that M satisfies the maximal (minimal) / increasing (decreasing) chain condition on submodules.

Definition 2.0.34. *Noetherian & Artinian ring.* R with ${}_R R$ is said to be a left Noetherian (or Artinian) ring if ${}_R R$ is Noetherian (or Artinian) as a left module. (Similar for “right”)

Lemma 2.0.35. *Properties of Noetherian & Artinian modules.*

- (1) Let $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ be a short exact sequence. Then M is Noetherian (or Artinian) $\Leftrightarrow M'$ and M'' are Noetherian (or Artinian).
- (2) Let $M = \sum_{i=1}^n M_i$. Then M is Noetherian (or Artinian) $\Leftrightarrow M_i$ is Noetherian (or Artinian) for any i .

Theorem 2.0.36. *R left Noetherian $\Rightarrow$ f.g. R -modules Noetherian.*

If R is a (left) Noetherian (or Artinian) ring, then every finitely generated (left) R -module M is a Noetherian (or Artinian) module.

Proof. $M = Rm_1 + \cdots + Rm_n$ for $m_1, \dots, m_n \in M$. We have $\varphi_i : R \rightarrow Rm_i$ given by $r \mapsto rm_i$, so every Rm_i is a homomorphic image of ${}_R R$. Hence Rm_i is Noetherian. Hence M is Noetherian by lemma. $\square$

Definition 2.0.37. *Semisimple module.* A module that is the direct sum of simple submodules is called semisimple or completely reducible.

Lemma Let R be a ring with ${}_R R$. Suppose that $U = S_1 + \cdots + S_n$ is an R -module that can be expressed as a finite sum of simple submodules. If V is any submodule of U , then there is a subset $I \subseteq \{1, \dots, n\}$ such that

$$U = V \oplus \left(\bigoplus_{i \in I} S_i \right).$$

In particular:

- (1) V is a direct summand of U .
- (2) U is a direct sum of some subset of the S_i 's and hence semisimple.

Proof: Choose a subset I of $\{1, \dots, n\}$ maximal subject to the condition that the sum $W = V \oplus \left(\bigoplus_{i \in I} S_i \right)$ is a direct sum.

Claim: $W = U$. If $W \neq U$, then $S_j \not\subseteq W$ for some j . Note: $S_j \cap W = 0 \Rightarrow S_j + W = S_j \oplus W$, which contradicts the maximality of I . Hence $W = U$.

Corollary 2.0.38. *Equivalent definitions of semisimple R -modules. Let U be an R -module. TFAE:*

- 1. U is semisimple (true for infinite sum).

2. U is a sum of simple R -submodules. (Use Zorn's lemma)
3. Any R -submodule of U is a direct summand of U .

Proof. (1) $\Rightarrow$ (3): Same proof as above but apply Zorn's lemma.

(3) $\Rightarrow$ (2): 1° Every nonzero submodule of V has a simple submodule. Take $x \neq 0$ in U . Rx has a maximal submodule, say M (R has 1_R). Then $U = M \oplus M'$ for some $M' \subset U$. Then

$$Rx = Rx \cap U = Rx \cap (M \oplus M') = M \oplus (Rx \cap M')$$

where $Rx \cap M' \simeq Rx/M$ is simple. So Rx has a simple module.

2° Let N be the sum of all simple submodules of U . Then $U = N \oplus N'$ for some N' . Since $N \cap N' = 0$, N' has no simple submodules $\Rightarrow N' = 0$. So $U = N$.

(2) $\Rightarrow$ (1): Apply $V = 0$ in the proof of (1) $\Rightarrow$ (3). □

Definition 2.0.39 (Socle of U). *Prove by lemma $\rightarrow$ requires finite sum. Prove by theorem $\rightarrow$ holds for infinite sum. Let U be an R -module (with composition series of finite length).*

1. The sum of all simple submodules of U is a semisimple module that is the unique largest semisimple submodule of U (up to isomorphism), called the **socle** of U , denoted by $\text{Soc}(U)$.
2. The sum of all submodules of U isomorphic to some given simple module S is a submodule isomorphic to a direct sum of copies of S . It is the unique largest submodule with this property.

Corollary 2.0.40. *Let $U = S_1^{a_1} \oplus \cdots \oplus S_n^{a_n}$ be a semisimple module where the S_i 's are nonisomorphic simple R -modules. Then each submodule $S_i^{a_i}$ is uniquely determined and characterized as the unique largest submodule of U expressed as a direct sum of copies of S_i .*

Proof. It suffices to show that $S_i^{a_i}$ contains every submodule of U isomorphic to S_i .

Let T be any nonzero submodule of U such that $T \not\subseteq S_i^{a_i}$. Let $\pi_i : U \rightarrow S_i^{a_i}$ be the projection. There exists $j \neq i$ such that $\pi_j(T) \neq 0$. Further consider the projection $T \rightarrow S_j$. If T is simple, $T \simeq S_j \neq S_i$. □

Theorem 2.0.41. *Let R be a ring with 1. Let ${}_R V = V_1 \oplus V_2 \oplus \cdots \oplus V_n$ be a direct sum of simple R -modules V_i such that $V_i \simeq V_j$ for all i, j . Then $\text{End}_R(V)$ is isomorphic to the full matrix ring of degree n over the division ring $\text{End}_R(V_1)$.*

Proof. Let $\pi_i : V \rightarrow V_i$ be the projection.

Fix $\theta_i : V_1 \xrightarrow{\sim} V_i$ an isomorphism. For $b \in \text{End}_R(V)$, let $\varphi_{ij}(b) = \theta_i^{-1} \pi_i b \theta_j \in \text{End}_R(V_1)$.

$$V_1 \xrightarrow{\theta_j} V_j \xrightarrow{b} V \xrightarrow{\pi_i} V_i \xrightarrow{\theta_i^{-1}} V_1$$

Then we have $\varphi : \text{End}_R(V) \rightarrow M_n(E_1)$ where $E_1 = \text{End}_R(V_1)$:

$$b \mapsto (\varphi_{ij}(b))_{n \times n}$$

Check: φ is a ring homomorphism.

$$\text{For } b\beta : V_i \xrightarrow{\pi_i} V \xrightarrow{b} V \xrightarrow{\beta} V \xrightarrow{\pi_k} V_k$$

$$\sum_j \theta_k^{-1} \pi_j \beta \pi_i b \theta_i = \sum_j \theta_k^{-1} \pi_j \beta \theta_j \theta_j^{-1} \pi_i b \theta_i = \sum_j \varphi_{jk}(\beta) \varphi_{ij}(b)$$

Next, φ is an isomorphism.

1° Injective: If $\varphi(b) = 0$, then $\theta_i^{-1} \pi_i b \theta_j = 0$ for all i, j , which implies $\pi_i b(V_j) = 0$ for all i, j . Then $b(V_j) = 0$ for all j , so $b = 0$.

2° Surjective: Let $(b_{ij}) \in M_n(E_1)$. □

Remark 2.0.42. In $\varphi_{ij}(b) = \theta_i^{-1} \pi_i b \theta_j$, let $\tau_i : V_i \hookrightarrow V$ be the inclusion. Then $V_i \simeq (0, \dots, V_i, \dots, 0)$ in V . Set $b = \sum_{i,j} \tau_i \theta_i b_{ij} \theta_j^{-1} \pi_j \in \text{End}_R(V)$ (no need for inclusion). Hence we might not need τ_i .

Definition 2.0.43. *essential* An epimorphism of modules $f : U \rightarrow V$ is called essential if no proper submodule of U is mapped surjectively onto V by f .

Equivalently, whenever $g : W \rightarrow U$ is a map such that fg is an epimorphism, then g is an epimorphism.

(Proof: Let W be a proper submodule. Then $W \xrightarrow{g} U \xrightarrow{f} V$ with fg epimorphism $\Rightarrow$ induces $\text{im } g \rightarrow V \Rightarrow \text{im } g = U$.)

Definition 2.0.44. A projective cover of a module U is an essential epimorphism $P \rightarrow U$ where P is projective.

Proposition 2.0.45. “Minimality” of a projective cover.

1. Suppose that $f : P \rightarrow U$ is a projective cover of a module U and $g : Q \rightarrow U$ is an epimorphism where Q is projective. Then $Q = Q_1 \oplus Q_2$ so that g has the component $g = (g_1, 0)$ with respect to the direct sum decomposition and $g_1 : Q_1 \rightarrow U$ satisfies that the diagram commutes where γ is an isomorphism.
2. If it exists, the projective cover of a module U is unique up to isomorphism, by isomorphisms that commute with the essential epimorphism.

Proof. (1) Consider: Q projective $\Rightarrow \exists \beta$ such that $g = f\beta$. P projective $\Rightarrow \exists \alpha$ such that $f = g\alpha$. $f = g\alpha = f(\beta\alpha)$ epimorphism $\Rightarrow \beta\alpha$ is an epimorphism. In particular, β is an epimorphism.

$$Q \xrightarrow{\beta} P \rightarrow 0 \quad \text{exact}$$

P projective $\Rightarrow \beta$ splits.

$$Q = Q_1 \oplus Q_2, \quad Q_2 = \ker \beta, \quad \beta|_{Q_1} : Q_1 \rightarrow P \text{ isomorphism.}$$

$$g = (f\beta|_{Q_1}, 0), \quad \gamma = \beta|_{Q_1}.$$

(2) $f : P \rightarrow U$, $g : Q \rightarrow U$ projective covers.

$$Q = Q_1 \oplus Q_2, \quad g = (g_1, 0), \quad g \text{ essential} \Rightarrow Q_2 = 0.$$

□

Lesson 5 Radical

R : ring with 1

Definition 2.0.46. The radical of a module M is defined as

$$\text{Rad}(M) = \bigcap_N \{\text{maximal submodules of } M\}.$$

If M has no maximal submodule, we set $\text{Rad}(M) = M$.

Lemma 2.0.47 (Existence of maximal submodule). Let W be a proper R -submodule of an R -module V . If V/W is finitely generated over R , then there exists a maximal R -submodule of V containing W .

In particular, if V is finitely generated, then there exists a maximal submodule of V containing W .

Corollary: For every proper left ideal I of R , there exists a maximal left ideal of R containing I .

Proof. Let $\bar{V} = V/W = \sum_{i=1}^n R\bar{v}_i$, where $v_i \in V$.

Let θ denote the set of R -submodules of V containing W .

Claim: An arbitrary totally ordered subset in θ , say $\{W_\lambda \mid \lambda \in \Lambda\}$, has an upper bound.

Let $U = \bigcup_{\lambda \in \Lambda} W_\lambda$. If $U = V$, then for each $1 \leq i \leq n$, there exists W_{λ_i} such that $v_i \in W_{\lambda_i}$. Let W_α be the largest among W_{λ_i} . Then $W_\alpha = V$, since $v_i \in W_\alpha$ for all i , which contradicts $W_\alpha \in \theta$.

So $U \neq V$, and $U \in \theta$ is an upper bound for $\{W_\lambda\}_{\lambda \in \Lambda}$.

Then we can apply Zorn's lemma. □

Lemma 2.0.48. Let M, N be R -modules and $f : M \rightarrow N$ an R -morphism.

1. $f(\text{Rad}(M)) \subseteq \text{Rad}(N)$.

2. If f is an epimorphism with $\ker(f) \subseteq \text{Rad}(M)$, then $\text{Rad}(N) = f(\text{Rad}(M))$.

Lemma 2.0.49. *Let U be an R -module.*

1. *Suppose that $M_1, \dots, M_n$ are maximal submodules of U . Then there is a subset $I \subseteq \{1, 2, \dots, n\}$ such that*

$$U/(M_1 \cap \dots \cap M_n) \simeq \bigoplus_{i \in I} U/M_i.$$

2. *Suppose that U is Artinian. Then $U/\text{Rad } U$ is semisimple and $\text{Rad } U$ is a submodule of U with semisimple quotient.*

Proof. (1) Let I be a subset of $\{1, 2, \dots, n\}$ maximal with the property that the quotient homomorphisms

$$U/\bigcap_{i \in I} M_i \rightarrow U/M_i$$

induce an isomorphism

$$U/\bigcap_{i \in I} M_i \rightarrow \bigoplus_{i \in I} U/M_i.$$

We shall show: $\bigcap_{i \in I} M_i = M_1 \cap \dots \cap M_n$.

If not, there exists j such that $\bigcap_{i \in I} M_i \not\subseteq M_j$.

Consider the homomorphism $f: U \rightarrow (\bigoplus_{i \in I} U/M_i) \oplus (U/M_j)$ defined by

$$u \mapsto ((u + M_i)_{i \in I}, u + M_j).$$

□

f has kernel $M_j \cap (\bigcap_{i \in I} M_i)$. Then it suffices to show f is surjective.

Let $g: U \rightarrow (U/\bigcap_{i \in I} M_i) \oplus (U/M_j)$.

Observe that $M_j + \bigcap_{i \in I} M_i = U$.

$\forall x \in U$, then $x = y + z$ for some $y \in \bigcap_{i \in I} M_i$, $z \in M_j$.

$g(y) = (0, x + M_j)$, $g(z) = (x + \bigcap_{i \in I} M_i, 0)$.

So g is surjective $\Rightarrow f$ is surjective.

(2) U is Artinian then $\text{Rad } U$ is the intersection of finitely many maximal submodules so $U/\text{Rad}(U)$ is semisimple.

If V is a submodule of U with semisimple U/V . Then $U/V \simeq S_1 \oplus \dots \oplus S_n$ where S_i is simple.

Let M_i be the kernel of the composition $U \rightarrow U/V \rightarrow S_i$. Then M_i is maximal since S_i is simple.

And $V = M_1 \cap \dots \cap M_n \Rightarrow \text{Rad}(U) \subseteq V$.

Theorem 2.0.50. *(Nakayama's lemma)*

Let U be Noetherian, then $U \rightarrow U/\text{Rad } U$ is essential. Equivalently, if V is a submodule of U with $V + \text{Rad } U = U$, then $U = V$.

Proof: *If $V \neq U$, then $\exists$ a maximal submodule M of U such that $V \subseteq M \subseteq U$.*

Now $V + \text{Rad } U \subseteq M$ and so $V \subseteq M/\text{Rad } U$ which is not equal to $U/\text{Rad } U$ since $(U/\text{Rad } U)/(M/\text{Rad } U) \simeq U/M \neq 0$.

Proposition 2.0.51. (Sufficient conditions of essential epimorphism for Noetherian modules)

(1) Suppose that $f : U \rightarrow V$, $g : V \rightarrow W$ are two R -morphisms of R -modules. If two of f, g, gf are essential epi, then so is the third.

(2) Let $f : U \rightarrow V$ be a homomorphism of Noetherian modules. Then f is an essential epimorphism if and only if the radical quotient $U/\text{Rad } U \rightarrow V/\text{Rad } V$ is an essential epimorphism. (Semisimple case: isomorphism)

(3) Let $f_i : U_i \rightarrow V_i$ be homomorphisms of Noetherian modules for $i = 1, 2, \dots, n$. Then f_i are all essential epimorphisms $\Leftrightarrow \bigoplus f_i : \bigoplus U_i \rightarrow \bigoplus V_i$ is an essential epimorphism.

Proof. (1) 1° f, g essential $\Rightarrow gf$ essential: easy.

2° f, gf essential $\Rightarrow g$ essential.

g is surjective since gf is surjective.

Let $V_0 \subset V$, then $f^{-1}(V_0) \not\subseteq U$ since f is essential.

Then $g(V_0) \not\subseteq W$ since gf is essential.

3° g, gf essential $\Rightarrow f$ essential.

f is surjective since gf is surjective and g is essential.

Let $U_0 \subseteq U \Rightarrow gf(U_0) \not\subseteq W \Rightarrow f(U_0) \not\subseteq V$.

(2) Consider the commutative diagram

$$\begin{array}{ccc} U & \xrightarrow{f} & V \\ \phi_1 \downarrow & & \downarrow \phi_2 \\ U/\text{Rad } U & \xrightarrow{\bar{f}} & V/\text{Rad } V \end{array}$$

ϕ_1, ϕ_2 essential by Nakayama's Lemma, then f essential $\Leftrightarrow \bar{f}$ essential by (1).

(3) $\bigoplus U_i/\text{Rad } (\bigoplus V_i) \longrightarrow \bigoplus V_i/\text{Rad } (\bigoplus V_i)$

$\bigoplus (V_i/\text{Rad } V_i) \rightarrow \bigoplus (V_i/\text{Rad } V_i)$ □

Corollary 2.0.52. If P and Q are projective Noetherian modules over a ring, then $P \simeq Q$ iff $P/\text{Rad } P \simeq Q/\text{Rad } Q$.

Corollary 2.0.53. If P and Q are projective Noetherian modules over a ring, then $P \simeq Q$ iff $P/\text{Rad } P \simeq Q/\text{Rad } Q$.

Proof: $P \rightarrow P/\text{Rad } P$ and $Q \rightarrow Q/\text{Rad } Q$ are essential.

Then P (resp. Q) is a projective cover of $P/\text{Rad } P$ (resp. $Q/\text{Rad } Q$).

The rest follows by the uniqueness of projective covers.

Definition 2.0.54. A ring R (with 1) is called **semisimple** if ${}_R R$ is semisimple as a left module (i.e., it is a direct sum of simple left ideals).

Theorem 2.0.55. *Equivalent definition of semisimple Artinian ring. TFAE for an Artinian ring R :*

- (1) R is a semisimple Artinian ring.
- (2) Every R -module is semisimple.

Proof: (1) $\Rightarrow$ (2)

Let ${}_R R = \sum_{\lambda} I_{\lambda}$ be a sum of simple left R -submodules.

Then $V = \sum_{v \in V} \sum_{\lambda} I_{\lambda} v$ and let $f : I_{\lambda} \rightarrow I_{\lambda} v$ be an R -homomorphism

$$i \mapsto iv.$$

Then f is surjective, hence $I_{\lambda} v \simeq I_{\lambda} / \ker f = 0$ or I_{λ} .

Then $I_{\lambda} v$ is 0 or simple.

Lemma 2.0.56. $M \simeq \text{Hom}(R, M)$.

Let M be a left R -module. Then $M \simeq \text{Hom}_R(R, M)$ as left R -modules.

Proof. $\text{Hom}(R, M) \rightarrow M$

$$f \mapsto f(1)$$

$$f \mapsto f'(1)$$

$$f_m : r \mapsto rm \longleftrightarrow m$$

Corollary: $\text{End}_R({}_R R) \simeq R^{\text{op}}$

For any ring R (with 1), one has $\text{End}_R({}_R R) \simeq R^{\text{op}}$ as rings. □

Proof. $\text{End}_R(R) \rightarrow R^{\text{op}}$

$$f \mapsto f(1)$$

$$f_1 f_2(1) = f_1(f_2(1)) = f_1(f_2(1) \cdot 1) = f_2(1) f_1(1)$$

□

Lemma 2.0.57. $M_n(R)^{\text{op}} \simeq M_n(R^{\text{op}})$

Proof. Define $\varphi : M_n(R)^{\text{op}} \rightarrow M_n(R^{\text{op}})$

$$A \mapsto A^{\top}$$

Then $\varphi(A * B) = (BA)^{\top} = A^{\top} *' B^{\top} = \varphi(A) *' \varphi(B)$.

When R is not commutative:

$$(BA)^\top \neq A^\top B^\top$$

in fact $(BA)^\top = A^\top * B^\top$

□

Theorem 2.0.58 (Wedderburn-Artin). *Let A be a semisimple Artinian ring (with 1). Then A is a direct sum of matrix rings over division rings.*

Specifically, if $A \simeq S_1^{n_1} \oplus \cdots \oplus S_r^{n_r}$ where $S_1, \dots, S_r$ are non-isomorphic simple modules occurring with multiplicities $n_1, \dots, n_r$, then

$$A \simeq M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r)$$

where $D_i = \text{End}_A(S_i)^{\text{op}}$.

This is a ring isomorphism. Each $M_{n_i}(D_i)$ is a 2-sided ideal.

Proof. By Schur's lemma, D_i is a division ring.

$$\begin{aligned} {}_A A &\simeq S_1^{n_1} \oplus \cdots \oplus S_r^{n_r} \\ \text{End}_A({}_A A) &\simeq \text{End}_A(S_1^{n_1}) \oplus \cdots \oplus \text{End}_A(S_r^{n_r}) \\ \text{End}_A({}_A A) &\simeq \text{End}_A(S_1^{n_1}) \oplus \cdots \oplus \text{End}_A(S_r^{n_r}) \\ &\quad \left(\text{Hom}_A(S_i^{n_i}, S_j^{n_j}) = 0 \text{ if } i \neq j \right) \\ \text{End}_A(S_i^{n_i}) &\simeq M_{n_i}(D_i)^{\text{op}} \simeq M_{n_i}(D_i^{\text{op}}) \end{aligned}$$

□

Definition 2.0.59. *A ring is called simple if it has no nontrivial 2-sided ideals.*

Example 2.0.60. $M_n(D)$ is a simple ring if D is a division ring while

$$M_n(D)M_n(D) = \left\{ \begin{pmatrix} * & 0 & \cdots & 0 \\ \vdots & 1 & \vdots & \\ * & 0 & \cdots & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & * & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & * & 1 & \\ 0 & * & 0 & 0 \end{pmatrix} \right\} \oplus \cdots \oplus \begin{pmatrix} 0 & \cdots & * \\ 1 & 1 & * \\ 1 & \vdots & 1 \\ 0 & \cdots & 0 \end{pmatrix}$$

Simple left $M_n(D)$ -module

Definition 2.0.61. *R -algebra*

Let R be a commutative ring with 1 and A be a ring with 1. We call A an R -algebra if A is an R -module such that

$$r(xy) = (rx)y = x(ry) \quad \forall x, y \in A, \quad r \in R.$$

In particular, $(r \cdot 1)y = r(1 \cdot y) = r(y \cdot 1) = y(r \cdot 1)$.

Equivalently there is a ring homomorphism $R \xrightarrow{\varphi} A$ such that the image of R lies in the center of A and $R \times A \rightarrow A$,

$$(r, a) \mapsto \varphi(r)a.$$

If moreover R is a field and A is finite dimensional over R , then we say that the algebra is finite dimensional $\Rightarrow$ Noetherian & Artinian.

Example 2.0.62. (1) Group algebra: R commutative ring with 1; G finite group.

$$RG = \left\{ \sum_{g \in G} r_g g \mid r_g \in R \right\}$$

free R -module with basis $\{g \mid g \in G\}$.

(2) Polynomial ring $F[x]$ over a field.

(3) Matrix algebra over a commutative ring.

$$\begin{aligned} \varphi : R &\rightarrow M_n(R) \\ r &\mapsto \begin{pmatrix} r & & \\ & \ddots & \\ & & r \end{pmatrix} \end{aligned}$$

Let A be a k -algebra, where k is a field. Let S be a simple A -module. Then $\text{End}_A(S)$ is a division ring which is a k -algebra. If A is finite dimensional over k , so is $\text{End}_A(S)$. Furthermore, if A is finite dimensional and k is algebraically closed, then $\text{End}_A(S) \simeq k$. ($f : S \rightarrow S$ has an eigenvalue, $f - \lambda I : S \rightarrow S$ is singular since S is simple, $f - \lambda I = 0 \Rightarrow f = \lambda I$)

Lesson 6: Idempotents, Jacobson radical, Artinian ring Remark. Let M be an R -module, $x \in M$. $\text{Ann}_R(x) = \{r \in R \mid rx = 0\}$ is called the annihilator of x in $R \Rightarrow$ left ideal of R . $X \subseteq M$: $\text{Ann}_R(X) = \bigcap_{x \in X} \text{Ann}_R(x)$. $N \subseteq M$ submodule: $\text{Ann}_R(N)$ is a two-sided ideal of R .

Lemma 2.0.63. Let R be a ring with 1. Let S be a simple left R -module. Then S is a quotient of the left regular module ${}_R R$. Let $x \in S$, $x \neq 0$. Then $Rx = S$. Define $\varphi : R \rightarrow S$ by $r \mapsto rx$. Then $S \simeq R / \text{Ann}_R(x)$.

Theorem 2.0.64. A finite-dimensional semisimple k -algebra A over a field k with $k = \bar{k}$ is isomorphic to a direct sum of matrix algebras over k .

$$A \simeq M_{n_1}(k) \oplus \cdots \oplus M_{n_r}(k)$$

Corollary 2.0.65. *If A is a finite-dimensional K -algebra, then any simple A -module is finite-dimensional as a k -vector space.*

Proposition 2.0.66.

Proposition 2.0.67. *Let A be a finite-dimensional semisimple k -algebra over a field k . In any decomposition $A \simeq S_1^{n_1} \oplus \cdots \oplus S_r^{n_r}$ where the S_i are pairwise non-isomorphic simple A -modules, we have that $S_1, \dots, S_r$ is a complete set of representatives of the isomorphism classes of simple A -modules. If, moreover, $k = \bar{k}$, then $n_i = \dim_k S_i$ and $\dim_k A = \sum n_i^2$.*

Definition 2.0.68 (Idempotent). *Let R be a ring with identity. An idempotent in R is a nonzero element e with $e^2 = e$.*

Then eRe is a ring with identity e .

Two idempotents e_1, e_2 are called orthogonal if $e_1e_2 = 0 = e_2e_1$. More generally, n idempotents $e_1, \dots, e_n$ are called orthogonal if any two of them are orthogonal.

(Note that if $e_1, \dots, e_n$ are orthogonal, then $e_1 + \cdots + e_n$ is also an idempotent.)

An idempotent decomposition of an idempotent e is to decompose e into a sum of orthogonal idempotents $e = e_1 + \cdots + e_n$. If moreover $e_1, \dots, e_n$ are primitive, then it is called a primitive idempotent decomposition.

Remark 2.0.69. *Let $e \neq 0$ be an idempotent. Then $1 = e + (1 - e)$ is an idempotent decomposition. Note that e cannot be written as a sum $e = e_1 + e_2$ where e_1, e_2 are orthogonal idempotents unless one of them is zero.*

- $(1 - e)^2 = 1 - e$
- $(1 - e)e = 0 = e(1 - e)$

Theorem 2.0.70. *Let e be an idempotent of R . If $e = e_1 + \cdots + e_n$ is an idempotent decomposition, then $Re = Re_1 \oplus \cdots \oplus Re_n$ (as left ideals or left R -modules).*

Conversely, if Re is a direct sum of left ideals of R , $Re = I_1 \oplus \cdots \oplus I_n$, then there is an idempotent decomposition $e = e_1 + \cdots + e_n$ with $e_i \in I_i$ and $I_i = Re_i$.

That is, the idempotent decompositions of e are in bijection with the direct sum decompositions of the left ideal Re .

Proof. $\Rightarrow$ Let $e = e_1 + \cdots + e_n$ be an idempotent decomposition. Then $e_i = e_i e \Rightarrow Re_i \subseteq Re$. $\forall a \in R$, $ae = ae_1 + \cdots + ae_n \in Re_1 + \cdots + Re_n$ thus $Re = Re_1 + \cdots + Re_n$. Let $ae = a_1e_1 + \cdots + a_ne_n$. Then $ae_i = aee_i = a_ie_i$, i.e., the expression is unique. Hence $Re = Re_1 \oplus \cdots \oplus Re_n$.

$\Leftarrow$ Let $Re = I_1 \oplus \cdots \oplus I_n$. Let $e = e_1 + \cdots + e_n$. Then $e_i^2 = e_i$ and $e_ie_j = 0 = e_j e_i$ for $i \neq j$. $\square$

Corollary 2.0.71. *Equivalent definitions of primitive idempotent. Let e be an idempotent of R . TFAE:*

1. e is primitive.
2. Re is indecomposable as a left R -module.
3. e is the unique idempotent in eRe .

Proof. (1) $\Leftrightarrow$ (2) Immediate from the previous result.

(1) $\Rightarrow$ (3) Suppose $e' \in eRe$, $e' \neq 0$, $e'^2 = e'$, $e' \neq e$. Then $e = e' + (e - e')$ is an idempotent decomposition, contradicting (1).

(3) $\Rightarrow$ (1) If $e = e_1 + e_2$ is an idempotent decomposition, then $ee_1e = e_1$ and $ee_2e = e_2$, so $e_1, e_2 \in eRe$ are idempotents, contradicting (3). $\square$

Proposition 2.0.72. *Characterization of the direct summands of ${}_R R$. A left ideal I of a ring R is a direct summand of ${}_R R$ if and only if there is an idempotent $e \in R$ such that*

$$I = Re.$$

Moreover, if $e \in R$ is an idempotent, then so is $1 - e$, and Re and $R(1 - e)$ are direct complements of each other. That is,

$${}_R R = Re \oplus R(1 - e).$$

Proof. We have $\rho : R \rightarrow \text{End}({}_R R)$ an isomorphism. Moreover, for the projection $\pi : {}_R R \rightarrow I$, π is an idempotent in $\text{End}({}_R R)$ with $I = \text{im } \pi$. Let $e = \rho^{-1}(\pi)$, then we have $I = Re$. $\square$

Theorem 2.0.73. *Let e, f be idempotents of R and V be an R -module.*

1. $\text{Hom}_R(Re, V) \simeq eV$. In particular, $\text{Hom}_R(Re, Rf) \simeq eRf$ (as abelian groups).
2. $\text{End}_R(Re) \simeq (eRe)^{\text{op}}$ as rings.

Proof. (1) Let $\varphi \in \text{Hom}_R(Re, V)$, then $\varphi(e) = \varphi(ee) = e\varphi(e)$. Consider the map $g : \text{Hom}_R(Re, V) \rightarrow eV$,

$$\varphi \mapsto \varphi(e).$$

$\square$

$\varphi(ae) = a\varphi(e) \quad \forall a \in R \Rightarrow \varphi$ determined by $\varphi(e)$. Hence g is injective.

Let us define $\psi : Re \rightarrow V : xe \mapsto xeV$. Hence g is injective.

Let $V \in eV$, define $\psi : Re \rightarrow V : xe \mapsto xeV$. ψ is an R -homomorphism with $g(\psi) = \text{Id}$.

Hence g is surjective.

(2) $V = Re$ for $\sigma, z \in \text{End}_R(Re)$.

$$\sigma(z(e)) = \sigma(z(e)e) = z(e)\sigma(e)$$

So $\text{End}_R(Re) \simeq (eRe)^{\text{op}}$.

Theorem 2.0.74. *Let e, f be idempotents of R . TFAE:*

1. $Re \simeq Rf$ (as left modules)
2. $eR \simeq fR$ (as right modules)
3. $\exists b \in fRe, a \in eRf$ with $ab = e$ and $ba = f$.

Proof. WLOG, it suffices to prove $(1) \Leftrightarrow (3)$.

$(1) \Rightarrow (3)$: $Re \xrightarrow{\varphi} Rf$. Let $a = \varphi(e) = e\varphi(e) \in eRf$.

$$b = \varphi^{-1}(f) = f\varphi^{-1}(f) \in fRe.$$

Then $f = \varphi(\varphi^{-1}(f)) = \varphi(b) = \varphi(be) = b\varphi(e) = ba$.

$$e = \varphi^{-1}(\varphi(e)) = \varphi^{-1}(a) = \varphi^{-1}(af) = a\varphi^{-1}(f) = ab.$$

$(3) \Rightarrow (1)$: $eaf = eaba = ea = a$, $fbe = fbab = fb = b$. So $\varphi_a: Re \rightarrow Rf; xe \mapsto xf$, $\varphi_b: Rf \rightarrow Re; xf \mapsto xe$ satisfy $\varphi_a \circ \varphi_b = \text{id}_{Rf}$, $\varphi_b \circ \varphi_a = \text{id}_{Re}$. $\square$

Definition 2.0.75. *Jacobson radical of a ring.*

Let R be a ring. The (Jacobson) radical $J(R)$ of R is the intersection of all maximal left ideals of R .

Remark 2.0.76. $J(R) = \text{Rad}({}_R R)$.

Theorem 2.0.77. *Properties of $J(R)$.*

The following holds for the radical of a ring R .

1. $J(R)$ is the intersection of all the annihilator ideals of simple (left) R -modules. Consequently, $J(R)$ is a 2-sided ideal of R .
2. $a \in J(R) \Leftrightarrow 1 - xa$ has a left inverse for all $x \in R$, i.e., $\exists b \in R$ s.t. $b(1 - xa) = 1$.
3. $J(R)$ is the largest one among the 2-sided ideals I satisfying $a \in I \Leftrightarrow 1 - a$ is a unit in R .
4. $J(R)$ coincides with the intersection of all maximal right ideals of R .

Proof. (1) Let S be a simple R -module. Then $S \simeq R/\text{Ann}_R(x)$ for any $x \in S$. Thus $\text{Ann}_R(x)$ is a maximal left ideal of R . In particular, $J(R) \subseteq \text{Ann}_R(x)$. Hence $J(R) \subseteq \text{Ann}_R(S)$.

Conversely, we want to show that for any $x \in R$ such that $xS = 0$ for all simple modules S , then x is in every maximal left ideal. Let I be a maximal left ideal, then R/I is a simple R -module. $x \in \text{Ann}_R(R/I) \Rightarrow x \in I$.

(2) $\Rightarrow$ Let $a \in J(R)$. Suppose $1 - xa$ has no left inverse in R for some $x \in R$. Then $R(1 - xa) \neq R$. Hence there is a maximal left ideal M of R with $1 - xa \in M$. Since $a \in J(R) \subseteq M$, $xa \in M \Rightarrow 1 \in M$, contradiction.

$\Leftarrow$ If $a \notin J(R)$, then $a \notin M$ for some maximal left ideal M of R . Thus $M + Ra = R$. So $b + xa = 1$ for some $b \in M$, $x \in R$. Then $1 - xa = b \in M$ has no left inverse, contradiction.

(3) Let $a \in J(R)$. Then $\exists b$ such that $b(1 - a) = 1$. Now $b = 1 + ab$ has a left inverse, i.e., $\exists c \in R$ such that $cb = 1$. So $c = cb(1 - a) = 1 - a \Rightarrow 1 - a$ is a unit.

Let I be a 2-sided ideal with $a \in I \Leftrightarrow 1 - a$ is a unit of R . If $I \not\subseteq J(R)$ there exists a maximal left ideal M of R with $I \not\subseteq M$, whence $M + I = R$. $\exists b \in M$ such that $b + a = 1 \Rightarrow 1 - a \in M$, then $1 - a$ is not a unit, contradiction. $\square$

Theorem 2.0.78. *If I is a nil left ideal, then $I \subseteq J(R)$.*

Proof. Note that for any nilpotent element a , say $a^n = 0$, then $(1 - a)(1 + a + a^2 + \dots + a^{n-1}) = 1$; in particular, $1 - a$ is a unit.

Now, for any $a \in I$, $\forall x \in R$, $xa \in I$. Then $1 - xa$ is a unit $\Rightarrow a \in J(R)$. $\square$

Theorem 2.0.79. *If A is Artinian, then $J(A)$ is nilpotent.*

Proof. Consider $J(A) \geq J(A)^2 \geq \dots$

$\exists n$ such that $J(A)^n = J(A)^{n+1} = \dots$.

Assume $N = J(A)^n \neq 0$. Consider the set of left ideals I of A such that $NI \neq 0$ (nonempty since $N^2 \neq 0$). Let I_0 be a minimal member of this set. $\exists a \in I_0$ such that $Na \neq 0$ by construction. $Na \subseteq I_0$. $N(Na) = N^2a = Na \neq 0$. So $Na = I_0$ by the minimality of I_0 . In particular, $\exists b \in N$ such that $ba = a$, i.e., $(1 - b)a = 0$. Recall that $\forall b \in J(A)$, $1 - b$ is a unit $\Rightarrow a = 0$, contradiction. $\square$

Corollary 2.0.80. *Let A be an Artinian ring. Then $J(A)$ is the maximal nilpotent left ideal of A .*

Corollary 2.0.81. *Let A be an Artinian ring. Then $J(A)$ is the unique left ideal U of A with the property that U is nilpotent and A/U is semisimple.*

Theorem 2.0.82. *Let I be a 2-sided ideal of R with $I \subset J(R)$. Then $J(R/I) = J(R)/I$.*

Proof. $\subseteq$: Let $a + I \in J(R/I)$. Then for $x + I \in R/I$, $\exists b$ s.t. $b(1 - xa) + I = 1 + I$. Assume $b(1 - xa) = 1 - c$ for some $c \in I \subset J(R)$. Then $1 - c$ is a unit in $R \Rightarrow \exists d$ s.t. $db(1 - xa) = 1 \Rightarrow db$ is a left inverse of $1 - xa$. Hence $a \in J(R)$.

$\supseteq$: Let $a \in J(R)$. Then for any $x \in R$, $1 - xa$ has a left inverse in R , i.e. $\exists b$ s.t. $b(1 - xa) = 1$. In particular, $b(1 - xa) + I = 1 + I$. Thus we have shown for any $x + I \in R/I$, $(1 - xa) + I$ has a left inverse, i.e. $a + I \in J(R/I)$. $\square$

Theorem 2.0.83. *Let A be an Artinian ring. Then A is semisimple $\Leftrightarrow J(A) = 0$.*

Proof. Let M_λ , $\lambda \in \Lambda$ be all maximal left ideals of A . We consider the set of left ideals that are expressed as a finite intersection of some M_λ . Artinian $\Rightarrow$ This set has a minimal element, say $L = M_{\lambda_1} \cap \cdots \cap M_{\lambda_r}$. If $L \neq 0$, since $J(A) = 0$, there exists M_λ with $M_\lambda \cap L \subsetneq L$, contradicting the minimality of L . So $L = 0$.

Consider the map $A \rightarrow A/M_{\lambda_1} \oplus \cdots \oplus A/M_{\lambda_r}$. This is a monomorphism. Hence ${}_A A$ is semisimple.

$\Leftarrow$: $A = S_1 \oplus \cdots \oplus S_r$, S_i simple. $J(A) \leq \text{Ann}_R(S_i)$. Then $J(A)A = 0 \Rightarrow J(A) = 0$. $\square$

Proposition 2.0.84. *For a ring R , TFAE:*

- (a) R has a simple left generator, i.e. $\exists$ some simple module T and some $\varphi : T^n \rightarrow {}_R R$ for some n .
- (b) R is simple and Artinian.
- (c) For some simple ${}_R T$, ${}_R R \simeq T^{(n)}$ for some n .
- (d) R is simple and ${}_R R$ is semisimple.

Remark 2.0.85. *The four equivalent statements are equal to the according four equivalent statements for “right”, denoted by (a') , (b') , (c') , (d') .*

(a) $\Leftrightarrow$ (c)

(c) $\Rightarrow$ (d) Let I be a proper ideal of R . Then I is contained in a maximal left ideal L of R with $R/L \simeq T$. Since $R \simeq T^n$ for some n , $\text{Ann}_R(T) = 0$. Now, $IR \subseteq L$. Then

$$I \subseteq \text{Ann}_R(R/L) = \text{Ann}_R(T) = 0.$$

Hence R is simple.

(d) $\Rightarrow$ (b)

Lesson 7: Radical Series and Socle Series, Lifting Idempotents

Proposition 2.0.86. *Let A be an Artinian ring, U a f.g. A -module, $V \subseteq U$ a submodule.*

(1) TFAE:

- (i) $V = \text{Rad } U$
- (ii) V is the smallest submodule of U with semisimple quotient
- (iii) $J(A)U = V$

(2) TFAE:

- (i) $V = \text{Soc } U$
- (ii) V is the largest semisimple submodule of U

$$(iii) \ V = \{u \in U \mid J(A) \cdot u = 0\}$$

Proof. (1) Let $V = J(A)U$. Then U/V is an $A/J(A)$ -module, hence semisimple. On the other hand, if $V' \subseteq U$ with semisimple quotient U/V' , then $J(A)(U/V') = 0$, so $J(A)U \subseteq V'$.

(2) (iii) $\Rightarrow$ (i) $\{u \in U \mid J(A)u = 0\}$ is the largest submodule of U annihilated by $J(A)$. It is an $A/J(A)$ -module, hence semisimple. It is the largest since each simple module is annihilated by $J(A)$. $\square$

Definition 2.0.87 (Radical series & Socle series). *Let U be a f.g. A -module with A Artinian. Define inductively:*

$$\text{Rad}^n(U) = \text{Rad}(\text{Rad}^{n-1}(U)), \quad \text{Soc}^n(U)/\text{Soc}^{n-1}(U) = \text{Soc}(U/\text{Soc}^{n-1}(U)).$$

So $\text{Rad}^n(U) = J(A)^n U$ and

$$\text{Soc}^n(U) = \{u \in U \mid J(A)^n u = 0\}.$$

Chains of submodules (Loewy series):

$$\begin{aligned} U = \text{Rad}^0(U) \supseteq \text{Rad}(U) \supseteq \text{Rad}^2(U) \supseteq \cdots &\rightarrow \text{radical series} \\ 0 \subseteq \text{Soc}(U) \subseteq \text{Soc}^2(U) \subseteq \cdots &\rightarrow \text{socle series} \end{aligned}$$

Lemma 2.0.88.

$$\text{Rad}^n(U \oplus V) = \text{Rad}^n(U) \oplus \text{Rad}^n(V)$$

$$\text{Soc}^n(U \oplus V) = \text{Soc}^n(U) \oplus \text{Soc}^n(V)$$

Theorem 2.0.89 (Properties of radical series and socle series). *Let A be an Artinian ring. Let U be a f.g. A -mod.*

1. *The radical series of U is the fastest descending series of submodules of U with semisimple quotient.*
2. *The socle series of U is the fastest ascending series with semisimple quotient.*
3. *The two series terminate, and if m and n are the minimal integers with $\text{Rad}^m U = 0$ and $\text{Soc}^n U = U$, then $m = n$.*

Proof. (1) Suppose that $\cdots \subseteq U_2 \subseteq U_1 \subseteq U_0$ is a series of submodules of U with semisimple quotient. We show by induction on r that $\text{Rad}^r(U) \subseteq U_r$.

$r = 0$: clear.

Suppose $r > 0$, $\text{Rad}^{r-1}(U) \subseteq U_{r-1}$.

$$\text{Rad}^{r-1}(U)/\text{Rad}^{r-1}(U) \cap U_r \simeq \text{Rad}^{r-1}(U) + U_r/U_r \leq U_{r-1}/U_r$$

So $\text{Rad}^{r-1}(U) \cap U_r \supseteq \text{Rad}(\text{Rad}^{r-1}(U)) = \text{Rad}^r(U)$.

(3) $J(A)^r = 0$ for some r .

$$\text{Rad}^r(U) = J(A)^r U = 0$$

$$\text{Soc}^r(U) = \{u \in U \mid J(A)^r u = 0\} = U$$

□

Lemma 2.0.90. 1. Let A be a semisimple ring. Then A is Artinian if and only if it is Noetherian.

2. If A is an Artinian ring, then it is Noetherian.

Proof. (2) See A as a left module ${}_A A$. Let $J = J(A)$.

$$J^2 = J \cdot J = \text{Rad}(J^2), \quad J^{n+1} = \text{Rad}(J^n)$$

$$A \supseteq J \supseteq J^2 \supseteq \cdots \supseteq J^n = 0$$

where J^n/J^{n+1} is semisimple.

□

Definition 2.0.91. Lift an idempotent e in R/I to an idempotent $f \in R$.

Let I be a 2-sided ideal of R . Let $f \notin I$ be an idempotent of R . $e = f + I$ is an idempotent in R/I . We say f lifts e . If for every idempotent e of R/I , there is an idempotent $f \in R$ lifting e , then we can say lift idempotents from R/I to R .

Theorem 2.0.92. Lift idempotents from a nilpotent ideal.

Let I be a nilpotent ideal of R . Then we can lift idempotents.

Theorem 2.0.93. Lift idempotents from a nilpotent ideal.

Let I be a nilpotent ideal of R . Then we can lift idempotents from R/I to R . Moreover, if e is primitive, so is any lift f .

Proof. We define (inductively) idempotents $e_i \in R/I^i$ such that $e_i + I^{i-1}/I^i = e_{i-1}$ for each i and $e_1 = e$. (Use $R/I^i/I^{i-1}/I^i \simeq R/I^{i-1}$.) Suppose e_{i-1} is an idempotent of R/I^{i-1} . Pick any element $a \in R/I^i$ with $a + I^{i-1} = e_{i-1}$. $a^2 - a \in I^{i-1}/I^i$. Since $(I^{i-1})^2 \subseteq I^i$, we have $(a^2 - a)^2 = 0$ in R/I^i . Put $e_i = 3a^2 - 2a^3$, $e_i + I^{i-1} = e_{i-1}$. $e_i^2 - e_i = (3a^2 - 2a^3)^2 - 3a^2 + 2a^3 = -(3 - 2a)(1 + 2a)(a^2 - a)^2 = 0$ in R/I^i . If $I^r = 0$, let $f = e_r$. Suppose e is primitive and f has an idempotent decomposition $f = f_1 + f_2$. Then $e = e_1 + e_2$ with $f_i + I = e_i$.

$$e_1 e_2 = e_2 e_1 = 0$$

$$e_1^2 = e_1, \quad e_2^2 = e_2$$

e primitive $\Rightarrow e_1 = 0$ or $e_2 = 0$. Then $f_1 \in I$ or $f_2 \in I$. Contradiction, since I has no idempotent.

□

Corollary Let I be a nilpotent (2-sided) ideal of a ring R , and let $1 = e_1 + \cdots + e_n$ be an idempotent decomposition in R/I . Then we have an idempotent decomposition $1 = f_1 + \cdots + f_n$ in R with $e_i = f_i + I$. If e_i are primitive, so is f_i .

Proof. Induction on n . $n = 1$ ✓ Suppose $n > 1$. Suppose it holds for all $m < n$. Write $1 = e_1 + E$ in R/I , with $E = e_2 + \cdots + e_n \Rightarrow E$ is an idempotent orthogonal to e_1 . Then we lift e_1 to $f_1 \in R$, $e_1 = f_1 + I$. By induction hypothesis, we have $1 = f_1 + F$ in R where F is a lift of E . Now F is the identity of the ring FRF , FIF is a nilpotent ideal of FRF . The composition homomorphism $FRF \rightarrow R \rightarrow R/I$ has kernel $FRF \cap I = FIF$. Indeed, obviously $FIF \subseteq FRF \cap I$. On the other hand, $\forall x \in FRF \cap I$, $x = Fx \in FIF$. Then we get an inclusion $FRF/FIF \hookrightarrow R/I$ with image $E(R/I)E$ which has identity E . By induction hypothesis, $F = f_2 + \cdots + f_n$, $f_i + I = e_i$, $i \geq 2$. Hence $1 = f_1 + \cdots + f_n$. $\square$

Corollary 2.0.94. *Let f be an idempotent in a ring R that has a nilpotent ideal I . Then f is primitive in R if and only if $f + I$ is primitive in R/I .*

Lemma 2.0.95. *Existence of primitive decomposition in a semisimple Artinian ring.*

Let A be a semisimple Artinian ring.

1. There is a primitive idempotent decomposition $1 = e_1 + \cdots + e_n$. This corresponds to ${}_A A = Ae_1 \oplus \cdots \oplus Ae_n$ where Ae_i is simple.
2. For any simple A -module S , there is a primitive idempotent e with $S \simeq Ae$.
3. For any simple A -module S , there exists $1 \leq i \leq n$ such that $e_i S \neq 0$ (where e_i is from (1)), and if T is a simple A -module with $T \not\simeq S$, then $e_i T = 0$.

Proof. (3) $S = 1 \cdot S = (e_1 + \cdots + e_n)S \Rightarrow \exists i$ s.t. $e_i S \neq 0$. For any T simple, $T \simeq Ae$. We have $e_i Ae \simeq \text{Hom}_A(Ae, Ae_i) = 0$ if $Ae \not\simeq Ae_i$. $\square$

Theorem 2.0.96. *From semisimple Artinian to Artinian.*

Let A be an Artinian ring and S a simple A -module.

1. *There is an indecomposable projective A -module P_S with $P_S/\text{Rad}(P_S) \simeq S$ of the form $P_S \simeq Af$ where f is a primitive idempotent in A .*
2. *The idempotent f has the property that $fS \neq 0$ and if T is any simple module with $T \not\simeq S$, then $fT = 0$.*
3. *P_S is the projective cover of S ; it is uniquely determined up to an isomorphism by S .*

Proof. (1) Let $e \in A/J(A)$ be any primitive idempotent with $eS \neq 0$. Let f be any lift of e to A . Then e is primitive. $fS = eS \neq 0$. $fT = eT = 0$ for any simple module $T \not\simeq S$. Set $P_S = Af$: an indecomposable projective module. $Af \rightarrow (A/J(A))(f + J(A)) = S$ is

an A -module homomorphism $af \mapsto (a + J(A))(f + J(A))$ with kernel $J(A)Af = \text{Rad } Af$. Hence $P_S/\text{Rad } P_S \simeq S$. $\square$

Theorem 2.0.97. *Let A be an Artinian ring. Up to isomorphism, the indecomposable projective A -modules are exactly the modules P_S that are the projective covers of simple modules, and $P_S \simeq P_T \Leftrightarrow S \simeq T$. Each P_S appears as a direct summand of ${}_A A$ (left regular module), with multiplicity equal to the multiplicity of S as a summand of $A/J(A)$, i.e.,*

$${}_A A \simeq \bigoplus_S (P_S)^{n_S}$$

where S runs through simple A -modules up to isomorphism and $n_S = \dim_D S$ with $D = \text{End}_A(S)$.

Proof. (We only prove for finitely generated indecomposable projective modules, but true for all.) $\square$

Let P be an indecomposable projective module and write

$$P/\text{Rad } P \simeq S_1 \oplus \cdots \oplus S_n.$$

Then $P \rightarrow S_1 \oplus \cdots \oplus S_n$ is a projective cover. Now $P_{S_1} \oplus \cdots \oplus P_{S_n} \rightarrow S_1 \oplus \cdots \oplus S_n$ is also a projective cover, and by uniqueness of projective covers we have

$$P \simeq P_{S_1} \oplus \cdots \oplus P_{S_n}.$$

Since P is indecomposable we have $n = 1$ and $P \simeq P_{S_1}$.

Suppose that each simple A -module S occurs with multiplicity.

Let A be an Artinian ring, V a f.g. A -mod. Then V has a composition series (both Artinian and Noetherian). For any simple A -module S , the number of the composition factors isomorphic to S is called the multiplicity of S in V .

Theorem 2.0.98. *Let A be an Artinian ring, $f \in A$ be a primitive idempotent and $e = f + J(A) \in A/J(A) = \bar{A}$ a primitive idempotent. Then the multiplicity of $\bar{A}e$ in any f.g. A -module V is equal to the composition length of eV as an eAe -module.*

Proof. Let $V = V_0 \supseteq V_1 \supseteq \cdots \supseteq V_n = 0$ be a composition series of V . Consider fAf -modules:

$$fV = fV_0 \supseteq fV_1 \supseteq \cdots \supseteq fV_n = 0,$$

$$V_i/V_{i+1} \simeq \bar{A}e \Leftrightarrow f(V_i/V_{i+1}) \neq 0.$$

In this case, $e\bar{A}e \simeq f(V_i/V_{i+1}) = fV_i + V_{i+1}/V_{i+1} \simeq fV_i/fV_i \cap V_{i+1} = fV_i/fV_{i+1}$ which is also simple as fAf -module. $\square$

Lesson 8: Projective morphism, integral closed, localization and fractional modules 2025 年 4 月 13 日星期日

Definition 2.0.99 (Cartan matrix). Let A be an Artinian ring. Let $1 = e_1 + \cdots + e_n$ be a primitive idempotent decomposition in $A/J(A)$. Let $1 = f_1 + \cdots + f_n$ be the according primitive idempotent decomposition in A with $f_i + J(A) = e_i$. We denote by C_{ij} the multiplicity of $(A/J(A))e_i$ in Af_j , which is called the Cartan invariant in A . The $n \times n$ matrix $(C_{ij})_{n \times n}$ is called the Cartan matrix of A .

Note that we have

$$C_{ij} \neq 0 \Leftrightarrow f_i A f_j \neq 0.$$

For convenience, if S, T are two simple A -modules, we denote by C_{ST} the corresponding Cartan invariant, i.e., the multiplicity of S in P_T .

Proposition 2.0.100. Let A be a finite-dimensional k -algebra over a field k and S a simple A -module with projective cover P_S . Let U be a finite-dimensional A -module.

(1) If T is a simple A -module, then

$$\dim_k \operatorname{Hom}_A(P_S, T) = \begin{cases} \dim_k \operatorname{End}_A(S) & \text{if } S \simeq T \\ 0 & \text{otherwise} \end{cases}$$

(2) The multiplicity of S in U is $\dim_k \operatorname{Hom}_A(P_S, U) / \dim_k \operatorname{End}_A(S)$.

(3) If $e \in A$ is an idempotent, then $\dim_k \operatorname{Hom}_A(Ae, U) = \dim_k eU$.

(4) Let e_S and e_T be idempotents so that $P_S = Ae_S$, $P_T = Ae_T$ are projective covers of S and T . Then

$$C_{ST} = \dim_k \operatorname{Hom}_A(P_S, P_T) / \dim_k \operatorname{End}_A(S) = \dim_k e_S A e_T / \dim_k \operatorname{End}_A(S).$$

If moreover $k = \bar{k}$, then

$$C_{ST} = \dim_k \operatorname{Hom}_A(P_S, P_T) = \dim_k e_S A e_T.$$

Proof. (1) If $P_S \rightarrow T$ is nonzero, the kernel must be a maximal submodule containing $\operatorname{Rad}(P_S)$. But $P_S / \operatorname{Rad}(P_S) \simeq S \Leftrightarrow S \simeq T$. □

Now let R be a commutative ring with 1. For R -modules X, Y and M , we have an R -morphism

$$\operatorname{Hom}_R(X, M) \otimes_R \operatorname{Hom}_R(M, Y) \rightarrow \operatorname{Hom}_R(X, Y)$$

for R -modules X, Y and M , we have an R -morphism

$$\operatorname{Hom}_R(X, M) \otimes_R \operatorname{Hom}_R(M, Y) \rightarrow \operatorname{Hom}_R(X, Y)$$

$$\varphi \otimes \psi \mapsto \psi \circ \varphi$$

Proposition 2.0.101. *The image of this morphism is comprised of those morphisms from X to Y which factor through some finite power of M .*

$$X \rightarrow M^n \rightarrow Y$$

Proof. An element of the image is $\sum_{i \in I} \psi_i \circ \varphi_i$ for some finite set I .

This induces a morphism $\varphi: X \rightarrow M^{|I|}; x \mapsto (\varphi_i(x))_{i \in I}$ and $\psi: M^{|I|} \rightarrow Y; (m_i)_{i \in I} \mapsto \sum_{i \in I} \psi_i(m_i)$.

□

Definition 2.0.102. $\text{Hom}_R^f(X, Y)$ denotes the set of morphisms from X to Y which factor through a finite power of M .

Denote $X^* = \text{Hom}_R(X, R)$ dual. Taking $M := R$, we get a morphism the image factors through f -g. $\rightarrow$ projective modules (direct summand of a free module).

$$Z_{X,Y}: X^* \otimes_R Y \rightarrow \text{Hom}_R(X, Y) \quad (\text{Hom}_R(R, Y) \cong Y)$$

Denote by $\text{Hom}_R^{pr}(X, Y)$ the image of $Z_{X,Y}$. $f \mapsto f(1)$ The elements of $\text{Hom}_R^{pr}(X, Y)$ are called the projective morphisms from X to Y .

Lemma 2.0.103. *For any R -module M , $\text{Hom}_R^{Pr}(M, M)$ is a two-sided ideal of the ring $\text{End}_R(M) = \text{Hom}_R(M, M)$.*

Proof. Easy.

□

Theorem 2.0.104. *Let P be an R -module. TFAE:*

- (1) P is a f -g. projective module.
- (2) $\text{Hom}_R^{Pr}(P, P) = \text{End}_R(P)$.
- (3) For any R -module X , the morphism $Z_{X,P}: X^* \otimes P \rightarrow \text{Hom}_R(X, P)$ is an isomorphism.
- (4) For any R -module X , the morphism $Z_{P,X}: P^* \otimes_R X \rightarrow \text{Hom}_R(P, X)$ is an isomorphism.
- (5) The morphism $Z_{P,P}: P^* \otimes_R P \rightarrow \text{End}_R(P)$ is an isomorphism.

Proof. (1) $\Rightarrow$ (2), (3) $\Rightarrow$ (5), (4) $\Rightarrow$ (5) ✓

(1) $\Rightarrow$ (2): Since $\text{Hom}_R^{Pr}(P, P)$ is a two-sided ideal, it suffices to show $\text{id} \in \text{Hom}_R^{Pr}(P, P)$, which is obvious since P is a direct summand of R^n for some n .

(2) $\Rightarrow$ (3): Assume $\text{id}_P = Z_{P,P}(\sum_{i \in I} \varphi_i \otimes_R m_i)$, i.e., $\forall v \in P, v = \sum_i \varphi_i(v)m_i$.

Consider the morphism $\theta: \text{Hom}_R(X, P) \rightarrow X^* \otimes P$

$$\alpha \mapsto \sum_i \varphi_i \alpha \otimes m_i$$

Check: θ is the inverse of $Z_{X,P}$.

$$\text{Hom}_R(X, P) \xrightarrow{\theta} X^* \otimes_R P \xrightarrow{Z_{X,P}} \text{Hom}_R(X, P)$$

$$\alpha \mapsto \sum_i \varphi_i \alpha \otimes m_i \mapsto \left(x \mapsto \sum_i \varphi_i(\alpha(x)) m_i \right)$$

$$X^* \otimes_R P \xrightarrow{Z_{X,P}} \text{Hom}_R(X, P) \xrightarrow{\theta} X^* \otimes_R P$$

$$\psi \otimes v \mapsto Z_{X,P}(\psi \otimes v) \mapsto \sum_i \varphi_i Z_{X,P}(\psi \otimes v) \otimes m_i$$

where $Z_{X,P}(\psi \otimes v)(x) = \psi(x)v$, and $\varphi_i(Z_{X,P}(\psi \otimes v)(x)) = \varphi_i(\psi(x)v) = \varphi_i(\psi(x))v$. $\square$

$$= \varphi_i(v)\psi(x)$$

Hence $\varphi_i \circ \tau_x \circ p(\psi \otimes v) = \varphi_i(v)\psi$. Then $\psi \otimes V = \psi \otimes (\sum_i \varphi_i(V)m_i) = \sum_i \psi \otimes \varphi_i(V)m_i$

$$\begin{aligned} &= \sum_i \varphi_i(v)\psi \otimes m_i \\ &= \sum_i \varphi_i \circ \tau_x \circ p(\psi \otimes v) \otimes m_i \end{aligned}$$

(5) $\Rightarrow$ (1) Id p is projective. $p \xrightarrow{\text{Id}_p} p$

$$\begin{array}{ccc} & p & \\ f \searrow & & \swarrow \hat{g} \\ & R^n & \end{array}$$

We have $\text{Id}_p = gf \Rightarrow P$ is a direct summand of R^n .

Integrality Let S be an integral domain with 1.

Definition 2.0.105 (integral element over a subring R). *Let R be a subring of S containing 1_S . Let $x \in S$. We say α is an integral element over R if there exists a monic polynomial $f(x) \in R[x]$ such that $f(\alpha) = 0$.*

Proposition 2.0.106 (equivalent definition of an integral element). *Let S be an integral domain with 1. Let $R \subseteq S$ be a subring with 1_S . Let $\alpha \in S$. TFAE:*

1. α is an integral element over R .
2. $R[\alpha]$ is a finitely generated R -module.
3. There exists a subring T of S with $R \subseteq T$ such that T is finitely generated as

an R -module and $\alpha \in T$.

Proof. (1) $\Rightarrow$ (2): Obvious.

(2) $\Rightarrow$ (3): Take $T = R[\alpha]$.

(3) $\Rightarrow$ (1): Let $\varepsilon_1, \dots, \varepsilon_n$ be a set of generators of T as an R -module. Write

$$\alpha \varepsilon_i = a_{i1} \varepsilon_1 + \dots + a_{in} \varepsilon_n, \quad a_{ij} \in R.$$

Then

$$-a_{i1} \varepsilon_1 + \dots + (\alpha - a_{ii}) \varepsilon_i + \dots + (-a_{in}) \varepsilon_n = 0 \quad \forall i.$$

$$(\varepsilon_1, \dots, \varepsilon_n) \begin{pmatrix} \alpha - a_{11} & -a_{21} & \dots & -a_{n1} \\ -a_{12} & \alpha - a_{22} & & \vdots \\ \vdots & & \ddots & \\ -a_{1n} & \dots & \dots & \alpha - a_{nn} \end{pmatrix} = (0, \dots, 0)$$

Multiplying by A^* gives

$$(\varepsilon_1, \dots, \varepsilon_n) \begin{pmatrix} \det A & & \\ & \ddots & \\ & & \det A \end{pmatrix} = (0, \dots, 0)$$

$$\Rightarrow \sum_i \varepsilon_i \det A = 0$$

$$\Rightarrow (\det A)T = 0$$

$$\Rightarrow \det A = 0.$$

□

Lemma 2.0.107. *If M_1, M_2 are finitely generated R -modules, then $R[M_1, M_2]$ is finitely generated.*

Let S, R be as above. $R \subseteq M_i \subseteq S$ where $M_i, i = 1, 2$ are subrings of S . If M_1, M_2 are f.g. R -modules, then the ring $R[M_1, M_2]$ is f.g. as an R -module.

Pf: It suffices to show if $R[\alpha], R[\beta]$ is f.g. then $R[\alpha, \beta]$ is f.g.

Indeed, if $R[\alpha]$ is generated by $1, \alpha, \dots, \alpha^n$, $R[\beta]$ is generated by $1, \beta, \dots, \beta^m$, then $R[\alpha, \beta]$ is generated by $\alpha^i \beta^j$ ($0 \leq i \leq n, 0 \leq j \leq m$).

Theorem 2.0.108. *Integral elements form a subring.*

Let S, R be as above. The integral elements of S over R form a subring of S containing R .

Proof. Let α, β be integral elements $\Rightarrow R[\alpha], R[\beta]$ f.g.

$\Rightarrow R[\alpha, \beta]$ is f.g. by lemma.

$\Rightarrow \alpha - \beta, \alpha\beta \in R[\alpha, \beta]$ are integral elements.

□

Definition 2.0.109. *Integral closure.*

Definition 2.0.110. *Integral closed.*

Let S, R be as above.

1. The subring of S consisting of all integral elements of S over R is called the **integral closure** of R in S .
2. R is said to be **integrally closed** in S if every integral element of S over R is contained in R .
3. We say an integral domain R is **integrally closed** if it is integrally closed in its fraction field.
4. $\mathbb{Z} \subseteq \mathbb{C}$. If $z \in \mathbb{C}$ is integral over $\mathbb{Z}$, we call z an **algebraic integer**. All algebraic integers form a ring.

Definition 2.0.111. *Torsion and torsion free elements.*

R : ring. M : R -module. $m \in M$ is said to be

- a **torsion element** if $\text{Ann}_R(m) \neq 0$.
- a **torsion free element** if $\text{Ann}_R(m) = 0$.

Example 2.0.112. 1. $\mathbb{Z}$ -module $\mathbb{Q}$ has no torsion element but 0.

2. R commutative. R as an R -module has no torsion element but $0 \Leftrightarrow R$ is an integral domain.
3. If I is a proper ideal, all elements of R -module R/I are torsion.
4. R/I has no torsion element but $0 \Leftrightarrow I$ is prime.

Lemma 2.0.113. *Torsion module.*

If R is an integral domain, the subset of M

$$\text{Tor}(M) = \{m \in M \mid \text{Ann}_R(m) \neq 0\}$$

is a submodule of M , called the **torsion submodule** of M .

We say M is a **torsion module** if $M = \text{Tor}(M)$.

M is **torsion free** if $\text{Tor}(M) = 0$.

Example 2.0.114. (1) $\mathbb{Z}$ -modules: $\mathbb{Z}, \mathbb{Q}, \mathbb{Z}/2\mathbb{Z}$.

- $\text{tor}(\mathbb{Q}) = 0$

- $\text{tor}(\mathbb{Z}/2\mathbb{Z}) = 2\mathbb{Z}$

(2) R : integral domain, M an R -module. $\text{tor}(M^*) = 0$ where $M^* = \text{Hom}_R(M, R)$.

(3) R = integral domain. Free modules are torsion free.

Proposition 2.0.115. *Let R be an integral domain. Any finitely generated torsion free R -module M is isomorphic to a submodule of a free R -module of finite rank.*

Proof. Suppose M is generated by nonzero elements $x_1, \dots, x_n$.

Up to reordering, assume $x_1, \dots, x_r$ is a maximal linearly independent subset.

If $r = n$, M is free.

Assume $r < n$. Consider $M_0 = Rx_1 + \dots + Rx_r = Rx_1 \oplus \dots \oplus Rx_r$.

For each $j \geq r + 1$, $\exists \lambda_j \in R$ s.t. $\lambda_j x_j \in M_0$. Let $\lambda = \lambda_{r+1} \dots \lambda_n$.

$\Rightarrow \lambda M \leq M_0$. Consider $M \rightarrow M_0$ injective:

$$m \mapsto \lambda m$$

□

Localization

Definition 2.0.116 ($S^{-1}R$). *Let R be an integral domain with fraction field F . S : nonempty multiplicative closed subset of $R \setminus \{0\}$. Then*

$$S^{-1}R = R[S^{-1}] = \left\{ \frac{\lambda}{s} \mid s \in S, \lambda \in R \right\}$$

is a subring of F containing R .

$i: R \rightarrow S^{-1}R$ the natural injection.

Definition 2.0.117 ($S^{-1}M$). *Let M be an R -module. $S^{-1}M$ is the quotient of the set $M \times S$ by the equivalence relation*

$$(m_1, s_1) \sim (m_2, s_2) \Leftrightarrow \exists s \in S \text{ s.t. } s(s_2 m_1 - s_1 m_2) = 0.$$

It is an $S^{-1}R$ -module with

$$\left(\frac{\lambda}{s} \right) (m, s') = (\lambda m, ss').$$

For $s \in S$, $m \in M$, denote by $\frac{m}{s}$ the equivalence class of (m, s) .

Lemma 2.0.118. $S^{-1}R \otimes_R M \simeq S^{-1}M$

The map $f: \frac{\lambda}{s} \otimes m \mapsto \frac{\lambda m}{s}$ defines an isomorphism of $S^{-1}R$ -modules between $S^{-1}R \otimes_R M$ and $S^{-1}M$.

Proof. It is easy to see f is a surjective $S^{-1}R$ -morphism. □

It suffices to show f is injective.

First, show any element of $S^{-1}R \otimes M$ can be written as $\frac{1}{s} \otimes m$ for some $s \in S, m \in M$.

Let $\sum_i \frac{\lambda_i}{s_i} \otimes m_i \in S^{-1}R \otimes M$. Let $s = \prod_i s_i$. For all i , let $s'_i = \prod_{j \neq i} s_j$. Then

$$\sum_i \frac{\lambda_i}{s_i} \otimes m_i = \sum_i \frac{\lambda_i s'_i}{s} \otimes m_i = \sum_i \frac{1}{s} \otimes \lambda_i s'_i m_i = \frac{1}{s} \otimes \sum_i \lambda_i s'_i m_i.$$

If $f(\frac{1}{s} \otimes m) = 0$, then $\frac{m}{s} = 0$, so $\exists t \in S$ such that $tm = 0$. Then $\frac{1}{s} \otimes m = \frac{t}{st} \otimes m = \frac{1}{st} \otimes tm = 0$.

Corollary 2.0.119. $F \otimes M$

Set $FM = F \otimes M = S^{-1}M$ where $S = R \setminus \{0\}$.

The kernel of $R \rightarrow FM, m \mapsto 1 \otimes m$ is $\text{Tor}(M)$.

1. $F \otimes \text{Tor}(M) = 0$.

2. TFAE:

(i) The natural R -morphism $M \rightarrow FM, m \mapsto 1 \otimes m$ is injective;

(ii) M is torsion free.

3. If M is a f.g. projective R -module, then $M \rightarrow FM$ is injective.

Definition 2.0.120. rank of a f.g. torsion free module

The rank of a f.g. torsion free R -module is the dimension of the F -space FM .

Remark 2.0.121. 1. A f.g. torsion free R -module M of rank one is isomorphic to a f.g. R -submodule of F .

Indeed, if $FM = Fe$, then $I = \{\lambda \in F \mid \lambda e \in M\}$ is an R -submodule of F , and we have $M = Ie$. Embed M into FM ; M is f.g. $\xrightarrow{\text{iso}}$ I is f.g.

2. Let M be a f.g. torsion free R -module. The map $M \rightarrow FM, m \mapsto 1 \otimes m$ is injective. So M can be viewed as an R -submodule of FM . (Note: M f.g. $\Rightarrow FM$ f.d.)

Given an F -basis $(e_i)_{1 \leq i \leq r}$ of FM , assume M is generated by $m_1, \dots, m_n$. Then

$$m_i = \frac{s_1}{t_1} e_1 + \dots + \frac{s_r}{t_r} e_r$$

for some $s_i \in R, t_i \in R \setminus \{0\}$ (so $\frac{s_i}{t_i} \in F$).

$\exists \lambda \in R \setminus \{0\}$ s.t. $\lambda M \subseteq Re_1 \oplus \dots \oplus Re_r$.

Additionally, M generates FM as an F -vector space.

Definition 2.0.122. fractional R -module Let V be an F -vector space of dimension r . A fractional R -module M in V is an R -submodule of V such that

1. M generates V as a vector space.

2. given any basis $(e_i)_{1 \leq i \leq r}$ of V , there exists $\lambda \in R$, $\lambda \neq 0$, such that $\lambda M \subseteq \bigoplus_{i=1}^r Re_i$.

In the case $V = F$, fractional modules are called fractional ideals of R . It is an R -submodule I of F such that $\exists \lambda \in R$, $\lambda \neq 0$, with $\lambda I \subseteq R$ (hence λI is an ideal of R).

Example 2.0.123. 1. Every finitely generated R -submodule M of F is a fractional ideal (torsion free; R -submodule of $F \Rightarrow FM$ is of dimension 1). Moreover if R is Noetherian, then these are all fractional ideals.

2. If I is a fractional ideal, $\{\lambda \in F \mid \lambda I \subseteq R\}$ is again a fractional ideal (see below).

Lemma 2.0.124 (Properties of fractional modules). Let M be a finitely generated torsion-free R -module, let $V = FM$. (So M is identified with an R -submodule of FM .)

1. If $\lambda M \subseteq \bigoplus_{i=1}^r Re_i$ for some basis $(e_i)_{1 \leq i \leq r}$ of V and some $\lambda \in R$, $\lambda \neq 0$, then whenever $(f_i)_{1 \leq i \leq r}$ is a basis of V there exists $\mu \in R$, $\mu \neq 0$, such that $\mu M \subseteq \bigoplus_{i=1}^r Rf_i$.
2. Condition (1) in the above definition of fractional modules can be replaced by
(1') given a basis $(e_i)_{1 \leq i \leq r}$ of V , there exists $\mu \in R$, $\mu \neq 0$, such that $\bigoplus_{i=1}^r Re_i \subset M$.
3. If M_1 and M_2 are fractional modules in V , then $M_1 + M_2$ and $M_1 \cap M_2$ are also fractional modules.

Proof. (2) is obvious. (3) obvious from (2).

(1) We have $(e_1, \dots, e_r) = (f_1, \dots, f_r)A$ for some $A = (a_{ij})$ with non-zero determinant. Then

$$e_i = \sum_{j=1}^r a_{ji} f_j \text{ where } a_{ij} = \frac{b_{ij}}{c_{ij}} \text{ for some } b_{ij} \in R, c_{ij} \in R \setminus \{0\}.$$

Then set $C = \prod_{i,j}$. We have $ce_i^{T_1, j} \in \bigoplus_{j=1}^r Rf_j$. Since $\lambda MC \subseteq \bigoplus_{i=1}^r Re_i$, we have $GM \subseteq C \bigoplus_{i=1}^r Re_i C \bigoplus_{j=1}^r Rf_j$. $\square$

Corollary 2.0.125. connecting fractional ideals with fractional modules

If $(e_i)_{1 \leq i \leq r}$ is a basis of V and $(\alpha_i)_{1 \leq i \leq r}$ is a family of fractional ideals then

$$M = \bigoplus_{i=1}^r \alpha_i e_i$$

is a fractional module.

$M = \bigoplus \alpha_i e_i$ is a fractional module.

Lemma 2.0.126. *constructing fractional ideals*

Let I_1, I_2 be fractional ideals.

1. $I_1 + I_2$ (the R -submodule of F generated by $I_1 \cup I_2$)
2. $I_1 \cap I_2$
3. $I_1 I_2$ the R -submodule of F generated by all $x_1 x_2$, $x_1 \in I_1$, $x_2 \in I_2$

are all fractional ideals.

Proposition 2.0.127. *characterization of I^**

(1) For any fractional ideal, there is a natural identification

$$I^* = \{x \in F \mid xI \subset R\}$$

which is again a fractional ideal. ($I^* = \text{Hom}_R(I, R)$)

(2) Let V be an r -dimensional F -space with a basis $(e_i)_{1 \leq i \leq r}$, and let $(I_i)_{1 \leq i \leq r}$ be a family of fractional ideals. Set $M = \bigoplus_{i=1}^r I_i e_i$.

(a) The duality between V and V^* induces

$$M^* = \bigoplus_{i=1}^r I_i^* e_i^*$$

(b) The fractional module M is a finitely generated projective R -module iff for all i ($1 \leq i \leq r$), we have $I_i^* I_i = R$. In particular a fractional ideal I is a finitely generated projective R -module iff $I = R$.

Proof. (1) Let $0 \neq \lambda \in R$ s.t. $\lambda I \subset R$. Let $\varphi \in \text{Hom}_R(I, R)$. For $x, y \in I$, $\varphi(\lambda xy) = \lambda \varphi(xy) = \lambda x \varphi(y) = \lambda y \varphi(x)$. Since R is an integral domain, we then have $\varphi(xy) = x \varphi(y) = y \varphi(x)$ (in F). If $x, y \neq 0$, then $y^{-1} \varphi(y) = x^{-1} \varphi(x)$. So we may define $h_\varphi = x^{-1} \varphi(x)$ for any $x \in I \setminus \{0\}$. Easy to see the map $\text{Hom}_R(I, R) \rightarrow \{x \in F \mid xI \subset R\}$ is an isomorphism

$$\varphi \mapsto h_\varphi = x^{-1} \varphi(x)$$

Moreover, I^* is a fractional ideal since $\forall x \in I, xI^* \subset R$.

(2)(a) By assumption, $M \simeq \bigoplus_{i=1}^b I_i$. Then $M^* \simeq \bigoplus_{i=1}^b I_i^*$.

(b) The module M is f.g. projective $\Leftrightarrow I_i$ is f.g. projective for all i .

So it suffices to prove a fractional ideal I is f.g. projective $\Leftrightarrow II^* = R$.

From above, I is f.g. projective $\Leftrightarrow I^* \otimes_R I \simeq \text{End}_R(I)$.

$$\begin{aligned}
&\Leftrightarrow \exists \sum_{i=1}^n b_i \otimes a_i \in I^* \otimes I \text{ s.t. } \sum_{i=1}^n x b_i a_i = x \quad \forall x \in I \\
&\Leftrightarrow \sum_{i=1}^n a_i b_i = 1 \\
&\Leftrightarrow I^* I = R
\end{aligned}$$

□

Definition 2.0.128. *Dedekind domain & principal ideal domain*

- (1) A Dedekind domain is an integral domain R all ideals of which are finitely generated projective.
- (2) A principal ideal domain is an integral domain R all ideals of which are free (hence of rank one). ($\Leftrightarrow$ all ideals are principal)

Definition 2.0.129. *Local ring (probably non-commutative)*

We say a ring R (probably non-commutative) is local if the set of nonunits forms an ideal of R .

An equivalent definition of a local ring

Let R be a ring with 1. TFAE:

- 1. R is a local ring.
- 2. The set of nonunits of R coincides with $J(R)$. That is, $J(R)$ is a unique maximal left (or right) ideal of R .
- 3. $R/J(R)$ is a division ring.

Proof. Let M denote the set of nonunits of R .

(1) $\Rightarrow$ (2), (2) $\Rightarrow$ (3).

(3) $\Rightarrow$ (1): If $\bar{a} \neq 0$ in $R/J(R)$, then $\exists b \in R, c \in J(R)$ s.t. $ba = 1 - c$ has a left inverse $\Rightarrow a$ has a left inverse. Similarly, a has a right inverse. Hence a is a unit. □

Proposition 2.0.130. *R is a commutative local ring $\Rightarrow$ any f.g. projective module is free.*

Let R be a commutative ring. Assume R is local with unique maximal ideal $\mathfrak{m}$. Then any f.g. projective R -module is free.

Proof. Let X be a f.g. projective R -module. Then X/IX is a finite-dimensional (R/I) -vector space, say of dimension r . Choose a minimal basis $w_1, \dots, w_n$ of X . Then $\bar{w}_1, \dots, \bar{w}_r$ is an R/I -basis of X/IX .

Remark: an R/I -basis of X/IX .

Define a surjective map $\varphi : R^r \rightarrow X$

$$(a_1, \dots, a_r) \mapsto \sum_{i=1}^r a_i w_i.$$

If we set $K = \ker \varphi$, then from the minimal basis property, $\sum_{i=1}^r a_i w_i = 0 \Rightarrow a_i \in I$ for all i . Hence $K \subset I^r$.

Since X is projective, there exists $\psi : X \rightarrow R^r$ such that $\varphi\psi = 1_X$. Then we have $R^r = \psi(X) \oplus K$. Then

$$K = K \cap I^r = K \cap I(\psi(X) \oplus K) = K \cap IK.$$

Hence $K = IK$. Since K is a quotient of R^r , K is finitely generated over R ; thus applying Nakayama's lemma again we get $K = 0$. $\Rightarrow X \simeq R^r$. □

Corollary 2.0.131. *A local Dedekind domain is a local PID.*

Remark 2.0.132. (1) *Minimal basis: If W is a set of generators of an A -module M which is minimal, in the sense that any proper subset of W does not generate M , then W is said to be a minimal basis of M . Two minimal bases do not necessarily have the same number of elements.*

(2) *Local ring: $\{\text{minimal basis of } M\} \Leftrightarrow \{\text{basis of } \bar{M}\}$, hence any minimal basis has the same number of elements.*

Proof. Use Nakayama's lemma.

Let $\bar{u}_1, \dots, \bar{u}_n$ be a basis of $\bar{M}$, then $M = \sum_{i=1}^n Au_i + \mathfrak{m}M$. Since M is finitely generated, by Nakayama's lemma, $M = \sum_{i=1}^n Au_i$. The other way around is obvious. □

Proposition 2.0.133. *Localization.*

Let R be an integral domain. Let S be a nonempty multiplicatively closed subset of $R \setminus \{0\}$.

- *If R is a PID, so is $S^{-1}R$.*
- *If R is a Dedekind domain, so is $S^{-1}R$.*

Definition 2.0.134. *Invertible (fractional ideal).*

A finitely generated projective fractional ideal is called invertible. (Recall that a fractional ideal is finitely generated projective iff $II^ = R$.)*

Lemma 2.0.135. *An integral domain R is a Dedekind domain iff all fractional ideals are invertible.*

Proof. ($\Leftarrow$) If all fractional ideals are finitely generated projective, so are in particular the ideals of R . $\leftarrow \pm 1$ all fractional ideals are invertible, so are in particular the ideals of R .

$\Rightarrow$ Given a fractional ideal I , if $\lambda I \subseteq R$ for some $\lambda \neq 0$, the map $I \rightarrow \lambda I; a \mapsto \lambda a$ is an isomorphism of R -modules, hence I is invertible. $\square$

Proposition 2.0.136. *The set of fractional ideals of a Dedekind domain is an abelian group. Let R be a Dedekind domain. The set of fractional ideals of R is an abelian group under multiplication. The identity element is R , and the inverse of a fractional ideal is I^* (from now on denoted by I^{-1}). This group is called the ideal group of R .*

Definition 2.0.137 (Integral Ideal; I/J). (1) A fractional ideal I of R is called an integral ideal if $I \subseteq R$. (2) Let I and J be fractional ideals in R . We say that J divides I if there is an integral ideal K s.t. $I = JK$.

Lemma 2.0.138 ($J \mid I \Leftrightarrow I \subseteq J$). Let I and J be fractional ideals in R . TFAE:

1. $I \subseteq J$
2. $J \mid I$

Proof. (2) $\Rightarrow$ (1): clear.

(1) $\Rightarrow$ (2): Let $K = J^{-1}I$. Then $I = JK$ with $K \subseteq J^{-1}J = R$. $\square$

Corollary 2.0.139. *Every nonzero prime ideal of a Dedekind domain is maximal.*

pf: Assume that P is a nonzero prime ideal and $P \subseteq I$ for some ideal I . WTS: $I = P$ or R .

By above lemma, $\exists$ an ideal J s.t. $P = IJ$. Since P is prime, we have $I \subseteq P$ or $J \subseteq P$.

- $I \subseteq P \Rightarrow I = P$.
- $J \subseteq P \Rightarrow P = PI \Rightarrow I = R$ (multiply by P^{-1} on both sides).

Corollary 2.0.140. *A Dedekind domain is integrally closed.*

Proof. Let R be a Dedekind domain and let F be its field of fractions.

Let $x \in F$ be integral over R :

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 = 0$$

for some $a_i \in R$.

Then x belongs to the fractional ideal I generated by $1, x, \dots, x^{n-1}$.

Check: $I^2 = I \Rightarrow I = R$ (multiply by I^{-1} on both sides). $\square$

Theorem 2.0.141 (Factorization into Prime Ideals in Dedekind Domain). *Let R be a Dedekind domain. Every nonzero ideal I of R can be written uniquely as a product of prime ideals:*

$$I = P_1^{e_1} P_2^{e_2} \cdots P_r^{e_r}$$

where P_i are distinct prime ideals and $e_i \geq 1$.

Let R be a Dedekind domain. Every nonzero proper ideal of R can be written in a unique way as a product of prime ideals.

Proof:

Step 1: Prove any proper ideal is a product of prime ideals. If not, then there exists an ideal I which is maximal among proper ideals which are not products of prime ideals (by Noetherian property of R). In particular, I is not prime, hence is not maximal. So $\exists$ an ideal J s.t. $I \subset J \subset R$. So $I = JK$ for some ideal K of R , $I \subseteq K$. Moreover, $I \neq K$ otherwise $J = R$. Hence $I \subsetneq K \subsetneq R$. By maximality of I , J and K are products of prime ideals, so is I , a contradiction.

Step 2: Uniqueness. Assume $I = \mathfrak{p}_1 \cdots \mathfrak{p}_m = \mathfrak{q}_1 \cdots \mathfrak{q}_n$ where $\mathfrak{p}_i, \mathfrak{q}_j$ are prime. We prove by induction on m . Since $\mathfrak{q}_1 \cdots \mathfrak{q}_n \subset \mathfrak{p}_1$, $\exists j$ s.t. $\mathfrak{q}_j \subset \mathfrak{p}_1$. WLOG, assume $j = 1$. Then $\mathfrak{q}_1 = \mathfrak{p}_1$ by maximality of prime ideals. Then we have $\mathfrak{p}_2 \cdots \mathfrak{p}_m = \mathfrak{q}_2 \cdots \mathfrak{q}_n$ and it follows by induction hypothesis.

Corollary 2.0.142. *Factorization of fractional ideals in a Dedekind Domain.*

Let R be a Dedekind Domain and I be any fractional ideal. Then there exist integral ideals J and K such that $I = JK^{-1}$. Thus for every fractional ideal I , we can write

$$I = \prod_{\mathfrak{p} \text{ prime}} \mathfrak{p}^{v_{\mathfrak{p}}(I)}, \quad v_{\mathfrak{p}}(I) \in \mathbb{Z}$$

We call $v_{\mathfrak{p}}(I)$ the valuation of I at $\mathfrak{p}$.

Proposition 2.0.143. *Let I be a fractional ideal. Then there exist integral ideals J and K such that $I = JK^{-1}$.*

Proof. Let $a \in (R \cap I^*) \setminus \{0\}$. Then $aI \subset R$ and $Ra \subset I$. We can set $J = aI$ and $K = Ra$. $\square$

Definition 2.0.144. *Valuation of a field. Let k be a field. An (exponential) valuation of k is a real-valued function v defined on $k^x = k \setminus \{0\}$ satisfying*

1. $v(ab) = v(a) + v(b)$
2. $v(a + b) \geq \min\{v(a), v(b)\}$

Denote by (k, v) the valuation field. Then $v: K^x \rightarrow (\mathbb{R}, +)$ is a group homomorphism. The image $v(K^x)$ is called the value group of v . We set $v(0) = \infty$. We have:

1. $v(1) = 0, v(-1) = 0, v(a) = v(-a), v(a^{-1}) = -v(a)$

$$2. v(a) > v(b) \Rightarrow v(a+b) = v(b)$$

Let $R = \{a \in K \mid v(a) \geq 0\}$. R is a subring of K , called the valuation ring of v . Let $R^{-1} = \{a^{-1} \mid a \in R \setminus \{0\}\} = \{a \in K \mid v(a) \leq 0\}$. Then $K = R \cup R^{-1}$ and K is the fractional field of R . The set $P = \{a \in K \mid v(a) > 0\}$ is an ideal of R , called the valuation ideal of v . The group of units of R is $\{a \in R \mid v(a) = 0\}$. $\rightsquigarrow R$ is a local ring with $P = J(R) = \{\text{nonunits of } R\}$. R/P is a field, called the residue field of v .

Definition 2.0.145. *equivalent valuation & discrete valuation*

Let $\lambda \in \mathbb{R}$, $\lambda > 0$, define

$$v'(a) = \lambda v(a), \quad \forall a \in K^x \quad (v \sim v')$$

Then v' is also a valuation. We say v' is equivalent to v .

A valuation v of K is called discrete if its value group is isomorphic to $\mathbb{Z}$. In this situation, R is called a discrete valuation ring. In the case $v(K^x) = \mathbb{Z}$, v is said to be normalized.

Theorem 2.0.146. *R is PID and is integrally closed for k . Let (k, v) be a normalized discrete valuation field. Let R denote the valuation ring and P denote the valuation ideal.*

1. Let $\pi \in R$ with $v(\pi) = 1$. Then any $\alpha \in K$ is expressed as $\alpha = \pi^i u$ for some $u \in R \setminus P$ where $i = v(\alpha)$.
2. Any nonzero ideal of R is of the form $P^i = (\pi^i)$ for some $i \geq 0$. In particular, R is a PID.
3. R is integrally closed.

Proof. (1) Let $\alpha \in K$ with $v(\alpha) = i$. Then $v(\alpha\pi^{-i}) = 0$, so $u := \alpha\pi^{-i} \in R \setminus P$ and $\alpha = \pi^i u$.

(2) Let I be a nonzero ideal of R . Let $i = \min\{v(a) \mid a \in I\}$. Let a be an element in I such that $v(a) = i$. Then $a = \pi^i u$ for some unit u , hence $\pi^i = au^{-1} \in I$. For any element $b \in I$ with $v(b) = j$, we have $b = \pi^j u' = \pi^i (\pi^{j-i} u')$, hence $b \in (\pi^i)$. In particular, $P = (\pi)$ and $I = P^i$.

(3) Let $x \in K$ be integral over R . Then

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 = 0$$

for some $a_i \in R$. If $x \notin R$, then $x^{-1} \in R$, hence $x = -(a_{n-1} + a_{n-2}x^{-1} + \cdots + a_0x^{1-n}) \in R$. $\square$

Corollary 2.0.147. *Two discrete valuations v and v' of k are equivalent if and only if they have the same valuation ideal.*

Proof. It suffices to prove $\Leftarrow$.

$R = k \setminus P^{-1}$ is the valuation ring where $P^{-1} = \{x^{-1} \mid x \in P \setminus \{0\}\}$. Hence they have the same valuation ring. Then $\alpha = \pi^i u$ for some i and some $u \in R \setminus P$. Hence if v and v' are normalized, they will have the same value on R . $\square$

Theorem 2.0.148. *A discrete valuation ring is a local PID.*

Proof. It suffices to show $\supseteq$.

Let R be a local PID. Let $P = J(R) = (\pi)$.

Claim: $\bigcap_{i=1}^{\infty} P^i = 0$.

If $0 \neq a \in \bigcap_{i=1}^{\infty} P^i$, then $a = \pi a_1 = \pi^2 a_2 = \dots$ with $a_i \in R$. Then $a_i = \pi a_{i+1}$ (since R is an integral domain). This induces an ascending chain

$$(a_1) \subset (a_2) \subset \dots$$

contradicting the Noetherian property (since π is not a unit).

For any $0 \neq a \in R$, there exists $i \in \mathbb{Z}_{\geq 0}$ such that $a \in P^i$, $a \notin P^{i+1}$. Then $a = \pi^i u$ with $u \in R \setminus P$. Set $v(a) = i$. Let K be the fraction field of R . Define $v\left(\frac{a}{b}\right) = v(a) - v(b)$. Then v is a discrete valuation. $\square$

Definition 2.0.149. *Multiplicative valuation* Let v be an exponential valuation of a field k . Fix a real number γ , $0 < \gamma < 1$. Define $\varphi : k \rightarrow \mathbb{R}; a \mapsto \gamma^{v(a)}$.

Then:

1. $\varphi(a) \geq 0$ and $\varphi(a) = 0 \Rightarrow a = 0$.
2. $\varphi(ab) = \gamma^{v(ab)} = \gamma^{v(a)+v(b)} = \gamma^{v(a)}\gamma^{v(b)} = \varphi(a)\varphi(b)$.
3. $\varphi(a+b) \leq \max\{\varphi(a), \varphi(b)\}$.

with $\varphi(a) = \varphi(-a)$,

$$\varphi(a) = \varphi(a^{-1})^{-1},$$

$$\varphi(a) < \varphi(b) \Rightarrow \varphi(a+b) = \varphi(b).$$

The valuation ring $R = \{a \in K \mid \varphi(a) \leq 1\}$. The valuation ideal $P = \{a \in k \mid \varphi(a) < 1\}$.

Conversely, if we are given a function $\varphi : K \rightarrow \mathbb{R}$ satisfying (1), (2), (3) and define $v(a) = \log_{\gamma} \varphi(a)$ for any γ , then v is a valuation on K . φ is called a multiplicative valuation of k .

Two multiplicative valuations φ_1 and φ_2 are called equivalent if $\exists \gamma \in \mathbb{R}$, $\gamma \geq 0$, s.t. $\varphi_2(a) = \varphi_1(a)^{\gamma} \forall a \in K$.

Definition 2.0.150. *Metric space induced by multiplicative valuation.*

Define $d(a, b) = \varphi(a - b)$, a metric on k . K is a Hausdorff space. Moreover,

$$\begin{aligned} k \times k &\rightarrow k, & (a, b) &\mapsto a + b, \\ k^\times \times k^\times &\rightarrow k, & (a, b) &\mapsto ab^{-1} \end{aligned}$$

are continuous.

We say $\lim_{n \rightarrow \infty} a_n = a$ if $\forall \varepsilon > 0, \exists N$ s.t. $d(a_n, a) < \varepsilon$ for all $n \geq N$.

If every Cauchy sequence in K converges, we say K is complete, or a complete field.

The associated valuation ring is said to be a complete valuation ring.

Valuation ring. Given a valuation field K , let C denote the set of Cauchy sequences in K . Define

$$(a_n) + (b_n) = (a_n + b_n), \quad (a_n)(b_n) = (a_nb_n).$$

Then C is a commutative ring. Define $(a_n) + (b_n) = (a_n + b_n)$. Then C is a commutative ring with multiplication $(a_n)(b_n) = (a_nb_n)$.

If I denotes the subset of C consisting of the sequences converging to 0, then $\tilde{k} = C/I$ is a field. For any $(\widetilde{a_n}) \neq 0$, we may choose a representative such that $a_n \neq 0$ for all n , and then its inverse is $\left(\frac{1}{a_n}\right)$. Indeed, it suffices to find a lower bound for $\varphi(a_n)$: if $(\widetilde{a_n}) \neq 0$, then (a_n) does not converge to 0, i.e., there exists $\varepsilon > 0$ such that for all N , there exists $n > N$ with $\varphi(a_n) > \varepsilon$. On the other hand, there exists N_ε such that for all $n, m > N_\varepsilon$, $\varphi(a_n - a_m) < \frac{\varepsilon}{2}$. Take any $m > N_\varepsilon$ such that $\varphi(a_m) > \varepsilon$. Since $\varphi(a_m) > \varphi(a_n - a_m)$, we have $\varphi(a_n) = \varphi(a_m) > \varepsilon$.

Then for all $\gamma > 0$, there exists N such that for all $n, m > N$,

$$\varphi\left(\frac{1}{a_n} - \frac{1}{a_m}\right) = \varphi(a_n)^{-1} \varphi(a_m)^{-1} \varphi(a_m - a_n) < \frac{\gamma}{\varepsilon^2}.$$

In particular, I is a maximal ideal, and $\tilde{k}$ is called the *completion* of k .

If (a_n) is a Cauchy sequence, then $|\varphi(a_m) - \varphi(a_n)| \leq \varphi(a_m - a_n)$. Hence $(\varphi(a_n))$ is a Cauchy sequence in $\mathbb{R}$ and $\lim_{n \rightarrow \infty} \varphi(a_n)$ exists. Where $(x_n) \sim (y_n)$ if $(a_n) - (b_n) \in I$, then $\lim_{n \rightarrow \infty} \varphi(a_n) = \lim_{n \rightarrow \infty} \varphi(b_n)$ (i.e., if $d(x_n, y_n) \rightarrow 0$).

Define $\tilde{\varphi} : \tilde{k} \rightarrow \mathbb{R}$ by $\tilde{\varphi}(\widetilde{(a_n)}) = \lim_{n \rightarrow \infty} \varphi(a_n)$ for $(\widetilde{a_n}) = (a_n) + I$.

Check: $\tilde{\varphi}$ is a multiplicative valuation of $\tilde{k}$.

For $a \in k$, the sequence $(a) = (a, a, \dots)$ is a Cauchy sequence and $\tilde{a} \neq \tilde{b}$ if $a \neq b$, with $\tilde{\varphi}(\tilde{a}) = \varphi(a)$. Thus we can regard k as a subfield of $\tilde{k}$ by identifying a with $(\tilde{a})$, and $\tilde{\varphi}$ is an extension of φ . The field k is dense in $\tilde{k}$.

Lemma 2.0.151. $\tilde{K}$ is a complete field.

Theorem 2.0.152. $\tilde{R}/\tilde{P} \simeq R/P$

Let $\tilde{R}$ and $\tilde{P}$ be the valuation ring and valuation ideal of $\tilde{\varphi}$ respectively. Then $\tilde{R}/\tilde{P} \simeq R/P$.

Proof. It suffices to show $R + \tilde{P} = \tilde{R}$ (we have $R/P \rightarrow \tilde{R}/\tilde{P}; a + P \mapsto a + \tilde{P}$). Let $a \in \tilde{R}$ with $a \neq 0$. Then $0 \leq \tilde{\varphi}(a) \leq 1$. Since K is dense in $\tilde{K}$, $\exists a_n \in K$ such that $a = \lim a_n$. Then $\lim \tilde{\varphi}(a_n - a) = 0 \Rightarrow \exists m$ s.t. $\tilde{\varphi}(a_m - a) < \tilde{\varphi}(a)$.

$$\varphi(a_m) = \tilde{\varphi}(a_m - a + a) = \tilde{\varphi}(a) \leq 1 \Rightarrow a_m \in R.$$

$$\tilde{\varphi}(a_m - a) < \tilde{\varphi}(a) \leq 1 \Rightarrow a_m - a \in \tilde{P}.$$

$$\text{Hence } a = a_m - (a_m - a) \in R + \tilde{P}.$$

□

Proposition 2.0.153. *Let v be a normalized discrete valuation of K with valuation ring R and valuation ideal $P = (\pi)$. Then $\{P, P^2, \dots\}$ is a fundamental neighbourhood system of 0 since $\bigcap_{i=1}^{\infty} (\pi^i) = 0$.*

Therefore (a_n) is a Cauchy sequence $\Leftrightarrow$ for any $l \in \mathbb{Z}_{>0}$, $\exists N$ s.t. $a_m - a_n \in P^l$ for all $m, n \geq N$.

Also, $\lim_{n \rightarrow \infty} a_n = a \Leftrightarrow$ given l , $\exists N$ s.t. $a_n - a \in P^l$ for all $n \geq N$.

Lemma 2.0.154. *Let φ_1 and φ_2 be discrete multiplicative valuations of K . TFAE:*

1. φ_1 and φ_2 are equivalent.
2. $\{a \in K \mid \varphi_1(a) < 1\} = \{a \in K \mid \varphi_2(a) < 1\}$.
3. The topologies on K induced by φ_1 and φ_2 are the same.

Proof. (1) $\Leftrightarrow$ (2): Clear.

(2) $\Rightarrow$ (3): Clear.

(3) $\Rightarrow$ (2): Suppose $\varphi_1(a) < 1$. Then $\lim_{n \rightarrow \infty} \varphi_1(a^n) = \lim_{n \rightarrow \infty} \varphi_1(a)^n = 0$.

So $\lim_{n \rightarrow \infty} \varphi_2(a)^n = 0 \Rightarrow \varphi_2(a) < 1$.

Similarly $\varphi_2(a) < 1 \Rightarrow \varphi_1(a) < 1$.

□

Definition 2.0.155. *Norm on K -vector space w.r.t. φ .*

Let φ be a multiplicative valuation of K . Let V be a vector space over K . A real-valued function $\|\cdot\|$ defined on V is said to be a norm on V if

1. $\|v\| \geq 0$ and $\|v\| = 0 \Leftrightarrow v = 0$.
2. $\|av\| = \varphi(a)\|v\| \quad \forall a \in K, v \in V$.
3. $\|u + v\| \leq \max\{\|v\|, \|u\|\}$.

If we set $d(u, v) = \|u - v\|$, then V becomes a metric space.

$$\begin{cases} V \times V \rightarrow V, & (u, v) \mapsto u + v \text{ are continuous} \\ K \times V \rightarrow V, & (a, v) \mapsto av \end{cases}$$

Lemma 2.0.156. *Induced norm on K -vector space w.r.t. φ . Let $u_1, \dots, u_n$ be a basis of a finite-dimensional K -vector space V .*

For $v = a_1u_1 + \cdots + a_nu_n \in V$, define $\|v\|_0 = \max \{\varphi(a_i) \mid 1 \leq i \leq n\}$.

Then $\|\cdot\|_0$ is a norm on V . The following holds for the metric induced by $\|\cdot\|_0$:

1. Let $v_r = a_{r1}u_1 + \cdots + a_{rn}u_n$ be given for $r = 1, 2, \dots$. Then the sequence $\{v_r\}_r$ is a Cauchy sequence if and only if the sequence $(a_{ri})_r$ is a Cauchy sequence in K for all $i = 1, 2, \dots, n$. Also

$$\lim_{r \rightarrow \infty} v_r = \sum_{i=1}^n a_i u_i \Leftrightarrow \lim_{r \rightarrow \infty} a_{ri} = a_i \quad \forall i.$$

2. If K is complete, then so is V .

Proof. Immediate. □

Lemma 2.0.157. K complete $\Rightarrow$ unique norm on V (up to equivalence).

Suppose that k is complete. Then for any norm $\|v\|$ on V , there exists constants λ, μ , s.t. $\|v\| \leq \lambda\|v\|_0$, $\|v\|_0 \leq \mu\|v\|$ for all $v \in V$. Thus the topologies induced by these two norms are the same.

Proof. (1) λ : Let $\lambda = \max \{\|u_i\| \mid 1 \leq i \leq n\}$. Then for $v = \sum_{i=1}^n a_i v_i$, $\|v\| \leq \max \{\varphi(a_i) \|u_i\|\} \leq \lambda\|v\|_0$.

(2) μ : We will show the existence of μ by induction on n .

$n = 1$: trivial.

Let $n > 1$. For $v = a_1u_1 + \cdots + a_nu_n$, set $\varphi_i(v) = \varphi(a_i)$.

Claim: $\varphi_i(v) \leq \mu_i\|v\|$ for some μ_i . If it is true, take $\mu = \max \{\mu_i\}$.

Let $V' = Ku_1 \oplus \cdots \oplus Ku_{n-1}$. By induction, $\|\cdot\|$ and $\|\cdot\|_0$ induce the same topologies on V' . In particular, V' is complete w.r.t. both $\|\cdot\|$ and $\|\cdot\|_0$.

Suppose otherwise there is no μ_n with $\varphi_n(v) \leq \mu_n\|v\|$, $\forall v \in V$. Then for any $j \in \mathbb{Z}_{>0}$, there exists $v_j \in V \setminus V'$ such that $\varphi_n(v_j) > j\|v_j\|$. This holds also if v_j is replaced by its nonzero scalar multiples. Hence we may assume $v_j = a_{j1}u_1 + \cdots + a_{j,n-1}u_{n-1} + u_n$. Thus $\|v_j\| < \left(\frac{1}{j}\right) \varphi_n(v_j) = \frac{1}{j} \Rightarrow \lim_j v_j = 0$ w.r.t. $\|\cdot\|$. So $\lim (v_j - u_n) = -u_n$. But $v_j - u_n \in V'$ for all j and V' is closed in V . This shows $-u_n \in V'$, a contradiction. □

Definition 2.0.158. $\text{extension } (K, \varphi) \subseteq (L, \Phi)$ (consider L as an K -space)

Definition 2.0.159. $\text{extension } (k, \varphi) \subseteq (L, \Phi)$ (consider L as an k -space)

Let k, L be discrete valuation fields with multiplicative valuations φ and Φ respectively. If $k \subseteq L$ and Φ is an extension of φ , then we say (L, Φ) is an extension of (k, φ) , and write $(k, \varphi) \subseteq (L, \Phi)$. Consider L as an K -space. Let $\|x\| = \Phi(x)$ for $x \in L$. $\|\cdot\|$ is a norm of L as a K -vector space.

Theorem 2.0.160. *Extension of a complete (K, φ) is unique up to equivalence and complete.*

Theorem 2.0.161. *Let φ be a complete discrete multiplicative valuation of K and L be a finite extension of K . If there exists an extension (L, Φ) of (K, φ) , then Φ is unique up to equivalence. Furthermore, (L, Φ) is complete.*

Remark 2.0.162. *In fact, existence can be proved.*

Proof. By above lemma (see Φ as a norm of L as a vector space over K). □

Definition 2.0.163 (Metric on an R -module). *Let R be a ring, V an R -module. Let I be an ideal of R . Assume that $\bigcap_i I^i V = 0$. Fix $r \in R$, $0 < \gamma < 1$. Define $\|\cdot\|_I$ on V :*

$$\begin{aligned}\|0\|_I &= 0 \\ \|v\|_I &= \gamma^i \text{ for } v \in I^i V \setminus I^{i+1} V\end{aligned}$$

Define $d_I(u, v) := \|u - v\|_I$.

Check: $d_I(u, v) \leq \max\{d_I(u, w), d_I(w, v)\}$. So d_I is a metric on V . In this situation, we say d_I is defined on V .

A sequence (u_n) is a Cauchy sequence $\Rightarrow$ for any integer $m > 0$, there exists $N > 0$ s.t. $v_i - v_j \in I^m V$ for all $i, j \geq N$.

In particular, $v = \lim_{n \rightarrow \infty} v_n \Leftrightarrow$ for any integer $m > 0$, there exists $N > 0$ s.t. $v_i - v \in I^m V$ for all $i \geq N$.

The maps

1. $V \times V \rightarrow V; (a, b) \mapsto a + b$
2. $R \times V \rightarrow V; (r, a) \mapsto ra$
3. $f \in \text{Hom}_R(V, W)$ where d_I is defined on V, W

are continuous.

Theorem 2.0.164. *R complete $\Rightarrow$ finitely generated V complete.*

Assume d_I is defined on R . Assume V is finitely generated. If R is complete (w.r.t. d_I), then V is complete (w.r.t. d_I).

Proof. Let $\{v_1, \dots, v_s\}$ be a set of generators of V . Let $\{w_i\}$ be a Cauchy sequence in V . There exists a sequence $\{m(n)\}$ of nonnegative integers s.t. $w_i - w_j \in I^{m(n)} V$ for all $i, j > n$, with $m(n) \leq m(n+1)$, $\lim m(n) = \infty$.

Let $w_1 = \sum_{t=1}^s b_{1t} v_t$ for $b_{1t} \in R$.

$w_{n+1} - w_n = \sum_{t=1}^s a_{nt} v_t$ with $a_{nt} \in I^{m(n)}$.

So $w_n = \sum_{t=1}^s b_{nt} v_t$ with $b_{nt} = b_{1t} + \sum_{j=1}^{n-1} a_{jt}$.

$b_{n+k,t} - b_{nt} = \sum_{j=n}^{n+k-1} a_{jt} \in I^{m(n)}$.

Then $(b_{nt})_n$ is a Cauchy sequence in $R \Rightarrow b_t = \lim_{n \rightarrow \infty} b_{nt}$ exists in R .

Let $w = \sum_{t=1}^s b_t v_t$.

Check: $\lim_{n \rightarrow \infty} w_n = w$. Hence V is complete. $\square$

Lemma 2.0.165. *Suppose that d_P is defined for all f.g. R -modules. R is complete w.r.t. d_P . If V is a f.g. R -module. Then every submodule of V is closed.*

Proof. Let W be a submodule of V .

$f : V \rightarrow V/W$ is continuous. $W = f^{-1}(0)$ is closed. $\square$

R -algebra where R is a complete discrete valuation ring.

Example 2.0.166. *Let $I = P$. Then $d_I = v$. Complete w.r.t. d_I .*

Let R be a complete discrete valuation ring with valuation v , and let $P = (\pi)$ and $F = R/P$ be the valuation ideal and residue field of v respectively. ($P = J(R)$).

Let V be a f.g. R -module. Then $V = Ru_1 \oplus \cdots \oplus Ru_r$ is a direct sum of cyclic submodules. (R is a PID $\Rightarrow$ f.g. R -module is a direct sum of cyclic R -modules).

$(\pi^n)V = (\pi^n)u_1 \oplus \cdots \oplus (\pi^n)u_r \Rightarrow \bigcap_{n=1}^{\infty} \pi^n V = 0$.

Then d_P is defined for all f.g. R -modules. R is complete $\Rightarrow V$ is complete. In particular,

In particular, $R \subseteq C(A)$.

Let A be an R -algebra, which is f.g. as an R -module.

Fact: $A \times A \rightarrow A, (a, b) \mapsto ab$ is continuous.

$A^* = A/\pi A$ is a f.d. F -algebra.

Theorem 2.0.167. *Let A be a f.g. R -algebra. Then*

1. $A\pi \subseteq J(A)$
2. $J(A)^n \subseteq A\pi$ for some $n > 0$

Proof. (1) Let V be a simple A -module.

Then either $\pi V = V$ or $\pi V = 0$. Since $V = Av$ for some $0 \neq v \in V$,

A is f.g. over R , V is f.g. over R .

Thus if $\pi V = V$, then $V = 0$ by Nakayama's lemma.

Hence $\pi V = (A\pi)V = 0 \Rightarrow A\pi \subset J(A)$.

(2) $A^* = A/A\pi$ is f.d. F -algebra $\Rightarrow A^*$ Artinian $\Rightarrow J(A^*) = J(A)/A\pi$ is nilpotent.

Hence $J(A)^n \subset A\pi$ for some $n \geq 1$. $\square$

Theorem 2.0.168. *Lifting idempotents in f.g. R -algebra (R complete discrete valuation ring). Let A be a f.g. R -algebra. Let I be an ideal of A with $I \subseteq J(A)$ and $\bar{A} = A/I$.*

1. Let $\bar{e}$ be an idempotent in $\bar{A}$ and let $\bar{e} = \bar{e}_1 + \cdots + \bar{e}_n$ be an idempotent decomposition in $\bar{A}$. Then $\bar{e}$ lifts to an idempotent e of A and there exist orthogonal idempotents $e_1, \dots, e_n$ of A satisfying $\bar{e}_i = \bar{e}_i$ and $e = e_1 + \cdots + e_n$.
2. An idempotent e of A is primitive $\Leftrightarrow \bar{e}$ is primitive in $\bar{A}$.
3. Let e, f be primitive idempotents of A , then

$$Ae \simeq Af \Leftrightarrow \bar{A}\bar{e} \simeq \bar{A}\bar{f}$$

Proof. $I \subseteq J(A) \Rightarrow I^m \subseteq Az$ for some m . Hence $I^{mr} \leq A\pi^r$ for all $r \geq 1$.

Set $\bar{A}^{(r)} = A/I^{mr}$.

(1) I^m is a nilpotent ideal in $\bar{A}^{(1)}$. By $A/I^m/I/I^m \simeq A/I$, $\bar{e}$ lifts to an idempotent of $\bar{A}^{(1)}$, namely, $\exists e^{(1)} \in A$ s.t. $(e^{(1)})^2 \equiv e^{(1)} \pmod{I^m}$, $e^{(1)} \equiv e \pmod{I}$.

Next, I^m/I^{2m} is a nilpotent ideal in $\bar{A}^{(2)}$. Similarly, $e^{(1)}$ lifts to $e^{(2)} \in A$ s.t. $(e^{(2)})^2 \equiv e^{(2)} \pmod{I^{2m}}$, $e^{(2)} \equiv e^{(1)} \pmod{I^m}$.

Continue—we get $e^{(r)} \in A$ for each r s.t. $(e^{(r)})^2 \equiv e^{(r)} \pmod{I^{rm}}$, $e^{(r)} \equiv e^{(r-1)} \pmod{I^{(r-1)m}}$.

Then $(e^{(r)})_r$ is a Cauchy sequence and $e = \lim_{r \rightarrow \infty} e^{(r)}$ exists since R is complete.

e is an idempotent of A that lifts $\bar{e}$ to A (use $(e^{(r)})^2 \equiv e^{(r)} \pmod{I^{rm}}$).

Similarly, there exists $e_i^{(r)} \in A$ for each r s.t. $e_i^{(r)} \equiv e_i^{(r-1)} \pmod{I^{(r-1)m}}$, $(e_i^{(r)})^2 \equiv e_i^{(r)} \pmod{I^m}$ and $e^{(r)} \equiv e_1^{(r)} + \cdots + e_n^{(r)} \pmod{I^{rm}}$ is an idempotent decomposition in $\bar{A}^{(r)}$.

Each $(e_i^{(r)})_r$ is a Cauchy sequence $\Rightarrow$ Let $e_i = \lim_{r \rightarrow \infty} e_i^{(r)} \Rightarrow e_1, \dots, e_n$ are the desired idempotents.

(2) Similarly as before (use $J(A)$ has no idempotents).

(3) Ae is the projective cover of $\bar{A}\bar{e}$. □

Remark 2.0.169. Let $I = Az$, $\bar{A} = A/I = A^*$.

$$\begin{cases} K \otimes_R A : K\text{-alg} & (K \text{ valuation field}) \\ A^* : K\text{-alg} \\ A^* : F\text{-alg} & (F \text{ residue field}) \end{cases}$$

Let A be an algebra over a commutative ring R . Let U be an A -module.

If $R \rightarrow R'$ is a homomorphism to another commutative ring R' , we can form the R' -algebra $R' \otimes_R A$, and $R' \otimes_R U$ becomes an $R' \otimes_R A$ -module.

- If R is a subring of R' , we say that the module $R' \otimes_R U$ is obtained from U by extending the scalars from R to R' .
- If $R' = R/I$ for some ideal I of R , then applying $R' \otimes_R -$ to an R -module U is the same as reducing U modulo I :

$$R/I \otimes_R U \simeq U/IU.$$

- When U is free as an R -module, we can identify U with $R \otimes_R U$, and an R -basis of U becomes an $R' \otimes_R U$ -basis of $R' \otimes_R U$.

Counterexample: $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$. We have

$$1 \otimes 1 = \frac{1}{2} \otimes 2 = 0.$$

If an $R' \otimes_R A$ -module V has the form $R' \otimes_R U$ for some A -module U , we say that V is defined over R and that V can be lifted to U (U is a lift of V).

Proposition 2.0.170.

w.r.t. E Let $F \subseteq E$ be fields where every element of E is algebraic over F . Let A be a finite-dimensional F -algebra. Let V be a finite-dimensional $E \otimes_F A$ -module. Then there exists a field K with $F \subseteq K \subseteq E$ of finite degree over F , such that V is defined over K .

Proof. Let $a^1, \dots, a^n$ be an F -basis of A and let a^t (identifying A with $F \otimes_F A$) act on V with matrix (a_{ij}^t) with respect to some E -basis of V . Let

$$K = F[a_{ij}^t \mid 1 \leq t \leq n, 1 \leq i, j \leq d]$$

(where $\dim_E V = d$). Then $[K : F] < \infty$ since E is algebraic over F .

Then A acts by matrices over the field K . □

Definition 2.0.171. absolutely simple Let A be a finite-dimensional F -algebra where F is a field. A simple A -module U is said to be absolutely simple if U is simple as an $E \otimes_F A$ -module for all extension fields E of F .

We say an extension field E of F is a splitting field for A if every simple $E \otimes_F A$ -module is absolutely simple. ($E \otimes_F A$ is the new “ A ” which is a finite-dimensional E -algebra; now take any extension K of E , then for any simple $A \otimes_F E$ -module U , $K \otimes_E U$ is simple as an $A \otimes_F K$ -module.)

Definition 2.0.172.

Proposition 2.0.173. Equivalent definition of absolutely simple.

Let A be a finite-dimensional F -algebra where F is a field. Let U be a simple A -module. TFAE:

1. U is absolutely simple.
2. $\text{End}_A(U) = F$.
3. The matrix algebra summand of $A/J(A)$ corresponding to U has the form $M_n(F)$ where $n = \dim_F U$.

Proof. (1) $\Rightarrow$ (2): Will not prove.

(2) $\Rightarrow$ (3): $\checkmark$

(3) $\Rightarrow$ (1): $A \rightarrow A/J(A) \simeq M_n(F) \oplus \dots$. Then $U = F^n$. Let $F \subseteq E$ be an extension. $E \otimes_F A$ acts on $E \otimes_F U \simeq E^n$.

$$E \otimes_F A \rightarrow E \otimes_F M_n(F) \simeq M_n(E).$$

E^n is a simple $M_n(E)$ -module $\Rightarrow E \otimes_F U$ is a simple $E \otimes_F A$ -module. $\square$

Representation theory of finite groups

Example 2.0.174. *The Galois group of $x^4 - 2$ over $\mathbb{Q}$.*

$x^4 - 2$ has four roots in $\mathbb{C}$: $\alpha = \sqrt[4]{2}$, $i\alpha$, $-\alpha$, $-i\alpha$.

$$\begin{cases} \sigma : \alpha \leftrightarrow -i\alpha, -\alpha \leftrightarrow i\alpha \Rightarrow \langle \sigma, \tau \rangle \simeq D_8 \\ \tau : i\alpha \leftrightarrow -i\alpha \end{cases}$$

preserve the square. D_8 is “represented” as a subgroup of $\text{GL}_2(\mathbb{R})$:

$$\sigma \mapsto \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \tau \mapsto \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

General representation Let $\mathcal{C}$ be a category. For example:

- $\mathcal{C}$ is the category **Top** of topological spaces.
- $\mathcal{C}$ is the category **Vect** $_k$ of vector spaces over a field k , or the full subcategory **vec** $_k$ of **Vect** $_k$ whose objects are of finite dimension.
- $\mathcal{C}$ is the category **Set** of sets, or its full subcategory **set** whose objects are of finite cardinalities.

If X is an object of $\mathcal{C}$, the group $\text{Aut}(X)$ of automorphisms of X consists of all morphisms from X to X which admit an inverse morphism.

Definition 2.0.175. *Let G be a group. A representation of G on $\mathcal{C}$ (or a $\mathcal{C}$ -representation of G) is a pair (X, ρ) where*

- X is an object of $\mathcal{C}$,
- $\rho : G \rightarrow \text{Aut}(X)$ is a group homomorphism.

Note that:

- A representation (X, ρ) is said to be faithful if ρ is injective.
- A morphism between two representations (X, ρ) and (X', ρ') of G on the same category $\mathcal{C}$ is a morphism $f : X \rightarrow X'$ in $\mathcal{C}$ such that for all $g \in G$, the following

diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{\rho(g)} & X \\ f \downarrow & & \downarrow f \\ X' & \xrightarrow{\rho'(g)} & X' \end{array}$$

We can define the category $\text{Rep}(G, \mathcal{C})$ of $\mathcal{C}$ -representations of G . We can further define isomorphisms between two representations. If (X, ρ) is a $\mathcal{C}$ -representation of G , then we say G “acts on” X .

Example 2.0.176. $\rho : D_8 \rightarrow GL_2(\mathbb{R}) \subset \text{Aut}(\mathbb{R}^2)$ where $\mathbb{R}^2 \in \mathbf{Vect}_{\mathbb{R}}$. i.e., $(\mathbb{R}^2, \rho)$ is a $\mathbf{Vect}_{\mathbb{R}}$ -representation of D_8 .

Lemma 2.0.177. 1. A morphism $f : (X, \rho) \rightarrow (X', \rho')$ in $\text{Rep}(G, \mathcal{C})$ is an isomorphism if and only if the morphism $f : X \rightarrow X'$ is an isomorphism in $\mathcal{C}$.

2. Two representations (X, ρ) and (X, ρ') on the same object X are isomorphic if there exists an automorphism f of X such that for all $g \in G$,

$$\rho'(g) = f\rho(g)f^{-1}.$$

We write $(X, \rho) \simeq (X, \rho')$ and say: ρ and ρ' are uniformly conjugate.

Definition 2.0.178. The group $\text{Aut}(X) \times \text{Aut}(Y)$ acts on the set $\text{Mor}_{\mathcal{C}}(X, Y)$ by

$$(\alpha, \beta) \cdot f = \beta \circ f \circ \alpha^{-1} \quad \text{for any } \alpha \in \text{Aut}(X), \beta \in \text{Aut}(Y), f \in \text{Mor}(X, Y).$$

Composed with the diagonal morphism $G \rightarrow \text{Aut}(X) \times \text{Aut}(Y)$, $g \mapsto (\rho(g), \sigma(g))$, we can define...

$$\frac{\text{of } G \text{ on } \text{Mor}(X, Y)}{g \cdot f = \rho(g)f\sigma(g)^{-1}} \quad \forall g \in G, f \in \text{Mor}(X, Y)$$

commutes.

Lemma 2.0.179. See $\text{Mor}(X, Y)$ from the action of G on $\text{Mor}(X, Y)$.

The set of morphisms $(X, \rho) \rightarrow (Y, \sigma)$ in $\text{Rep}(G, \mathcal{C})$ is the set of fixed points of G in its action on $\text{Mor}_{\mathcal{C}}(X, Y)$.

Representation in Set Category We consider the case: $\mathcal{C}$: objects containing elements, morphisms are maps.

(X, ρ) : $\mathcal{C}$ -rept of G .

For $x \in X$, write $gx = \rho(g)(x)$.

A morphism $(X, \rho) \rightarrow (Y, \sigma)$ is a morphism $f : X \rightarrow Y$ in $\mathcal{C}$ such that

$$f(gx) = gf(x) \quad \forall x \in X, g \in G.$$

Denote the fixator of x in G as $G_x = \{g \in G \mid gx = x\}$.

Lemma 2.0.180. *Fixator change regarding G -action and isomorphisms.*

Let (X, ρ) be a $\mathcal{C}$ -rept of G .

$$(1) G_{gx} = gG_xg^{-1}$$

(2) If $f : (X, \rho) \rightarrow (Y, \sigma)$ is an isomorphism of repts, then $G_{f(x)} = G_x$ for all $x \in X$.

Definition 2.0.181. $\text{Aut}(G)$ acts on G -repts.

Let $\text{Aut}(G)$ denote the group of automorphisms of G .

For $g \in G$, consider

$$\text{Inn}(g) : G \rightarrow G; \quad h \mapsto ghg^{-1} \quad (\text{inner automorphism}).$$

Denote by $\text{Inn}(G)$ the group of $\text{Inn}(g)$, $g \in G$.

We have $\text{Inn}(G) \triangleleft \text{Aut}(G)$ as $a^{-1}\text{Inn}(g)a = \text{Inn}(aga^{-1}) \quad \forall a \in \text{Aut}(G), g \in G$.

The group $\text{Aut}(G)$ acts on $\mathcal{C}$ -repts: ${}^a(X, \rho) = (X, {}^a\rho = \rho \circ a^{-1})$ where $a \in \text{Aut}(G)$, (X, ρ) is any $\mathcal{C}$ -rept.

If $f : X \rightarrow X'$ induces a morphism of repts $(X, \rho) \rightarrow (X', \rho')$ then it also induces a morphism of repts

$${}^a(X, \rho) \rightarrow {}^a(X', \rho').$$

So $\text{Aut}(G)$ acts on the set of isomorphism classes of $\mathcal{C}$ -repts. For $g \in G$, we have $\text{Inn}(g)(X, \rho) \simeq (X, \rho)$ through i.e. $\text{Inn}(G)$ acts trivially on the isomorphism classes of $\mathcal{C}$ -repts.

Thus the group $\text{Out}(G) = \text{Aut}(G)/\text{Inn}(G)$, of outer automorphisms, acts on the isomorphism classes of $\mathcal{C}$ -repts.

Definition 2.0.182. Set repts: $(X, \rho) \amalg (Y, \sigma)$ and $(X, \rho) \times (Y, \sigma)$.

Let $\mathcal{C}$ denote the Set Category.

Definition 2.0.183. $\text{Rep}(G, \mathbf{Set}) \begin{cases} \text{object: } (X, \rho), X \text{ is a set, } \rho : G \rightarrow \text{Sym}(X) \\ \text{morphism: } (X, \rho) \rightarrow (Y, \sigma) \quad X \rightarrow Y \end{cases}$

Given sets X and Y , we can define $X \sqcup Y, X \times Y$.

If Z is a set, then $Z \times (X \sqcup Y) = (Z \times X) \sqcup (Z \times Y)$.

Let $(X, \rho), (Y, \sigma)$ be $\mathbf{Set}$ -reps of G . Define $\mathbf{Set}$ -reps:

- $(X, \rho) \sqcup (Y, \sigma) = (X \sqcup Y, \rho \sqcup \sigma)$ where $(\rho \sqcup \sigma)(g)(z) = \begin{cases} \rho(g)(z) & \text{if } z \in X \\ \sigma(g)(z) & \text{if } z \in Y \end{cases}$
- $(X, \rho) \times (Y, \sigma) = (X \times Y, \rho \times \sigma)$ where $(\rho \times \sigma)(g)(x, y) = (\rho(g)(x), \sigma(g)(y))$

Definition 2.0.184. *Subrepresentation*

Let (X, ρ) be a **Set**-rep of G . If $X_1 \subseteq X$ is stable under $\rho(g) \forall g$, then (X_1, ρ_1) is a subrepresentation of (X, ρ) where $\rho_1(g)$ is the restriction of $\rho(g)$ to X_1 .

Let X'_1 be the complement of X_1 in X , i.e. $X'_1 = X \setminus X_1$. Then X'_1 is also stable by G . Let $\rho'_1(g)$ be the restriction of $\rho(g)$ to X'_1 , then $(X, \rho) = (X_1, \rho_1) \sqcup (X'_1, \rho'_1)$.

Example 2.0.185. *Orbit*

(X, ρ) : **Set**-rep of G .

For any $x \in X$, the subset of X defined by $X_1 = \{\rho(g)(x) \mid g \in G\}$ is stable by G .

Definition 2.0.186. *Irreducible & indecomposable rep (same in **Set**-rep)*

Let (X, ρ) be a **Set**-rep of G , $X \neq \emptyset$. TFAE:

1. (X, ρ) is “simple” or “irreducible”, i.e. if any subrep (X_1, ρ_1) satisfies either $X_1 = \emptyset$ or $X_1 = X$.
2. (X, ρ) is “indecomposable”, i.e. if $(X, \rho) \simeq (X_1, \rho_1) \sqcup (X'_1, \rho'_1)$ then either $X_1 = \emptyset$ or $X'_1 = \emptyset$.
3. (X, ρ) is “transitive”, i.e. for any $x, y \in X$, there exists $g \in G$ s.t. $y = \rho(g)(x)$.

Example 2.0.187. *Left multiplication on left cosets of some subgroup H*

Let H be a subgroup of G . Then G acts on the set G/H , the set of left H -cosets in G , by left multiplication, denoted by $(G/H, \rho_H)$:

$$\rho_H(g)(xH) = gxH \quad \forall g, x \in G.$$

1. $(G/H, \rho_H)$ is transitive.
2. The stabilizer of xH in G is xHx^{-1} .

Definition 2.0.188 (Equivalence relation (orbit)). Let (X, ρ) be a Set-representation of G . Define an equivalence relation on X : $x \sim_G y \Leftrightarrow \exists g \in G$ s.t. $\rho(g)(x) = y$. Denote by $X/\sim_G$ the set of equivalence classes of this relation. Then:

1. Any equivalence class Ω in $X/\sim_G$ defines a transitive subrepresentation $(\Omega, \rho|_\Omega)$ of (X, ρ) .
2. $(X, \rho) = \bigsqcup_{\Omega \in X/\sim_G} (\Omega, \rho|_\Omega)$.

Theorem 2.0.189 (Characterization of transitive representations). 1. Let

(X, ρ) be a transitive representation of G . For $x \in X$, let G_x denote the

stabilizer of x in G . Then the map $G/G_x \rightarrow X$

$$gG_x \mapsto \rho(g)(x)$$

defines an isomorphism from $(G/G_x, \rho_{G_x})$ to (X, ρ) .

2. For subgroups H and H' of G , the representations $(G/H, \rho_H)$ and $(G/H', \rho_{H'})$ are isomorphic if and only if H and H' are conjugate.

Definition 2.0.190 (Degree of a representation). If (X, ρ) is a Set-representation of G , the cardinality of X is called the degree of the representation (X, ρ) .

Corollary 2.0.191. If G is a finite group, then the degree of any transitive Set-representation divides the order of G .

Let k be a field.

k -linear representations and matrix representations

Definition 2.0.192 (k -linear representations). A vector space representation (M, ρ) of G is called a (f.d.) (k -linear) representation of G , that is, a pair (M, ρ) where M is a k -vector space of finite dimension and $\rho : G \rightarrow \text{GL}(M)$ is a group morphism, where $\text{GL}(M)$ is the set of invertible linear transformations $M \rightarrow M$. I.e.,

- Objects of $\text{Rep}(G, \text{vect}_k)$: (M, ρ) k -linear representations.
- Morphisms: $f : (M, \rho) \rightarrow (N, \psi)$ with $f(gm) = gf(m)$ for all $g \in G$.

The dimension of M is called the degree of the representation (M, ρ) . The trivial representation $\rho_{\text{tr}} : G \rightarrow \text{GL}(k) \cong k^\times, g \mapsto 1$.

Definition 2.0.193 (Matrix representations). Assume $\dim M = r$. Fix a basis of M . There is a group isomorphism $\text{GL}(M) \rightarrow \text{GL}_r(k)$, where $\text{GL}_r(k) = \{\text{invertible matrices of degree } r \text{ over } k\}$. Then a k -linear representation of G can be seen as a pair (r, ρ) where $r \in \mathbb{Z}_{\geq 1}$ and $\rho : G \rightarrow \text{GL}_r(k)$ is a group morphism.

Remark 2.0.194. Two matrix representations (r, ρ) and (s, ψ) are isomorphic if and only if $r = s$ and there is an element $f \in \text{GL}_r(k)$ s.t. $\psi(g) = f\rho(g)f^{-1}$ for all $g \in G$.

i.e. the diagram

Definition 2.0.195. Representation induced from extension of scalars

Assume k is a subfield of a field k' . A k -rep (M, ρ) of G , its extension of scalars is a k' -rep $(k' \otimes M, k' \otimes \rho)$ where $k' \otimes \rho$ is the morphism $G \rightarrow \text{GL}(k' \otimes M)$

$$(k' \otimes \rho)(g) = \text{Id}_{k'} \otimes \rho(g) \quad \forall g \in G.$$

Note that a basis of M (over k) can be identified with a basis of $k' \otimes M$ (over k'). Then from the point of view of a matrix rep $G \rightarrow GL_r(k)$, extending the scalar means viewing the matrix entries of $\rho(g)$ as elements of k' . Conversely, a k' -rep (M', ρ') of G is said to be rational over k if there is a k -rep (M, ρ) s.t. $(M', \rho') \simeq (k' \otimes M, k' \otimes \rho)$. From the point of view of matrix reps, a matrix rep (r, ρ') of G over k' is rational over k if there exists $U \in GL_r(k')$ s.t. for all $g \in G$, the entries of $U\rho'(g)U^{-1}$ lie in k .

Definition 2.0.196. k -reps: $(M, \rho) \oplus (N, \psi) \ \& \ (M, \rho) \otimes (N, \psi)$

Let $(M, \rho), (N, \psi)$ be k -reps of G . Analog of disjoint union in Set-rep:

$$(M, \rho) \oplus (N, \psi) = (M \oplus N, \rho \oplus \psi) : (\rho \oplus \psi)(g)(m, n) = (\rho(g)(m), \psi(g)(n))$$

$$(M, \rho) \otimes_k (N, \psi) = (M \otimes_k N, \rho \otimes_k \psi) : (\rho \otimes \psi)(g) = \rho(g) \otimes \psi(g)$$

Analog of product in Set-rep.

Definition 2.0.197. Subrepresentation

If M_1 is a subspace of M which is stable under the action of all $\rho(g)$, $g \in G$, we get a subrepresentation (M_1, ρ_1) of (M, ρ) where $\rho_1(g)$ is the restriction of $\rho(g)$ to M_1 . In this case, we have the quotient rep $(M/M_1, \bar{\rho})$ where $\bar{\rho}(g)$ is an automorphism of M/M_1 induced by $\rho(g)$.

Definition 2.0.198. Indecomposable & irreducible (M, ρ) is called indecomposable if $M \neq 0$ and $(M, \rho) \simeq (M_1, \rho_1) \oplus (M_2, \rho_2)$ implies either $M_1 = 0$ or $M_2 = 0$. (M, ρ) is called simple or irreducible if $M \neq 0$ and the only G -stable subspaces of M are M and 0 .

Theorem 2.0.199. (Krull-Schmidt Theorem)

Let (M, ρ) be a representation of G of finite degree. (1) It is a direct sum of indecomposable representations. (2) If G is finite and if $(M, \rho) \simeq \bigoplus_{i \in I} (M_i, \rho_i) \simeq \bigoplus_{j \in J} (M'_j, \rho'_j)$ where $(M_i, \rho_i), (M'_j, \rho'_j)$ are indecomposable, then there is a bijection $\alpha : I \rightarrow J$ such that for all $i \in I$, $(M_i, \rho_i) \simeq (M'_{\alpha(i)}, \rho'_{\alpha(i)})$.

Definition 2.0.200. from $\text{Rept}(G, \text{Set})$ to $\text{Rept}(G, \text{Vect}_k)$

Let (Ω, ρ) be a set-representation of G . Denote by $k\Omega$ the k -vector space with basis Ω . Then ρ induces a group morphism $k\rho : G \rightarrow GL(k\Omega)$, a Vect_k -representation of G : $(k\Omega, k\rho)$ which is called the permutation representation. This gives a functor: $\text{Rept}(G, \text{Set}) \rightarrow \text{Rept}(G, \text{Vect}_k)$. In particular, this construction transforms:

- disjoint union of set-representations into direct sum of Vect_k -representations

- product of set-representations into tensor product of $\mathbf{Vect}_k$ -representations

$$\begin{pmatrix} \text{tensor product} \rightarrow \text{linear} \\ \text{direct product} \rightarrow \text{bilinear} \end{pmatrix}$$

Remark 2.0.201. $(\Omega, \rho) \in \text{Rept}(G, \mathbf{Set})$ irreducible $\Rightarrow (k\Omega, k\rho) \in \text{Rep}(G, \mathbf{Vect}_k)$ irreducible ($|\Omega| \geq 2$). Indeed, (transitive)

$s\Omega := \sum_{\omega \in \Omega} \omega \in k\Omega$ is invariant under all $\rho(g), g \in G \Rightarrow s\Omega$ spans a G -stable line.

Definition 2.0.202. regular representation

The group G acts on itself by left multiplication. The corresponding k -linear representation is called the regular representation of G .

Lemma 2.0.203. Assume that k has characteristic p dividing the order of G . Then the line $k(s_G)$ has no G -stable complement in the regular representation of G .

Proof. Let $k\sigma = k(s\sigma) \oplus V'$ as k -vector spaces for some subspace V' . Then we have a projection $\pi : kG \rightarrow k(sG)$. In particular, $\pi(k(SG)) = k(SG)$. If V' is G -stable, then $\pi(gx) = g\pi(x)$ for all $x \in kG, g \in G$. This implies $\pi(g \cdot 1) = g\pi(1)$. Since $\pi(1) \in k(SO)$, we have $g\pi(1) = \pi(1)$ for all $g \in G$. Thus $\pi(SG) = |G| \cdot \pi(1) = 0$, a contradiction. $\square$

Definition 2.0.204 (Ways to obtain new representations). Let $(M, \rho), (N, \sigma)$ be k -representations of G .

- $(M \otimes N, \rho \otimes \sigma)$
- $(M \oplus N, \rho \oplus \sigma)$
- $\text{Hom}_k(M, N)$

For all $\alpha \in \text{Hom}_k(M, N), g \in G$, define $g \cdot \alpha = \sigma(g)\alpha\rho(g)^{-1}$. Then $\text{Hom}_k(M, N)$ becomes a representation of G .

An element $\alpha \in \text{Hom}_k(M, N)$ is invariant under all $g \in G$ if and only if α is a morphism $(M, \rho) \rightarrow (N, \sigma)$.

In particular, there is a representation associated with $M^* = \text{Hom}_k(M, k)$: for all $\alpha \in \text{Hom}_k(M, k), g \cdot \alpha = \alpha\rho(g)^{-1}$, which is called the contragredient representation of (M, ρ) .

Definition 2.0.205 (Inflation). Let (M, ρ) be a k -representation of G/N where $N \triangleleft G$. Then

$$\rho : G/N \rightarrow GL(M) \rightsquigarrow \text{Inf}_{G/N}^G \rho : G \rightarrow GL(M).$$

Denote the inflation by $(M, \text{Inf}_{G/N}^G \rho)$.

Proposition 2.0.206. $\text{Hom}(G, k^\times) \simeq \text{Hom}(G/[G, G], k^\times)$.

Consider representations of degree 1:

$$G \rightarrow GL(k) = k^\times.$$

Its kernel contains $[G, G]$. Then we have $\text{Hom}(G, k^\times) \simeq \text{Hom}(G/[G, G], k^\times)$ (group homomorphism).

Example 2.0.207. $G = D_8 = \langle \sigma, \tau \mid \sigma^2 = \tau^2 = (\sigma\tau)^2 = 1 \rangle$.

$$[G, G] = Z(G) = \langle (\sigma\tau)^2 \rangle \Rightarrow G/[G, G] \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}.$$

So G has 4 degree 1 representations over a field k with $\text{char } k \neq 2$:

$$\rho_1 : \sigma \mapsto 1, \tau \mapsto -1$$

$$\rho_2 : \sigma \mapsto 1, \tau \mapsto 1$$

$$\rho_3 : \sigma \mapsto -1, \tau \mapsto -1$$

$$\rho_4 : \sigma \mapsto -1, \tau \mapsto 1$$

Lemma 2.0.208. $(M, \rho) \otimes (N, \sigma)$

Let (S, ρ) be an irreducible representation of G and let (k, σ) be a representation of degree 1. Then the tensor product representation $(S, \rho) \otimes_k (k, \sigma) = (S \otimes_k k, \rho \otimes_k \sigma) \simeq (S, \rho \otimes \sigma)$ is still irreducible.

Proof. Scalar representation preserves invariant spaces. □

Remark 2.0.209. (S, ρ) and $(S, \rho \otimes \sigma)$ may not be isomorphic.

Definition 2.0.210. Group algebra

Let G be a finite group. Let k be a field. Define kG to be the group algebra of G over k , the k -vector space with basis G endowed with multiplication defined by the multiplication of G : for $x = \sum x_s s$, $y = \sum y_t t \in kG$,

$$xy = \left(\sum_{s \in G} x_s s \right) \left(\sum_{t \in G} y_t t \right) = \sum_{g \in G} \left(\sum_{st=g} x_s y_t \right) g$$

Lemma 2.0.211. Let $z : kG \rightarrow k$ be the linear form on kG defined by

$$z(g) = \begin{cases} 0 & g \neq 1 \\ 1 & g = 1 \end{cases}$$

1. z is a “class function”: for all $x, y \in kG$, $z(xy) = z(yx)$.

2. The map $\widehat{\text{im}ath} : kG \rightarrow kG^* = \text{Hom}(kG, k)$

$$x \mapsto (y \mapsto z(xy))$$

is a k -linear isomorphism.

Class function Identifying $\{\text{functions } G \rightarrow k\} \longleftrightarrow \text{Hom}_k(kG, k)$ linear forms. Define by $F(G, k)$ the k -vector space of functions $G \rightarrow k$:

$$\text{Hom}_k(kG, k) = (kG)^*$$

For $f \in F(G, k)$, the inverse image of f under the isomorphism $\widehat{\text{im}ath}$ is

$$f^\circ = \sum_{g \in G} f(g^{-1})g$$

Let $f \in F(G, k)$ be a class function, that is, $f(gh) = f(hg)$ for all $g, h \in G$. Identify f with its corresponding element in $\text{Hom}_k(kG, k)$: $f(xy) = f(yx)$ for all $x, y \in kG$. Denote by $CF(G, k)$ the space of class functions on G (or kG).

Remark 2.0.212. Let $a \in kG$, $f \in F(G, k)$. Denote by $a \cdot f$ the element of $F(G, k)$ defined by

$$(a \cdot f)(x) = f(xa) \text{ for all } x \in kG$$

Then $(a \cdot f)^\circ = a \cdot f^\circ$.

Definition 2.0.213. γ_C and S_C for a conjugacy class C Let $\text{Cl}(G)$ be the set of conjugacy classes of G , and for $C \in \text{Cl}(G)$ set γ_C to be the characteristic function of C , i.e.

$$\gamma_C(g) = \begin{cases} 0 & \text{if } g \notin C \\ 1 & \text{if } g \in C \end{cases}$$

Set S_C to be the sum of elements in C , i.e.

$$S_C = \sum_{g \in C} g \in kG.$$

For $f \in CF(G, k)$,

$$f = \sum_{C \in \text{Cl}(G)} f(g_C) \gamma_C$$

$$f^\circ = \sum_{C \in \text{Cl}(G)} f(g_C^{-1}) S_C$$

where g_C is an element in C .

Lemma 2.0.214. $\text{CF}(G, k) \cong Z(kG)$

Let $Z(kG)$ be the center of kG .

1. The family $(\gamma_C)_{C \in \text{Cl}(G)}$ is a basis of $\text{CF}(G, k)$ and the family (S_C) is a basis of $Z(kG)$. $\Rightarrow \dim_k \text{CF}(G, k) = |\text{Cl}(G)|$
2. The isomorphism $\text{im} \hat{\alpha}$ restricts to a k -linear isomorphism $\text{CF}(G, k) \rightarrow Z(kG)$. $\Rightarrow \dim_k Z(kG) = |\text{Cl}(G)|$

Proposition 2.0.215. $\text{Rep}(G, \text{Vect}_k) \leftrightarrow kG\text{-module}$

Let (M, ρ) be a k -representation of G . $\rho : G \rightarrow \text{GL}(M)$ extends linearly to an algebra morphism $\rho : kG \rightarrow \text{End}_k(M) \Rightarrow M$ becomes a kG -module. Conversely, given an algebra morphism $\rho : kG \rightarrow \text{End}_k(M)$ (sending 1 to 1), it restricts to a group morphism $\rho : G \rightarrow \text{GL}(M)$.

Thus $\text{Rep}(G, \text{Vect}_k) \leftrightarrow kG\text{-module}$.

Lemma 2.0.216. The regular representation is faithful.

Let $\rho_{kG} : kG \rightarrow \text{End}_k(kG)$ denote the morphism associated to the regular representation. Then ρ_{kG} is injective, i.e. the regular representation is faithful.

Lemma 2.0.217. (Schur's lemma)

Let (S, ρ) and (S', ρ') be irreducible representations of G .

1. If (S, ρ) and (S', ρ') are not isomorphic, then $\text{Hom}_{kG}(S, S') = 0$.
2. $\text{End}_{kG}(S)$ is a finite-dimensional division k -algebra.
3. If k is algebraically closed, then $\text{End}_{kG}(S) = k \cdot \text{Id}$.

Remark 2.0.218. The assumption that k is algebraically closed is in general too strong. We often replace the hypothesis by the hypothesis that k is splitting for G . (Recall that k is splitting for an algebra A if

$$\text{End}_A(S) \simeq K \text{ for any simple } A\text{-module } S$$

If K is splitting for the group algebra KG , then we say K is splitting for G .

Theorem 2.0.219. (Brauer)

If k contains an $\exp(G)$ -th root of unity, then k is splitting for G . ($\exp(G) = \text{lcm}\{o(g), g \in G\}$, so $\exp(G) \mid |G|$).

Proposition 2.0.220. If $\text{char } k \mid |G|$, then kG is not semisimple.

Proof. Let $w = \sum_{g \in G} g$. For any $x \in G$, $xw = w = wx$, i.e. $w \in Z(kG)$.

Now $w^2 = w \sum_{g \in G} g = |G|w = 0$ since $\text{char } k \mid |G|$. Then the ideal kGw satisfies $(kGw)^2 = (kG)^2 w^2 = (0)$. $kGw \subset J(kG)$ hence kG is not semisimple. $\square$

Theorem 2.0.221. (*Maschke*)

If $\text{char } k \nmid |G|$, then kG is semisimple.

Proof. Let V be a kG -module, W a kG -submodule of V .

Aim: there is a kG -submodule U of V with $V = W \oplus U$.

Let $\iota : W \hookrightarrow V$ be the inclusion. Then ι is a kG -homomorphism.

As k -vector space: $V = W \oplus W'$ for some subspace W' of V .

Let $p : V \rightarrow W$ be the projection. Then $p \circ \iota = \text{Id}_W$.

Consider $\pi : V \rightarrow W$

$$\pi(v) = \frac{1}{|G|} \sum_{g \in G} (\rho(g)^{-1} \circ p \circ \rho(g))(v)$$

$$\forall h \in G, \rho(h)^{-1} \pi \rho(h) = \frac{1}{|G|} \sum_{g \in G} \rho(gh)^{-1} \circ p \circ \rho(gh) = \pi.$$

Hence π is a kG -morphism, $\pi \in \text{Hom}_{kG}(V, W)$.

$$\pi \circ \iota = \frac{1}{|G|} \sum_{g \in G} (\rho(g)^{-1} \circ p \circ \rho(g)) \circ \iota = \frac{1}{|G|} \sum_{g \in G} \rho(g)^{-1} \circ (p \circ \iota) \circ \rho(g) = \text{Id}_W.$$

Let $U = \ker \pi$, then $V = W \oplus U$. $\square$

Corollary 2.0.222.

Corollary 2.0.223. Let $\text{char } k \nmid |G|$. Then any finite-dimensional k -representation of G is a direct sum of irreducible representations of G .

$\text{char } k \nmid |G| \Rightarrow$ ordinary representation.

$\text{char } k \mid |G| \rightarrow$ modular representation.

Proposition 2.0.224. decomposition of KG . Let G be a finite group. K a field with $\text{char } K \nmid |G|$. (1) As a ring, KG is a direct sum of matrix algebras over division rings. (2) Suppose in addition k is splitting for G (for example, $k = \bar{k}$). Let $S_1, \dots, S_r$ be pairwise non-isomorphic simple kG -modules and let $d_i = \dim_k S_i$. Then d_i equals the multiplicity with which S_i is a summand of the regular representation of G , and $|G| \geq d_1^2 + \dots + d_r^2$ with equality if and only if $S_1, \dots, S_r$ is a complete set of representatives of isomorphism classes of simple kG -modules.

Definition 2.0.225. Character

(M, ρ) representation of G . $\rho : G \rightarrow GL_r(k)$. $\chi_M : G \rightarrow \mathbb{C}$ is called the character afforded by (M, ρ) .

$$g \mapsto \text{tr}(\rho(g))$$

Proposition 2.0.226. *Properties of characters.*

(1) χ_M is a class function, i.e. $\chi_M(g) = \chi_M(h)$ if g, h are G -conjugate. (2)

$\chi_M(g^{-1}) = \overline{\chi_M(g)}$. (3) If $M \simeq N$ as $\mathbb{C}G$ -modules, then $\chi_M = \chi_N$. (4)

$\chi_{M \oplus N} = \chi_M + \chi_N$, $\chi_{M^*} = \overline{\chi_M}$, $\chi_{M \otimes N} = \chi_M \chi_N$, $\chi_{\text{Hom}(M, N)} = \overline{\chi_M} \chi_N$. (5)

$\chi_M(1) = \dim_{\mathbb{C}} M$. $|\chi_M(g)| \leq \chi_M(1)$ with equality iff the action of g on M is a scalar. $\chi_M(g) = \chi_M(1)$ iff g acts trivially on M .

Definition 2.0.227. $\langle \cdot, \cdot \rangle$

$$k = \mathbb{C}. \quad CF(G, \mathbb{C}) = CF(G)$$

$\forall f_1, f_2 \in CF(G)$, define $\langle f_1, f_2 \rangle := \frac{1}{|G|} \sum_{g \in G} f_1(g) \overline{f_2(g)}$

$$\begin{aligned} &= \frac{1}{|G|} \sum_{C \in \text{Cl}(G)} \frac{|G|}{|C_G(g_c)|} f_1(g_c) \overline{f_2(g_c)} \quad (g_c \in C) \\ &= \sum_{C \in \text{Cl}(G)} \frac{1}{|C_G(g_c)|} f_1(g_c) \overline{f_2(g_c)} \end{aligned}$$

$\langle \cdot, \cdot \rangle$ is a Hermitian form on the $\mathbb{C}$ -vector space $CF(G)$. If f_1 and f_2 are characters of G , then $\langle f_1, f_2 \rangle = \langle f_2, f_1 \rangle$.

Lemma 2.0.228. V^G

Let V be a $\mathbb{C}G$ -module. Define $V^G = \{v \in V \mid gv = v \text{ for all } g \in G\}$. Then V^G is a submodule of V and $\dim V^G = \langle \chi_V, 1_G \rangle = \frac{1}{|G|} \sum_{g \in G} \chi_V(g)$. Then V^0 is a submodule of V and $\dim V^0 = \langle \chi_V, 1 \mid G \rangle = \frac{1}{|G|} \sum_{g \in G} \chi_V(g)$

Pf: Let $\pi : V \rightarrow V$

$$v \mapsto \frac{1}{|G|} \sum_{g \in G} gv$$

- $\pi \in \text{Hom}_{\mathbb{C}G}(V, V)$

$$\forall h \in G, v \in V, \pi(hv) = \frac{1}{|G|} \left(\sum_{g \in G} gh \right) v = \pi(v) = \frac{1}{|G|} \sum_{g \in G} hgv = h\pi(v)$$

- $\pi^2 = \pi$

$$\pi^2 = \left(\frac{1}{|G|} \sum_{g \in G} g \right) \left(\frac{1}{|G|} \sum_{h \in G} h \right) = \frac{1}{|G|^2} \sum_{g, h \in G} gh = \pi$$

Then $V = \ker \pi \oplus \text{Im } \pi$

- $V = \ker \pi \oplus \text{Im } \pi$

If $v \in V^G$, $\pi(v) = \frac{1}{|G|} \sum_{g \in G} gv = v \Rightarrow V^G \subseteq \text{Im } \pi$.

On the other hand, $\forall h \in G, h \left(\frac{1}{|G|} \sum_{g \in G} gv \right) = \frac{1}{|G|} \sum_{g \in G} hgv$ hence $\text{Im } \pi \subseteq V^G$.

Thus $V^G = \text{Im } \pi$ and $V = \ker \pi \oplus V^G$

- $\text{tr}(\pi) = \dim V^G$
- $\text{tr}(\pi) = \text{tr} \left(\frac{1}{|G|} \sum_{g \in G} \rho(g) \right) = \frac{1}{|G|} \sum_{g \in G} \text{tr}(\rho(g)) = \frac{1}{|G|} \sum_{g \in G} \chi_V(g)$

Lemma 2.0.229. $\text{Hom}_k(V, W)^G = \text{Hom}_{kG}(V, W)$

Theorem 2.0.230. (*1st orthogonality relation*)

Let V, W be simple kG -modules. Then

$$\langle \chi_V, \chi_W \rangle = \begin{cases} 1 & \text{if } V \simeq W \\ 0 & \text{otherwise} \end{cases}$$

Proof. $\dim_{\mathbb{C}} \text{Hom}_{\mathbb{C}}(V, W) = \dim_{\mathbb{C}} \text{Hom}_{\mathbb{C}}(V, W)^G = \langle \chi_{\text{Hom}_{\mathbb{C}}(V, W)}, 1 \rangle$

$$\begin{aligned} &= \langle \bar{\chi}_V \chi_W, 1 \rangle \\ &= \frac{1}{|G|} \sum_{g \in G} \bar{\chi}_V(g) \chi_W(g) \\ &= \langle \chi_V, \chi_W \rangle. \end{aligned}$$

Then apply Schur's lemma.

Let $\text{Irr}(G)$ denote the set of irreducible complex characters of G . □

Corollary 2.0.231. $\text{Irr}(G)$ is $\mathbb{C}$ -linearly independent.

Corollary 2.0.232. $\langle \chi_V, \chi_V \rangle = 1 \Leftrightarrow V$ is irreducible (use complete reducibility)

Corollary 2.0.233. $\chi_V = \chi_W \Leftrightarrow V \simeq W$.

Theorem 2.0.234. $\text{Irr}(G)$ is a $\mathbb{C}$ -basis of $CF(G)$

Proof. $\mathbb{C}G \simeq M_{n_1}(\mathbb{C}) \oplus \dots \oplus M_{n_r}(\mathbb{C}), \quad r = |\text{Irr}(G)|$

$$Z(\mathbb{C}G) = Z(M_{n_1}(\mathbb{C})) \oplus \dots \oplus Z(M_{n_r}(\mathbb{C})) \simeq \underbrace{\mathbb{C} \times \dots \times \mathbb{C}}_r, \quad \dim Z(\mathbb{C}G) = r.$$

$$\dim Z(\mathbb{C}G) = |\text{Cl}(G)| = |\text{CF}(G)|$$

$$Z(\mathbb{C}G) = \bigoplus_{C \in \text{Cl}(G)} \mathbb{C}(\hat{C})$$

Therefore, $|\text{Irr}(G)| = |\text{Cl}(G)|$. □

Corollary 2.0.235.

Corollary 2.0.236. Let V be a $\mathbb{C}G$ -module with $V = S_1^{n_1} \oplus \dots \oplus S_k^{n_k}$ where $S_1, \dots, S_k$ are non-isomorphic simple $\mathbb{C}G$ -modules. Then $n_i = \langle \chi_V, \chi_{S_i} \rangle$.

Theorem 2.0.237. (2^{nd} orthogonal relation)

Let X be the character table of G , regarded as a matrix, i.e.

$$\begin{aligned} \text{Irr}(G) &= \{\chi_1, \dots, \chi_r\} \\ \text{Cl}(G) &= \{C_1, \dots, C_r\} \end{aligned}$$

Let $C = \begin{pmatrix} |C_G(x_1)| & & \\ & \ddots & \\ & & |C_G(x_r)| \end{pmatrix}$ where $x_1, \dots, x_r$ are representatives of conjugacy classes of the elements of G .

Then $\bar{X}^\top X = C$.

Namely, $\sum_{i=1}^r \chi_i(g) \overline{\chi_i(h)} = \begin{cases} |C_G(g)| & \text{if } g \text{ and } h \text{ are conjugate in } G \\ 0 & \text{otherwise} \end{cases}$

Proof. By 1st orthogonal relation,

$$X \left(\frac{1}{|G|} \text{diag}(|C_1|, \dots, |C_r|) \right) \bar{X}^\top = I$$

i.e. $XC^{-1}\bar{X}^\top = I$.

$$\Rightarrow C^{-1}\bar{X}^\top X = I$$

$$\Rightarrow \bar{X}^\top X = C$$

□

Theorem 2.0.238. Irreducible representations of abelian groups

TFAE for a finite group G :

1. G is abelian.
2. All irreducible complex representations of G have degree 1.

Proof. (1) $\Rightarrow$ (2): $\mathbb{C}G \simeq M_{n_1}(\mathbb{C}) \oplus \dots \oplus M_{n_r}(\mathbb{C})$. G abelian $\Rightarrow n_1 = \dots = n_r = 1$.

(2) $\Rightarrow$ (1): $|G| = \sum_{\chi \in \text{Irr}(G)} \chi(1)^2 = |\text{Irr}(G)| = |\text{Cl}(G)| \Rightarrow$ abelian. □

Lemma 2.0.239. Product of irreducible characters is still irreducible.

Let G_1, G_2 be two finite groups. Let V_i be a kG_i -module, $i = 1, 2$. We can define for all $g_i \in G_i, v_i \in V_i$:

$$(g_1, g_2)(v_1 \otimes v_2) = g_1 v_1 \otimes g_2 v_2.$$

Then $V_1 \otimes V_2$ is a $k(G_1 \times G_2)$ -module. If $\chi_1 \in \text{Irr}(G_1)$, $\chi_2 \in \text{Irr}(G_2)$, then $\chi_1 \times \chi_2$ (afforded by $V_1 \otimes V_2$) $\in \text{Irr}(G_1 \times G_2)$, where $(\chi_1 \times \chi_2)(g_1, g_2) = \chi_1(g_1)\chi_2(g_2)$ for all $g_i \in G_i$, $i = 1, 2$.

Proof. $\langle \chi_1 \times \chi_2, \chi_1 \times \chi_2 \rangle = \frac{1}{|G_1||G_2|} \sum_{g_i \in G_i} \chi_1(g_1)\chi_2(g_2)\overline{\chi_1(g_1)\chi_2(g_2)} = 1.$ $\square$

Corollary 2.0.240. *The character table of $G_1 \times G_2$ is the tensor product of their character tables.*

Proof. (1) The conjugacy classes in $G_1 \times G_2$ are precisely the products of conjugacy classes.

$$(2) \langle \chi_{I_1} \otimes \chi_{I_2}, \chi_{J_1} \otimes \chi_{J_2} \rangle = \langle \chi_{I_1}, \chi_{J_1} \rangle \langle \chi_{I_2}, \chi_{J_2} \rangle$$

$$= \begin{cases} 1 & I_1 \simeq J_1, \quad I_2 \simeq J_2 \\ 0 & \text{otherwise} \end{cases}$$

Hence the irreducible representations of $G_1 \times G_2$ are precisely the tensor products of irreducible representations. $\square$

The degree of an irreducible representation of G (over $\mathbb{C}$) divides $|G|$

Definition 2.0.241 (Integral over $\mathbb{Z}$). *Let R be a commutative ring and let $x \in R$. We say that x is integral over $\mathbb{Z}$ if there exists $n \geq 1$ and $a_1, \dots, a_n \in \mathbb{Z}$ such that*

$$x^n + a_1x^{n-1} + \dots + a_n = 0.$$

- A complex number which is integral over $\mathbb{Z}$ is called an algebraic integer.
- If $x \in \mathbb{Q}$ is an algebraic integer, then $x \in \mathbb{Z}$: write $x = \frac{q}{p}$ where $(p, q) = 1$. Then

$$p^n + a_1qp^{n-1} + \dots + a_nq^n = 0.$$

Thus $q \mid p^n \implies q = 1$.

- x is integral over $\mathbb{Z} \iff$ the subring $\mathbb{Z}[x]$ of R generated by x is finitely generated as a $\mathbb{Z}$ -module. (Since $\mathbb{Z}$ is Noetherian.) $\iff$ There exists a finitely generated $\mathbb{Z}$ -submodule of R which contains $\mathbb{Z}[x]$.

Proposition 2.0.242. *If $C_1, \dots, C_r$ are conjugacy classes of G and $\lambda_1, \dots, \lambda_r$ are algebraic integers in $\mathbb{C}$, then $\sum_{i=1}^r \lambda_i(SC_i)$ is integral over $\mathbb{Z}$.*

Proof. Note that each product $(SC_i)(SC_j)$ is a linear combination with integer coefficients of the SC_k . The subgroup $\mathbb{Z}SC_1 \oplus \dots \oplus \mathbb{Z}SC_r$ of $Z(CG)$ is hence a subring. Since it is finitely generated over $\mathbb{Z}$, each of its elements is integral over $\mathbb{Z}$. $\square$

Lemma 2.0.243. *Let $\rho_1, \dots, \rho_r$ be all simple representations (up to isomorphism). $\rho_i : G \rightarrow \text{GL}_{d_i}(\mathbb{C}) \implies \rho'_i : CG \rightarrow M_{d_i}(\mathbb{C})$ $CG \simeq M_{d_1}(\mathbb{C}) \oplus \dots \oplus M_{d_r}(\mathbb{C}) \implies p'_i : CG \rightarrow$*

$M_{d_i}(\mathbb{C})$ projection. Fix i , let $x \in Z(\mathbb{C}G)$, then $p_i(x) = \lambda I$ for some $\lambda \in \mathbb{C}$. In fact,

$$p_i(x) = \frac{1}{d_i} \text{tr}(p_i(x)) \cdot I$$

Write $x = \sum_{g \in G} a_g g$, then

$$p_i(x) = \frac{1}{d_i} \sum_{g \in G} a_g \chi_i(g) \cdot I.$$

Theorem 2.0.244. *The degree of a simple representation of G over $\mathbb{C}$ divides $|G|$.*

Proof. Let $x = \sum_{g \in G} \overline{\chi_i(g)} g \in Z(\mathbb{C}G)$ where $\overline{\chi_i(g)}$ is an algebraic integer. Then x is integral over $\mathbb{Z} \Rightarrow p_i(x)$ is integral over $\mathbb{Z}$

$$p_i(x) = \frac{1}{d_i} \sum_{g \in G} \overline{\chi_i(g)} \chi_i(g) \cdot I = \frac{|G|}{d_i} I \Rightarrow \frac{|G|}{d_i} \text{ is integral over } \mathbb{Z} \Rightarrow d_i \mid |G|$$

□

Induction & Restriction

Definition 2.0.245 (Restriction). *Let W be a KG -module. $H \leq G$. Then $H \hookrightarrow G \rightarrow \text{GL}(W)$ So W is also a kH -module, denoted by $\text{Res}_H^G(W)$: the restriction of W to H .*

Definition 2.0.246 (Induction). *Let V be a kH -module. Define a kG -module $\text{Ind}_H^G(V) = kG \otimes_{kH} V$: induced from H to G .*

Remark 2.0.247. $\text{Res}_H^G : KG\text{-module} \rightarrow KH\text{-module}$ functors. $\text{Ind}_H^G : KH\text{-module} \rightarrow KG\text{-module}$

Proposition 2.0.248. *Characterization of $\text{Ind}_H^G(V)$ Let $H \leq G$. Let V be an RH -module (R is a commutative ring with 1). $G/H = \{g_1H, \dots, g_rH\}$ where $r = [G : H]$ and $g_1 \in H$. Denote the set of representatives as $[G/H]$. Then $\text{Ind}_H^G(V) = \bigoplus_{i=1}^r (g_i \otimes V)$ as R -modules, where*

$$g_i \otimes V = \{g_i \otimes v \mid v \in V\} \simeq V \text{ as an } R\text{-module.}$$

If V is free, then $\text{rank}_R \text{Ind}_H^G(V) = [G : H] \text{rank}_R V$. If $x \in G$, then $x(g_i \otimes V) = g_j \otimes V$ where $xg_i = g_j h$ for some $h \in H$. So the R -submodules $g_i \otimes V$ of $\text{Ind}_H^G(V)$ are permuted transitively under the action of G , and $\text{Stab}_G(g_1 \otimes V) = H$.

Proof. $RG_{RH} = \bigoplus_{i=1}^r R(g_iH) = \bigoplus_{i=1}^r g_i RH \simeq (RH)^r$ as an R -module.

$$RG \otimes_{RH} V = \bigoplus_{i=1}^r g_i(RH) \otimes_{RH} V = \bigoplus_{i=1}^r (g_i RH \otimes_{RH} V) = \bigoplus_{i=1}^r (g_i \otimes V) \text{ as } R\text{-mod.}$$

If $x \in G$, $x(g_i \otimes V) = g_j \otimes hV$ (where $xg_i = g_jh$, $h \in H$), i.e., $x(g_i \otimes V) \subseteq g_j \otimes V$. Similarly $x^{-1}(g_j \otimes V) \subseteq (g_i \otimes V)$, hence $x(g_i \otimes V) = g_j \otimes V$. Thus G permutes transitively on $g_i \otimes V$, $i = 1, \dots, r$. If $x \in H$, $x(g_1 \otimes V) = xg_1 \otimes V = g_1(g_1^{-1}xg_1) \otimes V = g_1 \otimes V \Rightarrow \text{Stab}_G(g_1 \otimes V) = H$. If $x \notin H$, $xg_1 = g_ih$ for some $g_i \notin H$, $h \in H$. $\square$

Proposition 2.0.249. *Represented as an induced module. Let M be an RG -module that has an R -submodule V such that M is a direct sum of R -submodules $\{gV \mid g \in G\}$. Then $M \simeq \text{Ind}_H^G(V)$ where $H = \{g \in G \mid gV = V\}$.*

Proof. Define a map of R -modules: $RG \otimes_{RH} V \rightarrow M$, an RG -morphism

$$g \otimes v \mapsto gv$$

We have $\{gV \mid g \in G\} \longleftrightarrow G/H$

$$gV \longleftrightarrow gH$$

$\square$

$RG \otimes V$ and M are direct sums of $[G : H]$ R -submodules $g \otimes V$, gV respectively: $RH \cong gV \cong V \cong gV$ as R -modules

$$g \otimes V \leftrightarrow V \leftrightarrow gV$$

Proposition 2.0.250. *Character of induced module*
 $H \leq G$. Let χ be a character of H . Then

$$\begin{aligned} \text{Ind}_H^G(\chi)(g) &= \frac{1}{|H|} \sum_{\substack{t \in G \\ t^{-1}gt \in H}} \chi(t^{-1}gt) \\ &= \sum_{\substack{t \in [G/H] \\ t^{-1}gt \in H}} \chi(t^{-1}gt) \end{aligned}$$

Proof. (1) Second “=”: If $t^{-1}gt \in H$, $h \in H$, then $(th)^{-1}g(th) = h^{-1}(t^{-1}gt)h \in H$, hence $\chi(t^{-1}gt) = \chi((th)^{-1}g(th))$.

(2) First “=”: Recall $\text{Ind}_H^G(V) = \bigoplus_{i=1}^n (g_i \otimes V)$.

Idea: g acts on $\text{Ind}_H^G(V)$:

$$\begin{cases} gg_i \otimes V = g_i \otimes V \Rightarrow \text{consider how } g \text{ acts on } g_i \otimes V \\ gg_i \otimes V \neq g_i \otimes V \Rightarrow \text{no contribution to the trace} \end{cases}$$

$$g(t \otimes V) = t \otimes V \Leftrightarrow t^{-1}gt \otimes V = 1 \otimes V \Leftrightarrow t^{-1}gt \in \text{Stab}_G(1 \otimes V) = H$$

Note that $\chi: H$ acts on V . If $t^{-1}gt \in H$, then g acts on $t \otimes V$ by

$$g(t \otimes v) = gt \otimes v = t(t^{-1}gt) \otimes v = t \otimes t^{-1}gtv \Rightarrow g : t \otimes v \mapsto t \otimes t^{-1}gtv$$

Hence trace of g on $t \otimes V = \text{trace of } t^{-1}gt \text{ on } V = \chi(t^{-1}gt)$.

Recall:

- V : A -module, W : B -module, T : A - B -bimodule.

$$\text{Hom}_A(W \otimes_B T, V) \simeq \text{Hom}_B(W, \text{Hom}_A(T, V))$$

- A, B rings. $\exists \varphi : A \rightarrow B$ ring homomorphism, $1_A \mapsto 1_B$. If W is a B -module, then it can be viewed as an A -module through $aw := \varphi(a)w$, denoted by $W|_A$.

□

Lemma 2.0.251. *Let $A \rightarrow B$ be a ring homomorphism with $A \rightarrow B$. Let V be an A -module, W a B -module. Then*

1. $\text{Hom}_B(B \otimes_A V, W) \simeq \text{Hom}_A(V, W \downarrow_A)$
2. $\text{Hom}_A(W \downarrow_A, V) \simeq \text{Hom}_B(W, \text{Hom}_A(B, V))$
3. If $B \rightarrow C$ is another ring homomorphism, then $C \otimes_B (B \otimes_A V) \simeq C \otimes_A V$

Pf: (1) & (2) $\text{Hom}_B(B \otimes_A B, W) \simeq W \downarrow_A \simeq B \otimes_B W$ as A -modules, $\omega \mapsto 1 \otimes \omega$.

Corollary 2.0.252. *Let $H \leq K \leq G$. Let V be an RH -module, W an RG -module.*

1. (Frobenius reciprocity)

$$\text{Hom}_{RG}(\text{Ind}_H^G V, W) \simeq \text{Hom}_{RH}(V, \text{Res}_H^G W) \text{ as } R\text{-modules}$$

$$\text{Hom}_{RG}(W, \text{Ind}_H^G V) \simeq \text{Hom}_{RH}(\text{Res}_H^G W, V)$$

2. $\text{Ind}_K^G(\text{Ind}_H^K V) \simeq \text{Ind}_H^G V$ as RG -modules
3. $\text{Res}_H^K(\text{Res}_K^G W) \simeq \text{Res}_H^G(W)$ as RH -modules
4. $\text{Ind}_H^G(V) \otimes_R W \simeq \text{Ind}_H^G(V \otimes_R \text{Res}_H^G(W))$ as RG -modules
5. $\text{Ind}_H^G(V) \simeq \text{Hom}_{RH}(RG, V)$ as RG -modules

Proof. (4) $(RG \otimes_{RH} V) \otimes_R W \rightarrow RG \otimes_{RH} (V \otimes_R W)$

$$g \otimes v \otimes w \mapsto g \otimes v \otimes g^{-1}w$$

$$g \otimes v \otimes gw \leftarrow g \otimes v \otimes w$$

(5) $RG \otimes_{RH} V \simeq \text{Hom}_{RH}(RG, V)$

$$g \otimes v \mapsto \phi_{g,v} : x \mapsto \begin{cases} (xg)v & \text{if } xg \in H \\ 0 & \text{otherwise} \end{cases}$$

$$\sum_{g \in [G/H]} g \otimes \theta(g^{-1})$$

□

Corollary 2.0.253. *Frobenius reciprocity.*

$H \leq K \leq G$. χ : character of H . ψ : character of G . Note that

$$1. \text{ (Frobenius reciprocity) } \langle \chi_V, \chi_W \rangle = \dim_{\mathbb{C}} \text{Hom}_{\mathbb{C}G}(V, W)$$

$$\langle \text{Ind}_H^G \chi, \psi \rangle_G = \langle \chi, \text{Res}_H^G \psi \rangle_H, \quad \langle \psi, \text{Ind}_H^G \chi \rangle_G = \langle \text{Res}_H^G \psi, \chi \rangle_H.$$

$$2. \text{Ind}_K^G(\text{Ind}_H^K(\chi)) = \text{Ind}_H^G(\chi)$$

$$3. \text{Res}_H^K(\text{Res}_K^G(\psi)) = \text{Res}_H^G(\psi)$$

$$4. \text{Ind}_H^G(\chi) \cdot \psi = \text{Ind}_H^G(\chi \cdot \text{Res}_H^G(\psi))$$

Mackey decomposition formula $H, K \leq G$. V a K -module. $\text{Res}_H^G(\text{Ind}_K^G(V))$

If Ω is a left G -set, write $G \backslash \Omega$ for the set of G -orbits on Ω and $[G \backslash \Omega]$ for the representatives. Similarly for a right G -set Ω , write Ω / G , $[\Omega / G]$.

Definition 2.0.254. (H, K) -double coset

$$HgK = \{h g k \mid h \in H, k \in K\} \quad \forall g \in G.$$

Lemma 2.0.255. *Properties of (H, K) -double cosets.*

$H, K \leq G$.

1. Each (H, K) -double coset is a disjoint union of right H -cosets and left K -cosets.
2. Any two (H, K) -double cosets either coincide or are disjoint. The (H, K) -double cosets partition G .
3. The set of (H, K) -double cosets is in bijection with the orbits $H \backslash (G/K)$ under the mapping $HgK \mapsto H(gK) \in H \backslash (G/K)$.

Proof. (2) If $h_1 g_1 k_1 = h_2 g_2 k_2 \in Hg_1K \cap Hg_2K$, then $g_1 = h_1^{-1} h_2 g_2 k_2 k_1^{-1} \in Hg_2K \Rightarrow Hg_1K \subset Hg_2K$. □

Definition 2.0.256. $H \backslash G / K$

We denote the set of (H, K) -double cosets in G by $H \backslash G / K$, and the representatives by $[H \backslash G / K]$.

Lemma 2.0.257. *Let $H, K \leq G$, $g \in G$. Then*

1. $HgK/K \simeq H/H \cap {}^gK$ as left H -sets

$${}^gK = g^{-1}Kg$$

2. $H \backslash HgK \simeq (H^g \cap K) \backslash K$ as right K -sets

$$H^g = gHg^{-1}$$

$$\text{So } [G : K] = \sum_{g \in [H \backslash G / K]} [H : H \cap {}^gK]$$

$$[G : H] = \sum_{g \in [H \backslash G / K]} [H^g \cap K]$$

Proof. then $\Omega \simeq G / \text{Stab}_G(w)$ for $w \in \Omega$.

HgK is a single H -orbit on left K -cosets with $\text{Stab}_H(gK) = \{h \in H \mid hgK = gK\} = \{h \in H \mid g^{-1}hg \in K\} = H \cap {}^gK$. $\square$

Definition 2.0.258. gV (a gH -module)

$$\begin{cases} \text{as sets } {}^gV = V \\ \text{for } v \in V, h \in H, \underbrace{{}^gh \cdot v = hv}_{\substack{{}^gh \text{ on } {}^gV \text{ vs } h \text{ on } V}} \end{cases}$$

i.e. If $H \xrightarrow{\varphi} GL(V)$ is the rep, then the rep of gH is given by

$$\begin{aligned} {}^gH &\rightarrow H \xrightarrow{\varphi} GL(V) \\ {}^gh &\mapsto h \end{aligned}$$

$$gh(g \otimes v) = ghg^{-1}g \otimes v = g \otimes hv$$

When $g \otimes V$ is identified with V via $g \otimes v \leftrightarrow v$, the action of gH on V coincides with the action of gH on gV .

Theorem 2.0.259. (Mackey decomposition formula)

$H, K \leq G$. V a K -module. Then

$$\text{Res}_H^G (\text{Ind}_K^G(V)) \simeq \bigoplus_{g \in [H \backslash G / K]} \text{Ind}_{H \cap {}^gK}^H \text{Res}_{H^g \cap K}^K V$$

Proof. $\text{Ind}_K^G V = \bigoplus_{x \in [G/K]} x \otimes V$.

For a double coset HgK , $\bigoplus_{x \in HgK/K} x \otimes V$ is an H -invariant subspace. $x \in HgK$ $\square$

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Theorem 2.0.260. $\text{char } k = p$, G a p -group. Then

1. the regular module KG is an indecomposable projective KG -module that is the projective cover of the trivial module
2. every f.g. projective kG -module is free
3. the only idempotent elements in KG are 0 and 1.

Chapter 3

Homework

3.1 Homework week 1

Exercise 3.1.1. Let R be a ring, and let M be an R -module. Suppose that S is a ring and

$$\theta : S \rightarrow R$$

is a ring homomorphism satisfying $\theta(1_S) = 1_R$. Let S act on M by

$$sx = \theta(s)x,$$

where $x \in M$ and $s \in S$. Prove that M is an S -module.

Proof. This action is well-defined because $\forall s \in S$, $\theta(s) \in R$ and $\theta(s)x \in M$ for all $x \in M$. We need to verify the module axioms, for all $s, t \in S$ and $x, y \in M$:

- $s(x + y) = \theta(s)(x + y) = \theta(s)x + \theta(s)y = sx + sy$
- $(s + t)x = \theta(s + t)x = \theta(s)x + \theta(t)x = sx + tx$
- $(st)x = \theta(st)x = \theta(s)\theta(t)x = \theta(s)(\theta(t)x) = s(tx)$.
- $1_Sx = \theta(1_S)x = 1_Rx = x$

Hence M is an S -module. □

Exercise 3.1.2. Let R be a ring, and let M and N be two right R -modules. Let $\text{Hom}_{\mathbb{Z}}(M, N)$ be the set of group homomorphisms from M to N (as abelian groups). The action of R on $\text{Hom}_{\mathbb{Z}}(M, N)$ is given as follows: for any $a \in R$ and $\varphi \in \text{Hom}_{\mathbb{Z}}(M, N)$, define

$$(\varphi a)(x) = \varphi(xa), \quad \forall x \in M.$$

Prove that $\text{Hom}_{\mathbb{Z}}(M, N)$ is a left R -module.

Proof. This action is well-defined. $\forall r \in R$, $\varphi r \in \text{Hom}_{\mathbb{Z}}(M, N)$ for all $\varphi \in \text{Hom}_{\mathbb{Z}}(M, N)$ as $(\varphi r)(x) = \varphi(xr) \in N$ for all $x \in M$. We need to verify the module axioms, for all $r_1, r_2 \in R$ and $\varphi, \psi \in \text{Hom}_{\mathbb{Z}}(M, N)$, $\forall x \in M$:

- $((\varphi + \psi)r)(x) = (\varphi + \psi)(xr) = \varphi(xr) + \psi(xr) = (\varphi r)(x) + (\psi r)(x)$
- $(\varphi(r_1 + r_2))(x) = \varphi(x(r_1 + r_2)) = \varphi(xr_1 + xr_2) = \varphi(xr_1) + \varphi(xr_2) = (\varphi r_1)(x) + (\varphi r_2)(x)$
- $((\varphi r_1)r_2)(x) = (\varphi r_2)(xr_1) = \varphi(xr_2r_1) = (\varphi r_2r_1)(x)$
- $(\varphi 1_R)(x) = \varphi(x 1_R) = \varphi(x)$

Hence $\text{Hom}_{\mathbb{Z}}(M, N)$ is a left R -module. □

Exercise 3.1.3. Let R be a ring, and let M be an R -module. Denote

$$\text{Ann}_R M = \{r \in R \mid rm = 0, \forall m \in M\}.$$

We call M a faithful R -module if $\text{Ann}_R M = \{0\}$. Define the action of the quotient ring $R/\text{Ann}_R M$ on M by

$$\bar{r}x = rx,$$

where $x \in M$, $r \in R$, and $\bar{r} = r + \text{Ann}_R M$. Prove that M is a faithful $R/\text{Ann}_R M$ -module.

Proof. First we show that the action is well-defined. Suppose

$$\bar{r} = \bar{r}' \in R/\text{Ann}_R M.$$

Then $r - r' \in \text{Ann}_R M$, so for every $x \in M$,

$$(r - r')x = 0.$$

Hence $rx = r'x$, so $\bar{r}x$ does not depend on the choice of representative.

The module axioms follow immediately from those of the R -module structure on M .

Now we prove faithfulness. Suppose $\bar{r} \in R/\text{Ann}_R M$ acts trivially on M , that is,

$$\bar{r}x = 0 \quad \forall x \in M.$$

By definition, this means

$$rx = 0 \quad \forall x \in M.$$

Therefore $r \in \text{Ann}_R M$, so $\bar{r} = \bar{0}$ in the quotient ring. Hence

$$\text{Ann}_{R/\text{Ann}_R M}(M) = \{\bar{0}\},$$

and M is faithful as an $R/\text{Ann}_R M$ -module. □

Exercise 3.1.4. Let $\varphi : M \rightarrow M$ be a homomorphism of R -modules such that

$$\varphi\varphi = \varphi.$$

Prove that

$$M = \ker \varphi \oplus \text{im } \varphi.$$

Proof. Let $x \in M$. Then $x = \varphi(x) + (x - \varphi(x))$. Where $\varphi(x) \in \text{im } \varphi$ and $x - \varphi(x) \in \ker \varphi$. Hence $M = \text{im } \varphi + \ker \varphi$. We only need to show that $\ker \varphi \cap \text{im } \varphi = \{0\}$. Suppose $x \in \ker \varphi \cap \text{im } \varphi$. Then $x = \varphi(y)$ for some $y \in M$ and $\varphi(x) = 0$. Then $\varphi(x) = \varphi\varphi(y) = \varphi(y) = x = 0$, so $x = 0$. Hence $\ker \varphi \cap \text{im } \varphi = \{0\}$. $\square$

Exercise 3.1.5. Show that any R -module is a homomorphism image of some free R -module.

Proof. Let M be a left R -module. Consider the free R -module

$$F = \bigoplus_{x \in M} Re_x$$

with basis $\{e_x\}_{x \in M}$ indexed by the underlying set of M . Define an R -module homomorphism

$$\pi : F \rightarrow M$$

by

$$\pi(e_x) = x \quad (x \in M),$$

and extend R -linearly. Thus

$$\pi\left(\sum_{i=1}^n r_i e_{x_i}\right) = \sum_{i=1}^n r_i x_i.$$

This is well-defined and R -linear.

For any $m \in M$, we have

$$\pi(e_m) = m,$$

so π is surjective. Therefore M is a homomorphic image of the free module F . Equivalently,

$$M \cong F / \ker \pi.$$

$\square$

Exercise 3.1.6. Let $\mathbb{Q}$ be the field of rational numbers. Is the additive group $(\mathbb{Q}, +)$ a free $\mathbb{Z}$ -module?

Proof. Claim: any two non-zero element of $\mathbb{Q}$ are linearly dependent over $\mathbb{Z}$. Hence if $\mathbb{Q}$ is a free $\mathbb{Z}$ -module, then it must have a basis of cardinality 1.

Proof of claim: let $0 \neq x = \frac{a}{b}$ and $0 \neq y = \frac{c}{d}$ be two elements of $\mathbb{Q}$. Then $dcx - aby = 0$. If $a = 0$ or $c = 0$, then $x = 0$ or $y = 0$, which contradicts the assumption. Hence $a, c \neq 0$ which means dc, ab are two non-zero integer coefficients. Hence x and y are linearly dependent over $\mathbb{Z}$. And if $\mathbb{Q}$ is a free $\mathbb{Z}$ -module, then it must have a basis of cardinality 1.

But if $\mathbb{Q}$ is a cyclic $\mathbb{Z}$ -module, then it has only one generator. Suppose it is $z = \frac{e}{f}$. Then $\mathbb{Q} = \{n \frac{e}{f} \mid n \in \mathbb{Z}\}$. But this is not possible because we can easily pick $\frac{1}{2f} \notin \mathbb{Q}$, which is absurd.

Therefore $\mathbb{Q}$ is not a free $\mathbb{Z}$ -module. $\square$